

A Life-Cycle Model of Mortgage Debt Relief: Refinancing and Forbearance during the COVID-19 Pandemic

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Abstract

When a crisis impacts households' mortgage obligations, relief can work through *prices* or through *eligibility*; by allowing borrowers to modify their mortgage debt or defer their payments, respectively. The COVID-19 response used both: a policy-rate cut transmitted through refinancing, which requires contemporaneous re-underwriting, and CARES forbearance, which gave eligible borrowers payment relief on request without it. This paper asks which borrowers each instrument reaches, and whether relief changes *who* defaults or only *when*. We study a partial-equilibrium life-cycle heterogeneous-agent model of mortgage choice with underwriting frictions, a multi-period delinquency bridge, non-recourse default, and a forbearance option whose length is governed by a within-year re-employment hazard rather than chosen—disciplined to the 2019 Survey of Consumer Finances and the 2020 mortgage market. As under CARES, forbearance is self-certified and approval is granted on request; what limits it is that a household must be carrying a payment of at least a fifth of its current income. The two instruments are substantially but incompletely segmented: the forbearance cohort refinances at 17.8% against 27.6% for never-applicants, about two-thirds the rate. That cohort is also far more fragile, with 36.6% eventually defaulting against 3.4% of never-applicants, so the option is concentrated on owners for whom foreclosure is a live prospect rather than a remote one. Relaxing each underwriting restriction in turn, the value of forbearance to the households it reaches falls to zero once the payment-to-income test is removed, the maturity rule is flattened, or refinancing is made costless, and falls by half or less when the down-payment requirement is loosened or cash-in refinancing is permitted. What separates the two instruments is the requirement to be re-underwritten at all rather than the price of doing so. That observed foreclosure rates nonetheless stayed low after the grace period ended suggests these policies were not the only force holding them down.

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1 Introduction

Emergency mortgage policy is rarely a single instrument. In a downturn, governments typically combine a *price* instrument that lowers the cost of credit—a policy-rate cut transmitted to households through mortgage refinancing—with an *eligibility* instrument that suspends scheduled payments without re-underwriting the loan. Both target the same aggregate, mortgage default, but they operate through opposite channels: refinancing is available only to borrowers who clear underwriting—positive equity, an acceptable payment-to-income ratio, documentable income—while payment relief is available to anyone. Whether the two instruments are substitutes or complements therefore depends on *whose* default each one can prevent, and *when*. This paper characterizes that incidence structurally. It shows that the instruments are substantially, though not completely, segmented; that forbearance recipients enter the crisis with both sharply lower income and higher leverage than refinancers; and that forbearance shifts the *timing* of default for a structurally fragile segment more than it lowers that segment’s lifetime default risk.

The segmentation is visible in loan-level data before any model. In the Fannie Mae Single-Family Loan Performance file, the borrowers who entered CARES-Act forbearance during 2020–2021 and those who refinanced during the concurrent rate-cut-driven boom occupy non-overlapping regions of the origination characteristic space. Eventual-forbearance borrowers originated at a mean loan-to-value (LTV) of 77.5% and a debt-to-income (DTI) of 38.3%; 2020–2022 rate-and-term refinancers, at 64.8% and 32.1%. The gap is present at origination—before either the pandemic or the CARES Act existed—and is large enough that the two groups have materially different origination risk profiles. The underwriting profile that makes a borrower likely to need forbearance is the same profile that makes that borrower unable to refinance. The 2020 episode is the natural laboratory, but the object of interest is general: which of two concurrent relief instruments reaches the borrowers facing default-margin stress. Section 2 documents this gap, and the reduced-form forbearance-protection evidence that accompanies it, in the loan-level data.

The U.S. response combined three concurrent interventions: a CARES-Act forbearance entitlement granted on borrower self-certification; a policy-rate cut that pushed the 30-year fixed mortgage rate to a historic low and triggered a refinancing wave of roughly 14 million loans; and a federal foreclosure moratorium. The moratorium administratively suspended foreclosure for federally backed loans, which makes the recorded 2020–2021 default decline largely mechanical and uninformative about the underlying credit response. The substantive question is which of the two *operative* instruments—payment relief or the rate cut—reached the borrowers facing default-margin stress. Because that question concerns the counterfactual reach of an instrument a given borrower did *not* use, it cannot be answered from realized data alone; it requires a structural model of the household’s choice set.

I build a life-cycle heterogeneous-agent model of mortgage choice in the incomplete-markets tradition (Eichenbaum et al., 2022; Wong, 2016), which also include frictions around mortgage debt origination. Similar to Low (2022), the model also adds a delinquency-bridge state—a missed payment that can persist for up to a year before resolving through cure, foreclosure, or forbearance—

and a forbearance option that routes an approved applicant into a deferred-payment state under CARES-style repayment, with an *endogenous duration* drawn from a re-employment hazard subject to a HAMP-style affordability gate, so take-up is self-limiting and the deferred fraction of the year’s payment is an outcome rather than a free parameter. Two features distinguish the specification and are central to the exercise. First, default is *non-recourse*: a defaulting owner surrenders the home and walks away from any deficiency, so the default decision is governed by the option value of walking away from negative equity rather than by a recourse penalty. Second, the crisis is an unanticipated (MIT) event grafted onto the aggregate state space and reached with zero probability from the normal states, the appropriate treatment for a shock households did not price in advance. The partial-equilibrium structure is deliberate: isolating the *incidence* of each instrument requires holding fixed the price feedback that a general-equilibrium housing block would introduce.

The calibrated economy reproduces the joint signature of the 2020 mortgage market before any of the incidence results are read. The cut-to-zero short rate sets off a refinancing wave in which about 24% of mortgaged owners refinance on impact, against the 26.8% COVID-period share in the Fannie Mae loan-level data. That share is a calibration target rather than a test. Forbearance is not: the eligibility threshold is set rather than fitted, so take-up is an outcome, and the model undershoots it—3.2% of mortgaged owners are approved on impact against the 5.4% of loans that entered forbearance in 2020. A model that reaches fewer households than the program did understates what the program does, so the counterfactuals below should be read as a lower bound. What the calibration does *not* target either, and the model delivers anyway, is that the foreclosure margin never leaves the floor: crisis default choices stay negligible. The model itself is disciplined to match the cross-section of homeownership and leverage in the 2019 SCF, and the refinancing, forbearance, and foreclosure margins of 2020.

The central finding is that the two instruments are *substantially segmented*, and that what separates them is the requirement to be re-underwritten at all. Tracking each shock-period owner, forbearance-approved (FA) owners refinance at 18.5% against 27.5% for non-applicants (NA)—a refinancing rate ratio of 0.67, well below one. Both cohorts are leveraged well above the routine payer, and the forbearance-approved are somewhat the more leveraged of the two—a pre-shock LTV of 0.70 against 0.60—but they differ far more sharply on flow income and on age: the forbearance-approved are the owners whose crisis income collapses low enough that they fail the payment-to-income test that refinancing into the historic-low rate would require, while refinancers keep their income at 0.99 and exercise the option the rate cut opens. The approved are also close to two decades older—a mean age of 60 against 42—and Section 6.6 takes up how much of the aggregate gap that age difference accounts for. The rate cut, in other words, reaches the high-leverage owner *only when her income is intact*; the owner whose income the shock destroys is exactly the owner forbearance must catch, because the price instrument cannot.

Relaxing each underwriting restriction one at a time shows how much of that gap each one generates, and the answer is narrower than the informal reading but firmer (Section 6.7). Removing the payment-to-income test and permitting cash-in refinancing each close most of the gap, by

amounts that cannot be told apart; relaxing the loan-to-value cap does considerably less, and the maturity rule and the transaction cost do less still. The binding object is therefore not a single test but re-underwriting itself—whether on flow income or on the ability to convert liquid wealth into repaired collateral. Segmented incidence is a structural consequence of that requirement, not an artifact of who happened to use each program.

Forbearance shifts the *timing* of default more than its *level*, and the model can say so as a treatment effect rather than an inference from cohort shares. Solving the identical economy with the option unavailable and simulating both under common random numbers, each household can be compared with itself (Section 7.2). Of those the program reached, only 3.3% avoided a foreclosure they would otherwise have suffered; 21.3% merely postponed one. Relief delayed foreclosure some six times more often than it prevented it. How that looks depends almost entirely on the horizon: at twelve months 10.3% of the cohort appears saved and no other outcome has yet materialised, and three-quarters of that apparent prevention is borrowed from the future. Descriptively the same pattern is visible in the cohort shares—the forbearance-approved default at 43.5% eventually against 2.6% for the never-applicant control, concentrated within a few periods of approval. Forbearance is thus best read as insurance against the shock-period liquidity event for a structurally fragile segment—older, high-leverage, income-exposed owners near retirement—not as a durable reduction in that segment’s lifetime default risk. This suggests that forbearance alone cannot explain why foreclosure rates remained lower than pre-pandemic levels after the forbearance grace period ended. A consumption-equivalent welfare comparison over the age $\times$ LTV grid confirms the reading, and sharpens it. Both instruments beat continuing to pay at the median, and by similar margins. Against default they separate: refinancing dominates it for 95% of refinancers, while only 34% of the approved end the year after the shock in a better position than if they had surrendered the house. Since these are continuation values, that is not a claim that the approved chose wrongly—they bought consumption during the hardship year with a worse position after it—but it is the postponement result showing up in the value function as well as in realized foreclosure dates. What gains forbearance does deliver against default land in the high-LTV, near-retirement cells for which foreclosure is the binding alternative.

The structural finding is corroborated, reduced-form, in the loan-level data. Among loans that reach serious (90+ day) delinquency, the eventual foreclosure rate is 17.0% for the pre-COVID (2016–2019) cohort that had no access to CARES, 6.1% for the COVID cohort that did not use forbearance, and 1.0% for the COVID cohort that did. The gap between the two COVID groups isolates the forbearance channel from the moratorium and the house-price boom; a competing-risks hazard model that switches the exposure on at the observed forbearance entry month and treats prepayment as a competing risk rather than as censoring puts the estimated forbearance foreclosure hazard at 0.13 of the no-forbearance hazard. Because forbearance timing and take-up are endogenous, I read that estimate as an association rather than a treatment effect (Section 2.3). The structural model supplies the counterfactual the reduced-form cannot: it identifies the population each instrument reaches and shows that the margin between them is the income test.

The paper makes three contributions. To the literature on monetary transmission through mortgage refinancing (Eichenbaum et al., 2022; Wong, 2016; Beraja et al., 2019; Di Maggio et al., 2017) and on refinancing frictions in recessions (DeFusco and Mondragon, 2020; Greenwald, 2026; Chen et al., 2020), it adds a second, non-price instrument. Chen et al. (2020) estimate a structural model in which realistic borrowing constraints determine which households can refinance at all; this paper shows that the households those constraints exclude are the ones forbearance reaches, so that relief which requires no re-underwriting fills the structural hole the price instrument leaves—the income-stressed, high-leverage segment. To the empirical COVID-forgiveness literature (Goodman and Zhu, 2024; Cherry et al., 2021; Gerardi et al., 2022; An et al., 2022; Boctor, 2024), which estimates the marginal effect of forgiveness for the borrowers who used it, it supplies what an instrumental-variable design cannot: an identification of *who* the policy reaches and the lifetime trajectory of the treated cohort beyond the within-shock window. To the structural literature on mortgage design and foreclosure policy (Guren et al., 2021; Campbell et al., 2021; Mitman, 2016), which asks what contract or policy is optimal in general equilibrium, it takes the realized crisis instruments as given and asks instead about their incidence; the recession option of Campbell et al. (2021)—interest-only payment with maturity extension—is closely related in economic function to the deployed CARES forgiveness studied here, though the two differ in their institutional repayment arrangements.

Roadmap. Section 2 documents the motivating loan-level evidence. Section 3 situates the paper in the related literatures. Section 4 presents the model and Section 5 the calibration, the parameter-to-target mapping, and the model fit. Section 6 reports the policy-incidence analysis—what each instrument does, who each instrument reaches, and which underwriting constraint generates the difference. Section 7 separates prevention from delay and reports the welfare comparison. Section 8 collects the robustness exercises, the cost discussion, and the scope conditions under which the counterfactuals should be read. Section 9 concludes.

2 Motivating Evidence: Three Facts in the Loan-Level Data

This subsection documents three facts in Fannie Mae loan-level data that the structural model is built to reconcile. First, prospective forgiveness users and pandemic-era refiners have characteristically different *origination profiles*—in LTV, credit score, loan amount, and PTI at the time the loan was written (Section 2.1). The claim is about the profiles these groups were originated with, not that the underlying borrower populations are disjoint; the public file does not link a terminated loan to the refinance loan that replaced it outside the HARP mapping, so an overlap statement cannot be identified here. The structural model supplies the overlap diagnostic instead (Section 6.6). Second, the forgiveness population is, on the leverage and payment-stress dimensions, the same population that foreclosure draws from (Section 2.2). Third, among seriously delinquent borrowers, forgiveness is associated with a large reduction in foreclosure that survives controlling for the

COVID macro environment (Section 2.3). These are reduced-form facts; they are selection-laden and not, on their own, causal. The structural model is what supplies the counterfactual reach of each instrument.

2.1 Policy Context and the Segmented-Cohort Question

The U.S. COVID-19 response combined three concurrent mortgage-market interventions. The CARES Act (March 2020) granted forbearance rights to all households with federally backed mortgages on self-certification, with no underwriting; by mid-2020 roughly 8% of Fannie Mae loans were in active forbearance. The Federal Reserve cut the policy rate to zero, pushing the 30-year fixed mortgage rate to a historic low and triggering a refinancing wave of roughly 14 million loans. A federal foreclosure moratorium ran alongside both through July 2021. The first two instruments sit on opposite sides of an underwriting boundary: forbearance required no underwriting, while the refinance channel required positive equity, a passing payment-to-income ratio, and a credit history clearing post-2010 standards. The moratorium suspended the foreclosure process administratively, which makes the recorded default decline largely mechanical. The substantive question is which of the two operative instruments reached the borrowers facing default-margin stress—which requires knowing the segment each instrument actually serves.

Table 1 reports origination characteristics for a 2-percent random sample of the Fannie Mae Single-Family Loan Performance dataset, loans originated 2000–2024, partitioned by eventual outcome. Table 2 reports origination characteristics for loans with a refinance purpose originated during the 2020–2022 window.

The differences run in the expected direction and are large. Future forbearance users originated at a mean LTV of 77.5% and DTI of 38.3%, against 70.3% and 33.6% for the never-event population; rate-and-term refinancers in 2020–2022 sit at 64.8% and 32.1%. The underwriting profile that makes a borrower likely to use forbearance is precisely the profile that would make the same borrower unable to refinance under standard post-2010 rules. Universal CARES-style forbearance reaches a population that market-based relief misses entirely.

These gaps are not an artifact of pooling loans across very different origination eras. Figure 20 plots the annual mean of each origination characteristic, separately for the three eventual-outcome groups. At essentially every vintage for which a group is populated, eventual-forbearance and eventual-foreclosure loans carry higher LTV and DTI and lower credit scores than never-event loans: the ordering in Table 1 holds vintage by vintage, not only on average. The eventual-foreclosure cohorts thin out sharply after 2012—post-QM originations rarely foreclose, so the later points rest on a few dozen loans and the confidence band widens accordingly—but the level gaps persist to the end of the sample; the underlying composition shift is taken up in Section 2.2.

2.2 Selection into Distress, Forbearance, and Foreclosure

The disjoint-cohort gap is not driven by a single characteristic. Table 28 reports pairwise mean differences across the forbearance, foreclosure, and never-event populations in the larger 2-percent

Table 1: Summary Statistics at Origination by Eventual Loan Outcome

Variable	Forbearance loans		
	N	Mean	SD
Original UPB (\$)	24,359	251,178	135,695
DTI (%)	24,257	38.3	8.6
Original LTV (%)	24,359	77.5	15.7
Credit Score	24,338	725.5	50.0
Variable	Foreclosure loans		
	N	Mean	SD
Original UPB (\$)	13,464	160,537	94,716
DTI (%)	13,004	39.0	11.9
Original LTV (%)	13,464	79.0	12.1
Credit Score	13,353	692.6	57.0
Variable	Never-event loans (all other)		
	N	Mean	SD
Original UPB (\$)	1,095,118	217,252	129,079
DTI (%)	1,080,304	33.6	11.1
Original LTV (%)	1,095,118	70.3	17.8
Credit Score	1,091,759	748.4	50.8

Note: 2-percent random sample from the Fannie Mae Single-Family Loan Performance dataset, loans originated 2000–2024. “Foreclosure loans” comprises all loans reaching foreclosure; the two distressed groups are not mutually exclusive.

Table 2: Summary Statistics at Origination by Refinance Type, 2020–2022

Variable	Rate-and-term refinance		
	N	Mean	SD
Original UPB (\$)	172,571	258,870	136,197
DTI (%)	172,274	32.1	10.1
Original LTV (%)	172,571	64.8	17.0
Credit Score	172,520	765.5	43.2
Variable	Cash-out refinance		
	N	Mean	SD
Original UPB (\$)	118,654	228,578	129,254
DTI (%)	118,389	34.3	9.8
Original LTV (%)	118,654	62.7	15.6
Credit Score	118,634	746.0	50.0

Note: First occurrence of each loan with purpose R (rate-and-term) or C (cash-out), originated 2020–2022, from the same 2-percent Fannie Mae sample as Table 1.

all-purpose sample. The forbearance population looks like the foreclosure population on the *leverage and payment-stress* dimensions—LTV (77.5% vs. 78.1%) and DTI (38.3% vs. 39.0%)—but is a higher-credit slice (FICO 725 vs. 697). Relative to the never-event population, forbearance users are higher-LTV (+7.3 pp), higher-DTI (+4.8 pp), and lower-FICO (−22.9 points), all significant at $p < 0.001$. Forbearance reached the high-leverage, high-payment-stress segment from which foreclosures are drawn—the slice of it with enough credit quality to have been served by nothing else.

The selection shows up equally in the univariate gradient of default itself, with no regression at all. Figure 21 sorts the 2-percent sample into deciles of estimated income and of origination LTV and plots lifetime delinquency and foreclosure rates within each. Foreclosure falls almost monotonically with income—from 2.8% in the lowest income decile (median $\approx$ \$14k) to 0.25% in the highest ($\approx$ \$115k), a factor of roughly eleven—and 90-day delinquency from 7.6% to 1.5%. Against leverage the gradient runs the other way: foreclosure rises with origination LTV and peaks at the 90%-LTV decile (2.3%). The borrowers exposed to default-margin stress are thus disproportionately low-income *and* high-leverage—and because refinancing into the record-low rate required clearing a payment-to-income test, that is exactly the population the price instrument screens out. The income margin the structural model isolates is already legible in the raw default data.

A set of linear probability models confirms the selection pattern at the loan level. Origination LTV, DTI, and the note rate raise the probability of both serious delinquency and forbearance take-up, while a higher credit score lowers both (Figure 1); in the foreclosure-conditional-on-distress model an ever-forbearance indicator carries a large negative coefficient—−27.0 percentage points across all vintages—as does a COVID-onset indicator (Figure 2). Tables 21 and 22 in Appendix C report the full coefficient set behind both figures. Re-estimating all six specifications by logit leaves these conclusions intact: the average marginal effects agree in sign with the linear coefficients on every standardised origination characteristic and on both main effects of the foreclosure model (Tables 23 and 24). The one coefficient that does not survive the change of estimator is the forbearance $\times$ COVID interaction, which the linear model puts at +23.7 points across all vintages and +7.1 on the post-2010 book but which the logit cannot distinguish from zero in the first case and estimates as negative in the second. The cell that interaction identifies—COVID-onset delinquencies that entered forbearance—foreclosed at 1.0% (Table 3), so it sits exactly where a linear probability model is least trustworthy, and we read nothing into it. These associations are endogenous and selection-laden in any case, which is why the causal weight in this paper rests on the structural counterfactual.

One dimension *does* separate the forbearance and foreclosure populations sharply—not a borrower characteristic but an origination *vintage*. Figure 22 plots, for each group, how its loans distribute across origination years. The two distressed populations are nearly disjoint in time: ever-foreclosure loans are overwhelmingly the pre-crisis book—89% originated before 2010, peaking at the 2007 vintage—while ever-forbearance loans are a modern, post-crisis cohort—72% originated between 2015 and 2021, peaking in 2019. This composition fact bears directly on the next subsec-

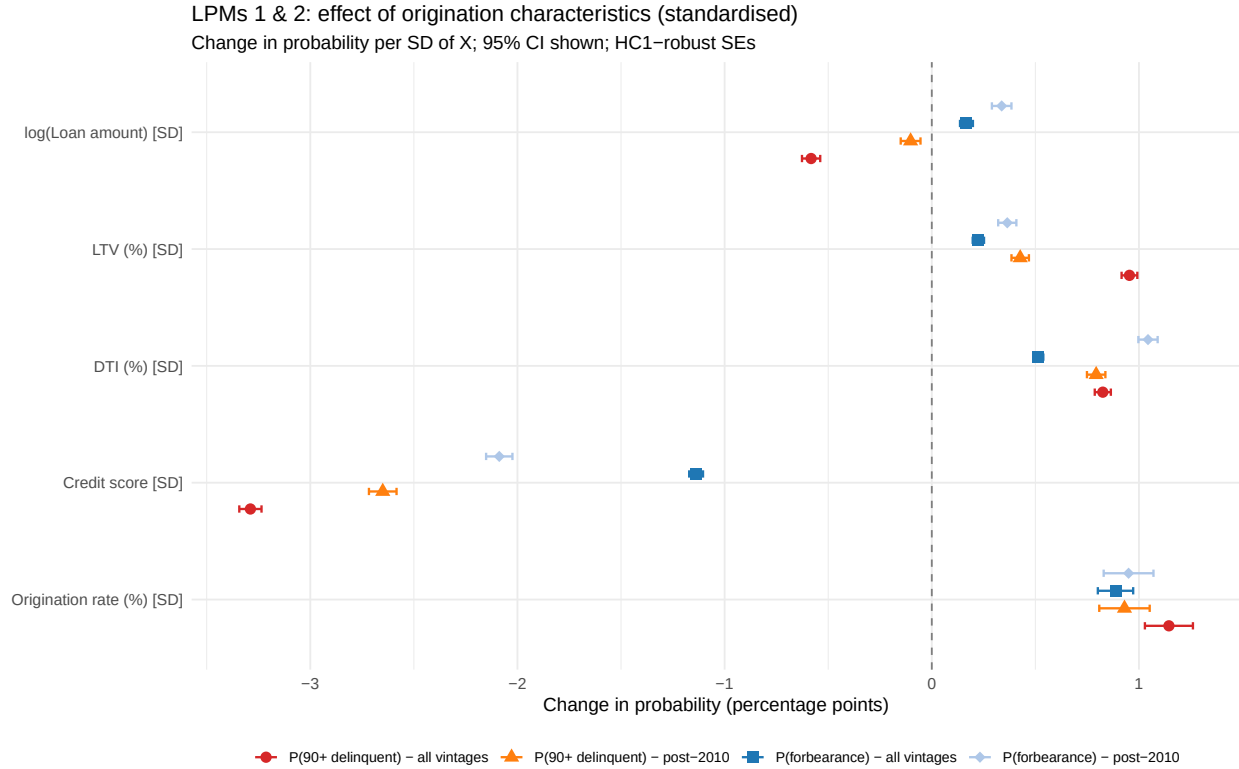


Figure 1: Linear probability models 1 and 2: per-standard-deviation effects of origination characteristics on $\Pr(\text{ever } 90+ \text{ delinquent})$ and $\Pr(\text{ever forbearance})$, in percentage points, each estimated on all vintages and on the post-2010 book, with HC1-robust 95% confidence intervals. All four models carry origination-year fixed effects; the full coefficient set, including the term and loan-purpose indicators the figure omits, is Table 21.

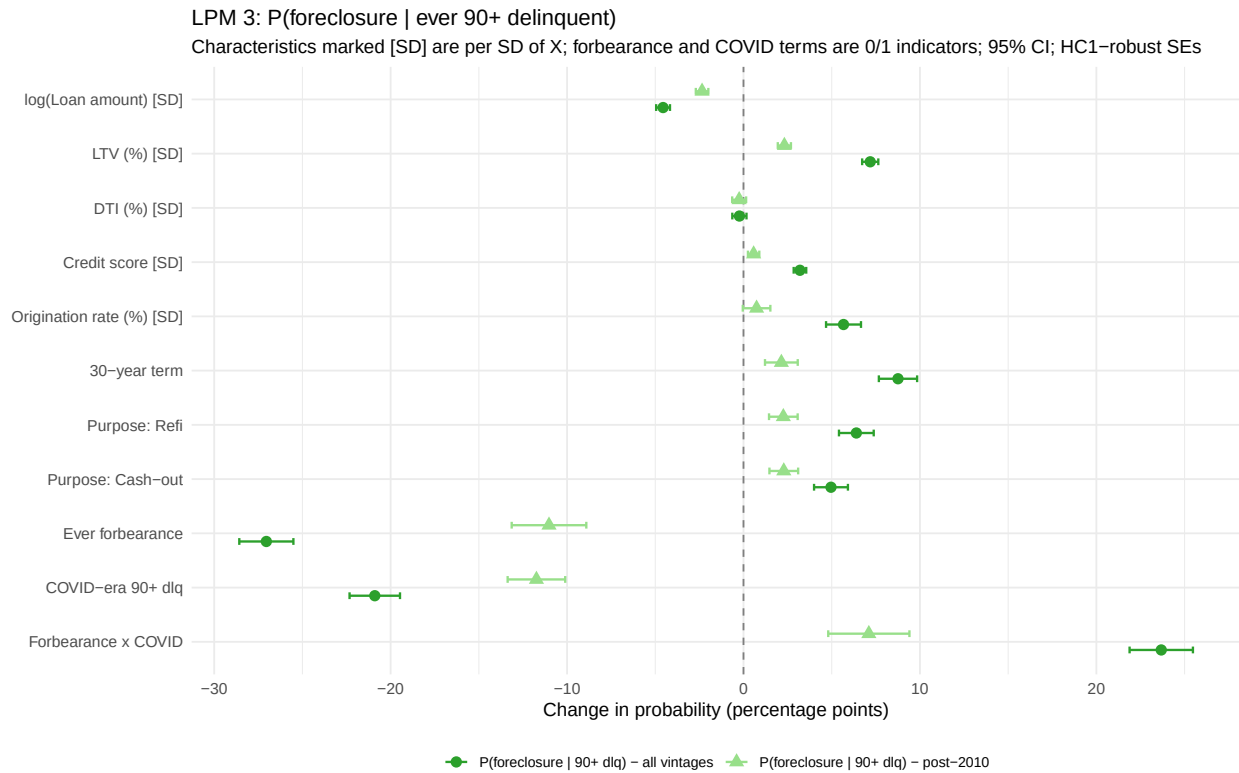


Figure 2: Linear probability model 3: $\Pr(\text{foreclosure} \mid \text{ever 90+ delinquent})$, adding an ever-forbearance indicator, a COVID-onset indicator, and their interaction to the origination characteristics of Figure 1, with HC1-robust 95% confidence intervals, shown for all origination vintages and for the post-2010 book. Coefficients are in percentage points; characteristics marked [SD] are per standard deviation, while the forbearance and COVID terms are indicators. The ever-forbearance and COVID-onset indicators both carry large negative coefficients, and that pattern is stable across the two vintage windows and across estimators. The forbearance $\times$ COVID interaction is positive in both windows here but does not survive re-estimation by logit (Table 24), and is discussed in the text. Table 22 reports these estimates with standard errors.

tion: because the two books are drawn from different origination eras, part of the raw 2020–2021 foreclosure decline reflects a younger, better-underwritten stock of loans rather than relief itself. The A/B/C comparison in Section 2.3 is built to difference that out, by holding the origination era fixed and comparing COVID-era loans that did and did not take forbearance.

2.3 Did Forbearance Protect, Net of the Macro Environment?

The reduced-form fact most directly relevant to the policy question is whether forbearance, as opposed to the moratorium and the house-price boom, drove the collapse in foreclosure among the distressed. Table 3 partitions loans that reached 90+ delinquency into three groups: (A) those whose first serious delinquency occurred pre-pandemic (2016–2019), with no access to CARES; (B) COVID-onset delinquencies that used forbearance; and (C) COVID-onset delinquencies that did not. Group C is the key control: if the macro environment alone drove the decline, C should look like B; if forbearance mattered, C should look like A.

Table 3: Foreclosure Outcomes Among Seriously Delinquent Loans

Group	N	Foreclosed	Fcl. rate (%)	Prepaid (%)
A: Pre-COVID (2016–2019)	4,166	709	17.0	46.7
B: COVID, with forbearance	13,766	133	1.0	41.2
C: COVID, without forbearance	3,531	214	6.1	25.3

Note: Loans reaching 90+ days delinquent, 2-percent sample. “Fcl. rate” is the share of the group that foreclosed over the observation window. All pairwise differences in foreclosure rate are significant at $p < 0.001$.

Group C foreclosed at 6.1%, far above Group B’s 1.0%: among COVID-onset delinquencies, the foreclosure rate of those *without* forbearance was six times that of those *with* it. Group C is a selected group—these are borrowers who were distressed yet declined free relief, so the comparison is suggestive rather than causal—but the pattern is striking: the macro environment alone—moratorium plus appreciation—did not, on its own, deliver the durable protection that Group B enjoyed. The cumulative-foreclosure curves (Figure 3) make the point sharply: by 36 months after onset, Group C had foreclosed at 32.9%—*above* the pre-COVID cohort’s 21.1% at the same horizon—while Group B remained at 1.2%. Among the loans that did eventually foreclose, forbearance also delayed the terminal event: the distribution of time from first 90+ delinquency to foreclosure shifts about two years to the right for Group B relative to Groups A and C (Figure 4).

A Cox proportional-hazard model of time-to-foreclosure among these borrowers, controlling for origination LTV, DTI, credit score, rate, balance, and loan purpose, puts the forbearance hazard at 0.117 of the no-forbearance hazard. Two features of that specification make the number unreliable as written, and both are worth confronting directly because the corrections turn out to matter for the design and not for the conclusion.

The first is *immortal time*. An ever-forbearance indicator in a model whose clock starts at first 90+ delinquency assigns to baseline information that only arrives later: a borrower who enters forbearance three months after the clock starts must survive those three months to be classified as

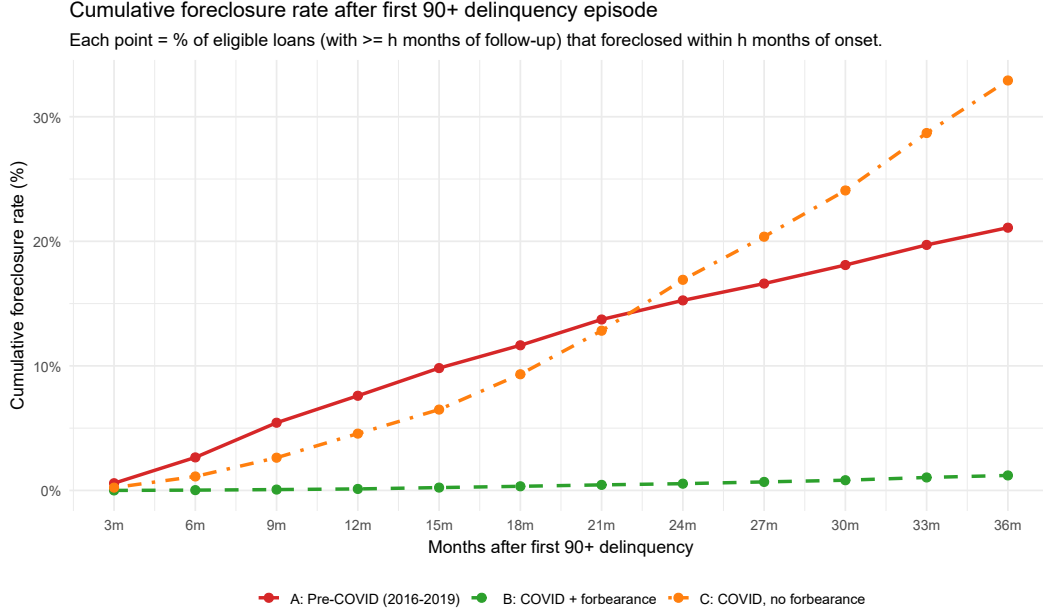


Figure 3: Cumulative foreclosure rate by months since first 90+ delinquency, for the three groups of Table 3. The COVID-without-forbearance group (C) overtakes the pre-COVID group (A) by roughly two years after onset; the COVID-with-forbearance group (B) stays near 1%.

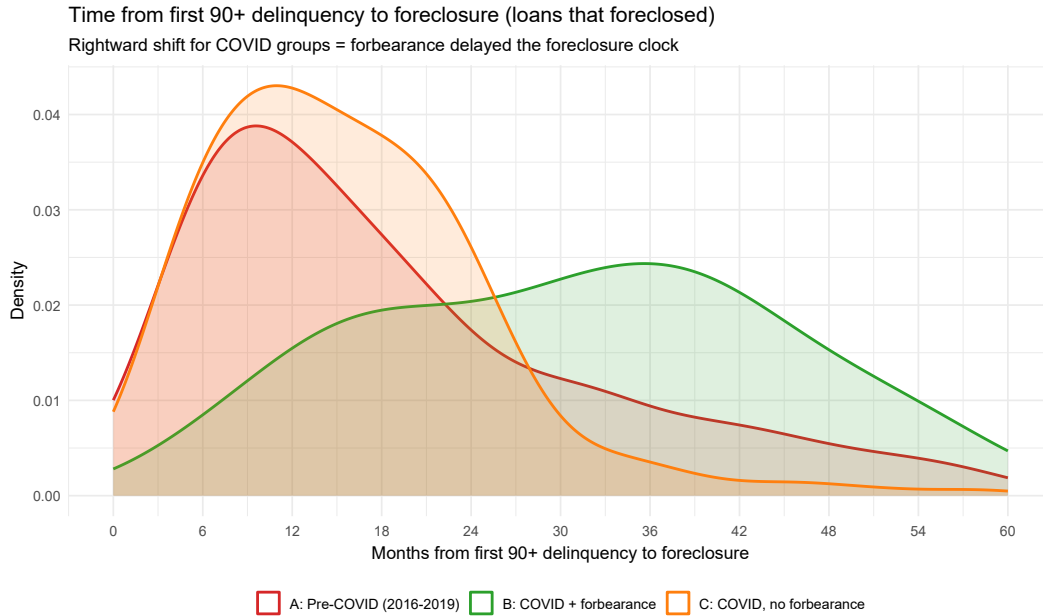


Figure 4: Timing of foreclosure among loans that foreclosed: kernel density of the number of months from first 90+ delinquency to foreclosure, for the three groups of Table 3. The COVID-with-forbearance group (B) is shifted markedly to the right, peaking near three years after onset, whereas the pre-COVID group (A) and the COVID-without-forbearance group (C) peak within the first year. For the loans that still foreclosed, forbearance postponed the terminal event by roughly two years.

treated, so survivorship is built into the treatment definition. The fix is to make the exposure time-varying, switching it on at the observed entry month. The second is *competing risks*. Prepayment removes a loan from the risk set for reasons unrelated to foreclosure, and it is not rare here: the cumulative incidence of prepayment reaches 35–40% within three years in both arms (Table 5), far too large to treat as independent censoring. The fix is the Fine–Gray subdistribution hazard, which models the cumulative incidence of foreclosure in the presence of prepayment rather than conditioning it away.

Table 4 reports the original specification alongside each correction and both together. The estimated protection is stable across all four: the hazard ratio moves from 0.117 under the original specification to 0.125 once exposure is aligned in time, to 0.137 once prepayment is treated as a competing risk, and to 0.126 under both corrections jointly. Every interval overlaps every other. The methodological objections are correct, in other words, and they attenuate the estimate by roughly a tenth without disturbing the finding: conditional on reaching serious delinquency, borrowers who entered forbearance foreclosed at a small fraction of the rate of those who did not.

Because the estimated hazard is not proportional—a Grambsch–Therneau test rejects proportionality for the forbearance term ($p < 0.001$), though not for the origination risk covariates—the single ratio should be read as an average over the follow-up window rather than a constant effect. The cumulative incidence functions in Table 5 are the more transparent statement of the same result and do not rest on proportionality. Three years after first 90+ delinquency, foreclosure had occurred for 1.5% of borrowers who entered forbearance within twelve months against 17.2% of those who did not, while prepayment reached 39.5% and 35.1% respectively. The gap in foreclosure is an order of magnitude; the near-equality of the prepayment columns is why the competing-risk correction, though necessary, changes so little.

None of this identifies a causal effect. Borrowers opt into forbearance, servicers differ in how readily they grant it, and distress severity drives both take-up and foreclosure, so the corrections above remove two specific design flaws without addressing selection. A third concern is more serious than either and is not a matter of estimator choice: `DLQ_STATUS` counts missed payments, and a borrower in forbearance is exercising a contractual right not to pay, so the 90+ threshold that defines the risk set is itself partly a consequence of treatment. Appendix H rebuilds the frame on a delinquency clock that does not accrue during forbearance. Roughly two-thirds of the association does not survive; what remains is a reduction of 1.3 to 1.8 percentage points, estimated on a frame with far better common support than the one behind Table 4. The counterfactual—whether the rate cut would have reached these same borrowers—is what the structural model is built to answer.

The same reversal is visible in how delinquencies resolved in the aggregate, beyond the selected 90+-day sample. Figure 23(a) decomposes the exit from every delinquency spell into its first terminal event. Before the pandemic, 12.3% of spells ended in foreclosure and almost none—0.3%—in forbearance; during COVID the foreclosure exit collapsed to 0.8% while forbearance became the modal alternative to curing, absorbing 22.2% of spells. Forbearance did not so much lower the foreclosure rate as *replace foreclosure as the resolution channel*. Panel (b) supplies the other half:

Table 4: Foreclosure hazard after first 90+ delinquency: the original specification and three corrections.

Specification	Exposure	Hazard ratio	95% CI
Original (prepayment censored)	Ever forbearance	0.117	[0.099, 0.139]
Time-varying exposure	Entry month	0.125	[0.105, 0.148]
Fine-Gray subdistribution	Ever forbearance	0.137	[0.116, 0.163]
Landmark ($L = 6$) + Fine-Gray	Entry by month 6	0.126	[0.105, 0.151]

Notes: Risk set is loans reaching 90+ delinquency during COVID; clock starts at first such delinquency. All specifications control for origination LTV, DTI, credit score, note rate, log balance, and loan purpose. Row 1 reproduces the design this paper previously reported: an ever-forbearance indicator with prepayment treated as independent censoring. Row 2 switches the exposure on at the observed forbearance entry month, removing immortal-time bias. Rows 3 and 4 estimate the Fine-Gray subdistribution hazard with prepayment as a competing risk; their working samples are the expanded risk-set representations that estimator requires and are not comparable to the roughly 15,000 loans underlying rows 1 and 2. All $p < 0.001$. A Grambsch-Therneau test rejects proportionality for the forbearance term ($\chi^2 = 14.1$, $p < 0.001$) and, more weakly, for loan purpose ($p = 0.027$); LTV, DTI, credit score, note rate, and balance satisfy it ($p > 0.08$ throughout). Estimates are associations, not causal effects. The risk set is defined on the raw `DLQ_STATUS` field, which continues to accrue while a loan is in forbearance; Appendix H re-estimates the same comparison on a clock that does not, and reports how much of the association survives.

Table 5: Cumulative incidence of foreclosure and prepayment, by forbearance status within twelve months.

Months since first 90+	Foreclosure (%)		Prepayment (%)	
	Forbearance	No forbearance	Forbearance	No forbearance
6	0.02	1.71	8.85	12.33
12	0.09	4.18	19.60	19.59
24	0.88	12.26	34.58	29.55
36	1.48	17.20	39.49	35.15

Notes: Aalen-Johansen cumulative incidence in the competing-risks model, stratified by whether the borrower entered forbearance within twelve months of first 90+ delinquency. Foreclosure and prepayment are the two absorbing states; the complement is still active or cured. Because these are cumulative incidences rather than hazards, they require no proportionality assumption. The prepayment columns are the reason prepayment cannot be treated as independent censoring; their near-equality across arms is why doing so nonetheless biased the foreclosure estimate only modestly.

before the pandemic the probability that a delinquency cured fell steeply with how deep it had gone—from 82% at 30–59 days to 37% once it reached 180+ days—so severity strongly predicted foreclosure. During COVID that gradient nearly vanished; even loans 180+ days delinquent cured about 90% of the time. Depth of distress stopped predicting whether the borrower kept the home—the signature of a universal, no-underwriting relief entitlement. These aggregate patterns are the data counterpart to the model’s central result that forbearance shifts the *timing* of default rather than its lifetime level (Section 7.1).

These reduced-form facts establish the pattern but cannot answer the counterfactual question—would the rate cut have reached the same borrowers?—because no borrower is observed under both instruments, and forbearance take-up is selected on exactly the unobservables that drive default. The structural model in Section 4 is designed to recover that counterfactual.

2.4 What the Loan-Level Evidence Does and Does Not Establish

Two scope conditions govern how the facts above should be read.

First, on *coverage*. Fannie Mae’s Single-Family Loan Performance file covers fully amortizing, full-documentation, conventional fixed-rate mortgages, and excludes adjustable-rate, government-insured, HARP/Refi Plus, interest-only, and other non-standard loans. The evidence therefore describes the conventional fixed-rate conforming segment, not the entire U.S. mortgage market. This is an important population—it is the population CARES forbearance and the 2020 refinancing wave both operated on—but the descriptive shares reported here should not be read as market-wide.

Second, on *interpretation*. Tables 1 and 2 compare the *origination* characteristics of loans that eventually entered forbearance with those of loans originated as rate-and-term refinances in 2020–2022. These are not two alternative outcomes for a common pre-pandemic population: the second group consists of loans originated during the window, and the public file does not link an old loan to the new loan that refinanced it outside the HARP mapping file. Accordingly the claim these tables support is that *origination risk profiles differ across the two groups*, not that a given borrower could or could not have refinanced. The counterfactual reach of an instrument a borrower did not use is what the structural model is for.

Finally, the borrower groups compared in Section 2.3 are self-selected. Borrowers opt in to forbearance, servicers differ in how readily they grant it, and distress severity influences both take-up and foreclosure. The comparisons below are therefore reported as associations throughout.

3 Related Literature

The paper relates to four strands: life-cycle heterogeneous-agent mortgage models, mortgage default and the delinquency-to-foreclosure pipeline, COVID-19 forbearance and payment relief, and monetary-policy transmission through the mortgage market. I describe what each has established and where the present paper extends or departs.

3.1 Life-Cycle Heterogeneous-Agent Mortgage Models

The framework combines the incomplete-markets tradition of Aiyagari (1994) and Krusell and Smith (1998), the bequest-motivated life-cycle structure of De Nardi (2004) and Ríos-Rull (1995), and the housing-finance life-cycle literature of Campbell and Cocco (2015) and Chambers et al. (2009). Campbell and Cocco (2015) develop the canonical structural model of mortgage choice—pay, re-finance, or default depending on income, equity, and the locked-versus-market rate gap—whose decision-tree structure underpins the policy-function architecture used here. Chambers et al. (2009) show that an LTV-constrained life-cycle model with transaction costs matches the rising home-ownership profile in U.S. data. Iacoviello and Pavan (2013) document how the fixed-rate-mortgage structure produces state-dependent exposure to interest-rate movements over the life cycle, and Li and Yao (2007) characterize the asymmetric distributional incidence of house-price changes across age cohorts—older equity-holders gain, younger would-be buyers lose—an asymmetry directly relevant to the appreciating-price COVID scenario, in which rising equity is what keeps the refinance channel open for older owners.

The most direct predecessor on the monetary-housing side is Eichenbaum et al. (2022), who quantify the state-dependent refinancing channel through which monetary policy reaches housing spending: the aggregate multiplier varies with the distribution of locked rates because the option to refinance is conditional on being above the prevailing market rate. Their paper does not have a default/foreclosure mechanism, which I follow from Low (2022). I adopt the discretized aggregate state-space and the life-cycle solution method from this literature; the departure is to add a forbearance instrument and a multi-period delinquency-bridge state, which together identify a default-margin channel that neither paper analyzes, and to make the object of study the *incidence* of two instruments rather than the transmission of one. A complementary heterogeneous-agent New Keynesian tradition (Kaplan et al., 2018, 2020) establishes that the direct intertemporal-substitution effects of monetary policy are small relative to indirect income effects—consistent with the present finding that the rate cut reaches default only indirectly, through the income-conditioned refinancing margin. Garriga and Hedlund (2020) model housing illiquidity and fire-sale externalities in a quantitative life-cycle setting; the delinquency-bridge state here captures a related household-level friction (cure costs deterring foreclosure exits) that the macro literature treats at the aggregate level.

3.2 Mortgage Default and the Delinquency Pipeline

The structural-default literature (Campbell and Cocco, 2015; Corbae and Quintin, 2015; Foote et al., 2008) identifies a small number of mechanical drivers of default. Foote et al. (2008) establish that negative equity is necessary but *not* sufficient: an income trigger is typically required. Ganong and Noel (2023) sharpen this with a causal decomposition—only about 6% of underwater defaults are driven purely by negative equity, roughly 70% by negative cash-flow events, and the remainder by their interaction—which motivates the joint income-and-LTV structure of the default decision here and is precisely why the income margin, rather than the equity margin, separates the two

instruments in this paper. Corbae and Quintin (2015) and Mitman (2016) study foreclosure and debt-relief policy in general-equilibrium structural models; Mitman (2016) in particular shows that debt-relief policy can sharply reduce foreclosures with modest aggregate price effects, the natural structural predecessor to the present policy-incidence exercise. Bhutta and Keys (2016) document how sensitive home-equity extraction is to interest rates, and Andersen et al. (2020) how far households depart from frictionless refinancing—between them establishing that the liquidity households can reach through their mortgage is governed by frictions rather than by fundamentals alone. A separate strand studies the delinquency-to-foreclosure pipeline: Loewenstein et al. (2025) model the cure-versus-foreclose choice, and this paper’s late-fee parameter follows the cure cost they assume; Collins and Urban (2018) show that the 2008–2010 foreclosure moratorium accelerated cure rates.

3.3 Forbearance, Payment Relief, and the CARES Act

The motivating result on the relative efficacy of payment reduction versus principal forgiveness is Ganong and Noel (2020): among underwater borrowers, a maturity extension that cuts the required payment while leaving the net present value of what is owed roughly unchanged sharply reduces default, whereas a principal reduction that raises borrower net worth without changing the near-term payment does not. This directly motivates the forbearance design, in which full payment deferral is available across the relevant state space. Fuster and Willen (2017) show that payment size moves default independently of equity, with an effect that survives even among deeply underwater borrowers; Agarwal et al. (2017) document that HAMP’s default-reduction effect was blunted by the limited renegotiation capacity of a few large servicers—an organizational and processing friction that the CARES self-certification process was designed to circumvent. The empirical COVID forbearance literature (Cherry et al., 2021; Goodman and Zhu, 2024; Gerardi et al., 2022; An et al., 2022; Bector, 2024) estimates the marginal effect of CARES forbearance for the borrowers who used it. Goodman and Zhu (2024) estimate that improved loss mitigation prevented roughly 536,000 foreclosures between April 2020 and March 2023; Gerardi et al. (2022) document that pandemic forbearance outperformed Great-Recession modification programs and that minority borrowers were both more likely to need forbearance and less likely to refinance; An et al. (2022) show that lower-income and minority borrowers took up relief at higher rates but defaulted more after exiting. These estimates are consistent with the present finding that forbearance was the active default-prevention margin during the shock; the structural decomposition adds the population each instrument reaches and the post-shock trajectory of the treated cohort.

3.4 Monetary-Policy Transmission through the Mortgage Market

A large empirical literature documents how rate movements propagate through refinancing to consumption. Beraja et al. (2019) document regional heterogeneity in refinancing responses driven by the housing-equity distribution—a geographic margin that parallels the life-cycle margin here. Di Maggio et al. (2017) provide causal evidence on rate-cut pass-through using ARM resets; Berger et al. (2021) show prepayment is path-dependent; DeFusco and Mondragon (2020) show that

employment-documentation requirements and out-of-pocket closing costs bind hardest in recessions and block refinancing for the income-stressed—the micro-empirical counterpart of the disjoint-incidence mechanism this paper derives structurally. The structural counterpart is Chen et al. (2020), who estimate a model in which refinancing is a liquidity-management device—households extract home equity to smooth consumption against countercyclical income risk—and in which a loan-to-value limit, a loan-to-income limit and a quasi-fixed origination cost jointly determine who can use it. Their finding is the premise of this paper: a household that cannot clear those restrictions gains little from a fall in interest rates, however large the fall. A rate cut is a conditional transfer, and the condition is underwriting. That is precisely the gap forbearance fills: it delivers payment relief to households who cannot afford to refinance—who cannot clear the income test, cannot raise the closing costs, or lack the equity—and it does so without re-underwriting them, which is why the two instruments reach different populations rather than the same one twice. A theoretical strand examines optimal mortgage-contract design (Campbell et al., 2021; Guren et al., 2021): Guren et al. (2021) characterize welfare-improving countercyclical contracts in a housing general equilibrium, and Campbell et al. (2021) show that an adjustable-rate contract with a recession option (interest-only with maturity extension) stabilizes consumption, shifts defaults to expansions, and is cheap to a risk-averse lender. This paper takes the existing crisis instruments as given and asks about their incidence.

3.5 What Is Already Known, and What This Paper Adds

It is worth stating plainly which parts of the mechanism this paper takes from the literature and which parts it contributes. That employment documentation, out-of-pocket closing costs, and liquidity constraints can close the refinancing channel—and that these frictions bind hardest in recessions—is established by DeFusco and Mondragon (2020); this paper does not rediscover the income-refinancing link. That the aggregate effect of a policy-rate cut depends on the distribution of refinancing gains across households is established by Eichenbaum et al. (2022). That a recession-contingent mortgage which temporarily reduces required payments can stabilize consumption and shift the timing of defaults is established by Campbell et al. (2021). The empirical COVID literature has likewise documented substantial heterogeneity in who took up forbearance and who refinanced.

The incremental contribution here is therefore *quantitative incidence and policy interaction*, not the discovery of an underwriting channel: given a borrower distribution and a set of institutional underwriting rules, how much of the observed gap between who refinances and who forbears is attributable to the income test, how much to collateral and cash-in restrictions, and how much to maturity and transaction-cost design. That decomposition (Section 6.7) is what the structural model is uniquely positioned to deliver, and it is what distinguishes this exercise from an estimate of the average effect of either instrument.

It is also worth being precise about the institutional boundary. CARES forbearance was not literally available to everyone: the entitlement covered borrowers with federally backed mortgages experiencing pandemic-related hardship who requested relief. Nor was refinancing literally closed to

forbearance users: FHFA explicitly permitted certain borrowers in forbearance who remained current, and borrowers who had exited forbearance and made three consecutive payments, to refinance, and introduced temporary alternative appraisal and employment-verification procedures during the pandemic.¹ The boundary between the two instruments is thus institutional and probabilistic rather than absolute—which is precisely why *segmentation* rather than disjointness is the right description of the result.

4 The Model

This section presents the model in full. It is a life-cycle, heterogeneous-agent, incomplete-markets model of housing tenure and mortgage choice, solved by backward induction and simulated forward on a panel of households hit by an unanticipated aggregate shock. Several features of the model follow directly from Eichenbaum et al. (2022), Wong (2016) and Low (2022). The presentation proceeds from the segmentation mechanism the model is built to deliver, to demographics and preferences, the income process and its estimation, the aggregate state and the crisis and the Markov process that governs them, the mortgage contract and its underwriting constraints, the delinquency bridge and the forbearance technology, the recursive problem written out for each household regime, and the computational implementation. Everything needed to state, solve, and reproduce the household’s problem is here; only the period-by-period cash-flow accounting of the individual regimes is deferred to Appendix A.

4.1 The Segmentation Mechanism

The model embeds the disjoint-population mechanism of Section 2 structurally. Three features bind asymmetrically across the life cycle and produce the empirical segmentation between refinance users and forbearance users. (i) A payment-to-income and loan-to-value ratio underwriting constraints gates new originations and refinances on flow income; a household whose income falls and/or mortgage debt rises loses refinance eligibility precisely when it most needs a lower payment. (ii) A no-cash-in refinance constraint $b_{\text{new}} \geq b$ requires positive amortization or equity headroom to reduce the balance, blocking high-LTV households from refinancing into a lower-payment contract when equity is scarce. (iii) A retirement-tied mortgage term $n = \min(30, T - a)$ caps the amortization window for new loans, raising the required payment for late-career originations. Forbearance, by contrast, requires no re-underwriting. As under CARES it is self-certified, so approval is granted on request during the shock; what limits it is a hardship condition, that the owner be carrying a scheduled payment of at least $\tau^{\text{Pti}} = 0.20$ of current income. The two channels therefore load

¹Federal Housing Finance Agency, “FHFA Announces Refinance and Home Purchase Eligibility for Borrowers in Forbearance,” news release, May 19, 2020, <https://www.fhfa.gov/news/news-release/fhfa-announces-refinance-and-home-purchase-eligibility-for-borrowers-in-forbearance>. Borrowers in forbearance who continued to make payments or reinstated the loan were eligible immediately; those who had exited became eligible three months after the forbearance ended, having made three consecutive payments under a repayment plan, payment deferral, or loan modification.

on different segments of the state space by construction, and the margin that separates them is flow income: the payment-to-income test is what shuts the refinance option for the income-stressed owner whom forbearance must catch.

4.2 The Aggregate State and the Crisis

The aggregate state is the vector $s = (r^{\text{short}}, Y, P^h)$: the short real rate, detrended output, and the house-price level. The normal component of the state space is a discretized empirical process—a finite grid of states with a transition matrix estimated externally from macroeconomic data (Section 4.3). Onto this normal process I graft crisis states that are reached with zero probability from the normal states, so each crisis is an unanticipated (MIT) event: the normal-times value functions and stationary distribution are unaffected by the existence of the crisis states, which are nonetheless solved so the simulator can drop households into them as a surprise. The mortgage rate available in state s is the endogenous OLS function

$$r^m(s) = \max \{ r_0^m + \beta_r r^{\text{short}}(s) + \beta_g \widetilde{\log Y}(s), \underline{r}^m \}, \quad (1)$$

estimated on the 30-year fixed rate against the short rate and detrended log output; through the loading $\beta_r \approx 0.55$, a policy-rate cut transmits to the modeled mortgage rate. Crucially, the rate is *locked* at origination, $r^{\text{lock}} = r^m(s_o)$, and does not update with the aggregate state; the only way to capture a lower market rate is to refinance. The baseline COVID crisis is a 10% output contraction accompanied by a cut of the short rate to zero—a 200 basis-point move which, through (1) and the loading $\beta_r \approx 0.55$, drives the modeled mortgage rate down by roughly 110 basis points—and by house prices *rising* 8%, with CARES forbearance available. It is an income shock against a strong housing market. Section 6 also imposes a 2008-type net-worth shock (a 30% house-price collapse) to trace how the incidence result depends on the housing backdrop.

4.3 The Estimated Aggregate Process

The normal component of the aggregate state is an eight-state grid $\mathcal{S}$ estimated outside the model and loaded at solve time. Each row of the 8×3 state matrix records a short-rate deviation, a detrended log-output level, and a log house-price deviation, and an 8×8 annual transition matrix P governs movement among them; both are estimated from FRED and BEA macroeconomic series. The short rate in a state is the recorded deviation plus the sample mean, in decimal; the house-price level is the exponential of the log deviation plus the sample mean, scaled to model units. At the annual frequency used throughout, the per-period transition is P itself; at higher frequencies it is the real principal matrix root of P , cleaned to a valid stochastic matrix.

The crisis states of Section 4.2 are appended to this grid, each carrying its own short rate, house price, output reduction, and forbearance-approval probability. Evaluating the rate rule (1) at every normal and crisis state yields a finite set of distinct mortgage rates, and that set is the locked-rate grid: an owner’s r^{lock} is a discrete state variable taking one of these values, pinned at origination

or refinance and thereafter invariant to the aggregate state.

4.4 Demographics, Timing, and Preferences

The model is solved at annual frequency; one model period is one calendar year. Households enter at age 25 and live at most $T = 60$ periods, to a terminal age of 84. Working life runs to age 64 and the household is retired from age 65 (Section 5).

The probability of surviving from age a to $a + 1$ is π_a , taken from a U.S. actuarial table (similar to Eichenbaum et al. (2022)), with certain death at T . Upon death the household values a bequest of terminal net worth W ,

$$\text{Beq}(W) = B \frac{\max\{W, 0\}^{1-\sigma} - 1}{1-\sigma}, \quad B > 0. \quad (2)$$

Flow utility is Cobb–Douglas over nondurable consumption c and housing services h , with constant relative risk aversion σ . A renter consuming c and h units of housing services receives

$$u^{\text{rent}}(c, h) = \frac{(c^\alpha h^{1-\alpha})^{1-\sigma}}{1-\sigma}, \quad (3)$$

and a homeowner receives an ownership premium $\theta^{\text{own}} \geq 1$ on housing services,

$$u^{\text{own}}(c, h) = \frac{(c^\alpha (\theta^{\text{own}} h)^{1-\alpha})^{1-\sigma}}{1-\sigma}. \quad (4)$$

The ownership premium is, as Section 5 shows, the single parameter that most tightly governs the homeownership rate, allowing us to match homeownership rates by age to the data.

4.5 Income, Retirement, and Shock Exposure

During working life, household labor income is the product of a permanent component and an aggregate-exposure term,

$$y(\eta, s, a) = \underbrace{\chi_a \exp(\eta)}_{\text{permanent, } \bar{y}} \cdot f(\bar{y}, Y(s)). \quad (5)$$

The permanent component is a deterministic age profile χ_a —hump-shaped over the working life and peaking near age 50, following the administrative-earnings estimates of Guvenen et al. (2021)—scaled by a persistent idiosyncratic component η that follows a first-order autoregression,

$$\eta' = \rho_\eta \eta + \varepsilon', \quad \varepsilon' \sim \mathcal{N}(0, \sigma_\eta^2), \quad (6)$$

discretized into a finite Markov chain by the method of Tauchen (1986). The exposure term loads aggregate output on households unequally,

$$f(\bar{y}, Y) = 1 + (Y - 1) (\chi_1 + \chi_2 \exp(\chi_3 \bar{y})), \quad \chi_3 < 0, \quad (7)$$

where $Y(s)$ is detrended aggregate output in aggregate state s (normalized to $Y = 1$ in the ergodic mean). Because $\chi_3 < 0$, the bracketed elasticity is larger for low-permanent-income households, so a recession ($Y < 1$) falls more heavily on the bottom of the income distribution—the pattern Guvenen et al. (2014) document across most of the earnings distribution, though not at its very top; the loadings are estimated directly and rescaled so that the age-40 cross-section satisfies an aggregate labor-income elasticity of about one. From age 65 the household receives a pension equal to a replacement ratio of its *own* labor income in the last working period,

$$y(\eta, s, a) = \varrho \cdot \chi_{64} \exp(\eta_{64}) \quad \text{for } a \geq 65, \quad (8)$$

so the pension inherits the household’s own permanent component and is thereafter constant. The idiosyncratic chain (6) is frozen at retirement—no further innovations are drawn—and the aggregate exposure term (7) is not applied to pension income, so the pension is invariant to the aggregate state in normal and crisis states alike. Two properties of the retirement phase matter for the crisis experiment. First, pension income is stable, so retirees do not face the labor-income contraction the GDP shock delivers to workers; I make this exemption explicit, so any distress among the old operates through the balance sheet, not through income. Second, because retirees hold little mortgage debt in the calibrated distribution, they are largely insulated from the crisis—which is what lets the age profile of incidence isolate which working-age cohort each instrument reaches.

4.6 Estimating and Discretizing the Income Process

Each of the three pieces of (5) is estimated outside the model and enters the solve as a fixed object; this subsection states them as implemented.

The deterministic age profile. The profile χ_a comes from an individual-fixed-effects regression of log earnings on age dummies in the PSID, which produces a hump-shaped dollar profile $\Gamma(a)$ over ages 25–64 rising from about \$40,600 at entry to a peak of \$70,982 at age 47 and easing to \$50,571 in the last working year. Model income is denominated in units of \$60,000, so the profile enters as $\chi_a = \Gamma(a)/\$60k$. From age 65 the pension is the replacement ratio—1.0 in the baseline—applied to the household’s own last-working-year income, $\Gamma(64)$ scaled by its $\exp(\eta)$ (equation 8); at a ratio of 1.0 this replaces 71% of peak deterministic earnings.

The persistent component. The autoregression (6) is estimated on the same PSID sample at annual persistence $\rho_\eta = 0.961$ with innovation standard deviation $\sigma_\eta = 0.131$. It is discretized into an $n_\eta = 11$ -node chain by the Tauchen (1986) method: nodes are equally spaced on $[-m\sigma_z, m\sigma_z]$ with $\sigma_z = \sigma_\eta / \sqrt{1 - \rho_\eta^2}$ and $m = 3$, and the transition probabilities integrate the innovation density over the midpoints between adjacent nodes. The permanent income level in (5) is then $\bar{y} = \chi_a \exp(\eta)$.

The exposure loadings. The bracketed loading in (7) is calibrated to $(\chi_1, \chi_2, \chi_3) = (0.807, 11.11, -3.31)$. These are estimated from the income-percentile sensitivity of aggregate output in the model’s own

units, then rescaled by a common factor so that the age-40 cross-section satisfies an aggregate labor-income elasticity of approximately one; the income-dependent exposure they encode is the regularity documented by Guvenen et al. (2014). A small income floor prevents non-positive realizations.

4.7 The Mortgage Contract, Underwriting, and Non-Recourse Default

Mortgages are fixed-rate and fully amortizing. A loan of balance b at locked rate r^{lock} with term n carries the constant annuity payment

$$M(b, r^{\text{lock}}, n) = b \frac{r^{\text{lock}} (1 + r^{\text{lock}})^n}{(1 + r^{\text{lock}})^n - 1}, \quad (9)$$

and, in a period in which the household pays, the balance evolves as

$$b' = b(1 + r^{\text{lock}}) - M(b, r^{\text{lock}}, n). \quad (10)$$

The term is tied to remaining life, $n = \min(30, \max(T - a, 1))$, so no new loan extends past the terminal age and the term shrinks with age near the end of the horizon, tightening the payment constraint on late-life originations. A new loan—at purchase, refinance, or trade-up—must satisfy a down-payment (LTV) constraint and a payment-to-income constraint,

$$b \leq (1 - \delta) P^h(s) h, \quad M(b, r^m(s), n) \leq \text{PTI}_{\max} \cdot y(\eta, s, a), \quad (11)$$

with minimum down payment δ and maximum payment-to-income ratio $\text{PTI}_{\max}$. A refinancing owner additionally faces the *no-cash-in* restriction

$$b_{\text{new}} \geq b, \quad (12)$$

so an owner already at the LTV cap whose income has fallen below the PTI requirement cannot refinance into a lower payment unless house-price appreciation lifts $P^h(s)h$ and relaxes the cap—precisely what the appreciating-price COVID path does and the collapsing-price 2008 path forecloses. Because (12) permits $b_{\text{new}} > b$, refinancing here is also a route to liquidity and not only to a lower rate, as in the structural estimates of Chen et al. (2020); the same three restrictions that govern the rate motive govern the liquidity motive, so a household shut out of one is shut out of both. Housing adjustment is *upsized-only*: an owner may hold or trade up to a larger home but may not sell and re-buy a smaller one, so shedding housing is not an alternative to default.² Default is *non-recourse*: a defaulting owner surrenders the home, is released from the balance, incurs a utility penalty ψ and a T_{lock} -period re-purchase lockout, and re-enters as a renter. Because deficiency does not follow the borrower, the default decision is governed by the option value of walking away from negative equity rather than by a recourse penalty.

²The exclusion of downward adjustment is defended in Section E.4: in the 2020 episode households did not trade down, and a diagnostic shows the owners who would use such a margin are equity-rich optimizers disjoint from the leveraged, income-stressed households the paper studies.

4.8 The Delinquency Bridge and the Forbearance Technology

Distress does not resolve instantaneously. An owner may *become delinquent*—skip a scheduled payment without immediate foreclosure, paying only maintenance while the balance accrues at the locked rate—entering a delinquent state with delinquency count $d' = d + 1$. A household may remain delinquent for at most one year before the state must resolve. A delinquent owner may *cure* (pay the accrued payment plus a late fee ϕ_{late} calibrated to Loewenstein et al. 2025, returning to good standing), *foreclose*, or, in a CARES/reset state, *apply for forbearance*. Applying requires that the household be eligible, which we write as

$$e(\Omega) = \mathbf{1}\{b > 0\} \cdot \mathbf{1}\{M(b, r^{\text{lock}}, n) \geq \tau^{\text{pti}} y(\eta, s, a)\}, \quad (13)$$

so that an owner with no balance left to defer, or with a scheduled payment below the hardship threshold τ^{pti} of current income, simply does not have the application in the choice set. When an eligible owner applies and the outcome is uncertain, the application value is

$$V^{\text{app}} = p^{\text{forb}}(s) V^{\text{forb}} + (1 - p^{\text{forb}}(s)) \max\{V^{\text{def}}, V^{\text{become-del}}\}, \quad (14)$$

where $p^{\text{forb}}(s)$ is the state-dependent approval probability. Under the CARES design approval is self-certified rather than underwritten, and we set $p^{\text{forb}}(s) = 1$ in the CARES and reset states and $p^{\text{forb}}(s) = 0$ elsewhere; (14) accordingly collapses to V^{forb} wherever the application is available, and the rejection branch is carried through the notation only so that the approval margin can be varied. The bridge is the structural margin on which forbearance acts: an approval routes a would-be defaulter into a deferred state, bypassing the cure-versus-foreclose resolution. Forbearance intensity is an *endogenous duration*, not a menu choice. An approved forbearer faces a two-regime re-employment hazard and, in each month of the year, compares remitting that month’s payment against deferring it, given the income its employment state delivers. The spell ends when paying becomes preferable—which is typically, but not mechanically, when income recovers. The realized number of deferred months $\mathcal{D}$ is therefore chosen rather than assigned, and the deferred fraction of the year’s payment is

$$1 - \xi = \frac{\mathcal{D}}{12}. \quad (15)$$

Deferred interest is capitalized and the term extended so that the post-forbearance payment matches the pre-forbearance payment, and the forbearance-used indicator is set, $f' = 1$, so each household draws on the option at most once. Eligibility, (13), is a restriction on the choice set rather than a servicer screen, which is the form the CARES design takes: approval is certain on request, but the owner must be carrying a payment burden of at least τ^{pti} of current income, and must have a balance left to defer. Beyond that, take-up is limited by what forbearance costs—deferred interest capitalizes into the balance, and the once-only option is spent. Both the application and its length are the household’s own choices; Section 6 varies the repayment regime that settles the deferred balance on cure. Appendix F reports the alternative in which ξ is instead picked once at application from

a discrete annual menu $\{0, 1/12, \dots, 1\}$, which leaves the aggregate take-up and default moments almost unchanged but shifts the duration distribution sharply, so take-up alone does not identify which mechanism generates the spells.

4.9 The Recursive Problem

State variables. A household is summarized by the state vector

$$\Omega = \left(\underbrace{\text{liquid assets}}_x, \underbrace{\text{mortgage balance}}_b, \underbrace{\text{house size}}_h, \underbrace{\text{locked rate}}_{r^{\text{lock}}}, \underbrace{\text{income shock}}_\eta, \underbrace{\text{aggregate state}}_s, \underbrace{\text{delinq. count}}_d, \underbrace{\text{forb. used}}_f, \underbrace{\text{lockout periods}}_\ell, \underbrace{\text{age}}_a \right). \quad (16)$$

Here $x \geq 0$ is liquid saving carried at the risk-free return r^f ; $b \geq 0$ is the mortgage balance and r^{lock} the rate locked at its origination (both zero for a non-owner); h is the owned house size; η and s are the idiosyncratic and aggregate states of Sections 4.5–4.2; $d \in \{0, 1\}$ is the delinquency count; $f \in \{0, 1\}$ indicates that the once-only forbearance option has been used; $\ell \in \{0, 1, \dots, T_{\text{lock}}\}$ is the number of post-default lockout periods still to run; and a is age. Tenure is read off the state rather than carried separately: a household is a *non-owner* when $b = h = 0$, an *owner in good standing* when it owns and $d = 0$, and a *delinquent owner* when it owns and $d = 1$.

The continuation operator. Let Ω' denote next period's state, split into the *endogenous* block $(x', b', h', r^{\text{lock}'}, d', f', \ell')$ chosen this period and the *exogenous* block (η', s') drawn from the income and aggregate Markov chains of Sections 4.5–4.2. Every continuation value is the survival-weighted expectation

$$\mathbb{E}V' \equiv \beta \left[\underbrace{\pi_a}_{\text{survive}} \underbrace{\mathbb{E}_{\eta', s' | \eta, s} V(\Omega', a+1)}_{\text{expected continuation value}} + \underbrace{(1 - \pi_a)}_{\text{die}} \underbrace{\text{Beq}(W')}_{\text{bequest}} \right], \quad \underbrace{W' = x' + P^h(s) h' - b'}_{\text{terminal net worth}}, \quad (17)$$

with discount factor β and age- a survival probability π_a . The tenure superscript on V inside (17)— N , O , or D —is the tenure the current choice transitions into. Every budget below is understood to also impose the consumption floor $c \geq c_{\min}$ and the no-borrowing condition $x' \geq 0$, which I leave implicit. New price and cost symbols are ρ^{rent} , the unit rental rate on rental services h^{rent} , which the household chooses subject to a floor $\underline{h}^{\text{rent}}$ and the ceiling $\bar{h}$ that also bounds owned houses; δ_m , the maintenance rate; the purchase fee Φ^{new} , the refinance cost $\Phi^{\text{refi}} = \phi_r P^h(s) h + F_r$ (a proportional plus a fixed component), and the selling cost ϕ^{sell} levied on the value of a home that is sold; the late fee ϕ_{late} ; the default utility penalty ψ ; and the lockout length T_{lock} . The floor $\underline{h}^{\text{rent}}$ binds only where a household could not otherwise afford any rental services; with Cobb–Douglas preferences the interior choice is $h^{\text{rent}} = (1 - \alpha)X / \rho^{\text{rent}} P^h(s)$ in disposable resources X , so it is the ceiling rather than the floor that binds over most of the state space.

Non-owner. A non-owner rents or, once the post-default lockout has expired ($\ell = 0$), may buy:

$$V^N(x, \eta, s, \ell, a) = \max \left\{ \overbrace{V^{\text{rent}}(x, \eta, s, \ell, a)}^{\text{keep renting}}, \mathbf{1}_{\{\ell=0\}} \overbrace{V^{\text{purch}}(x, \eta, s, a)}^{\text{buy a home}} \right\}. \quad (18)$$

A *renter* rents h^{rent} units of housing services and continues as a non-owner with the lockout counting down, $\ell' = \max\{\ell - 1, 0\}$:

$$\begin{aligned} V^{\text{rent}} &= \max_{c, x', h^{\text{rent}}} u^{\text{rent}}(c, h^{\text{rent}}) + \mathbb{E}V'^N \\ \text{s.t. } c + x' &= \overbrace{x(1 + r^f)}^{\text{cash on hand}} + \overbrace{y(\eta, s, a)}^{\text{income}} - \overbrace{\rho^{\text{rent}} P^h(s) h^{\text{rent}}}^{\text{rent}}, \\ \underline{h}^{\text{rent}} &\leq h^{\text{rent}} \leq \bar{h}. \end{aligned} \quad (19)$$

A *buyer* chooses a house h' and a loan b' at the current market rate, $r^{\text{lock}'} = r^m(s)$, pays the down payment and the purchase fee out of cash on hand, and enters next period as an owner in good standing ($d' = 0$, $f' = f$):

$$\begin{aligned} V^{\text{purch}} &= \max_{h', b', c, x'} u^{\text{own}}(c, h') + \mathbb{E}V'^O \\ \text{s.t. } c + x' &= \overbrace{x(1 + r^f) + y(\eta, s, a)}^{\text{resources}} - \overbrace{(P^h(s)h' - b')}^{\text{down payment}} - \overbrace{\Phi^{\text{new}} P^h(s)h'}^{\text{purchase fee}}, \\ &\quad \overbrace{b' \leq (1 - \delta) P^h(s) h'}^{\text{down-payment (LTV) limit}}, \quad \overbrace{M(b', r^m(s), n) \leq \text{PTI}_{\text{max}} \cdot y(\eta, s, a)}^{\text{payment-to-income limit}}, \\ &\quad n = \min\{30, \max(T - a, 1)\}. \end{aligned} \quad (20)$$

Owner in good standing. An owner with $d = 0$ chooses among paying, refinancing, trading up, applying for forbearance (only while the option is unused, $f = 0$, and the eligibility condition (13) holds), becoming delinquent, and defaulting:

$$V^O(x, b, h, r^{\text{lock}}, \eta, s, f, a) = \max \left(\{V^{\text{pay}}, V^{\text{refi}}, V^{\text{up}}, V^{\text{bdel}}, V^{\text{def}}\} \cup \{V^{\text{app}} : f = 0, e(\Omega) = 1\} \right). \quad (21)$$

Pay remits the scheduled payment and maintenance, amortizes the balance, and keeps the locked rate:

$$\begin{aligned} V^{\text{pay}} &= \max_{c, x', p} u^{\text{own}}(c, h) + \mathbb{E}V'^O \\ \text{s.t. } c + x' &= \overbrace{x(1 + r^f) + y}^{\text{resources}} - \overbrace{M(b, r^{\text{lock}}, n)}^{\text{mortgage payment}} - \overbrace{p}^{\text{prepayment}} - \overbrace{\delta_m P^h(s) h}^{\text{maintenance}}, \\ &\quad \overbrace{b' = b(1 + r^{\text{lock}}) - M(b, r^{\text{lock}}, n) - p}^{\text{amortized balance, net of prepayment}}, \quad 0 \leq p \leq b, \quad r^{\text{lock}'} = r^{\text{lock}}, \quad d' = 0. \end{aligned} \quad (22)$$

Refinance replaces the loan with a new balance b_{new} at the market rate, pays the refinance cost Φ^{refi} , and resets the locked rate; it must clear the LTV, payment-to-income, and no-cash-in tests:

$$\begin{aligned}
V^{\text{refi}} &= \max_{b_{\text{new}}, c, x'} u^{\text{own}}(c, h) + \mathbb{E}V'^O \\
\text{s.t.} \quad c + x' &= \overbrace{x(1 + r^f) + y}^{\text{resources}} + \overbrace{(b_{\text{new}} - b)}^{\text{net cash from refi}} - \overbrace{\delta_m P^h(s)h}^{\text{maintenance}} - \overbrace{\Phi^{\text{refi}}}^{\text{refi cost}}, \\
&\quad \overbrace{b_{\text{new}} \leq (1 - \delta)P^h(s)h}^{\text{LTV}}, \quad \overbrace{b_{\text{new}} \geq b}^{\text{no cash-in}}, \\
&\quad \overbrace{M(b_{\text{new}}, r^m(s), n) \leq \text{PTI}_{\text{max}} \cdot y}^{\text{payment-to-income, at the market rate}}, \quad r^{\text{lock}'} = r^m(s), \quad d' = 0.
\end{aligned} \tag{23}$$

Trade-up sells the current home, realizing its equity, and buys a strictly larger home $h' > h$ under the same origination constraints:

$$\begin{aligned}
V^{\text{up}} &= \max_{h' > h, b_{\text{new}}, c, x'} u^{\text{own}}(c, h') + \mathbb{E}V'^O \\
\text{s.t.} \quad c + x' &= \overbrace{x(1 + r^f) + y}^{\text{resources}} + \overbrace{((1 - \phi^{\text{sell}})P^h(s)h - b)}^{\text{equity released on sale, net of the selling cost}} \\
&\quad - \overbrace{(P^h(s)h' - b_{\text{new}})}^{\text{down payment on new home}} - \overbrace{\Phi^{\text{new}}P^h(s)h'}^{\text{purchase fee}}, \\
&\quad \overbrace{b_{\text{new}} \leq (1 - \delta)P^h(s)h'}^{\text{LTV}}, \quad \overbrace{M(b_{\text{new}}, r^m(s), n) \leq \text{PTI}_{\text{max}} \cdot y}^{\text{payment-to-income}}, \quad r^{\text{lock}'} = r^m(s).
\end{aligned} \tag{24}$$

Become delinquent skips the payment—remitting only maintenance while the balance accrues at the locked rate—and moves to the delinquent state:

$$\begin{aligned}
V^{\text{bdel}} &= \max_{c, x'} u^{\text{own}}(c, h) + \mathbb{E}V'^D \\
\text{s.t.} \quad c + x' &= \overbrace{x(1 + r^f) + y}^{\text{resources}} - \overbrace{\delta_m P^h(s)h}^{\text{maintenance}}, \quad \overbrace{b' = b(1 + r^{\text{lock}})}^{\text{balance accrues, no payment}}, \quad d' = d + 1.
\end{aligned} \tag{25}$$

Default surrenders the home and, because default is non-recourse, walks away from the balance b ; it bears the utility penalty ψ and the lockout $\ell' = T_{\text{lock}}$, and continues as a non-owner:

$$\begin{aligned}
V^{\text{def}} &= \max_{c, x', h^{\text{rent}}} u^{\text{rent}}(c, h^{\text{rent}}) - \overbrace{\psi}^{\text{default penalty}} + \mathbb{E}V'^N \big|_{\ell' = T_{\text{lock}}} \\
\text{s.t.} \quad c + x' &= \overbrace{x(1 + r^f) + y}^{\text{resources}} - \overbrace{\rho^{\text{rent}} P^h(s)h^{\text{rent}}}^{\text{rent}} \quad (\text{balance } b \text{ released, non-recourse}).
\end{aligned} \tag{26}$$

Delinquent owner. An owner with $d = 1$ cures, forecloses, or—in a CARES/reset state, if the option is unused and the eligibility condition (13) holds—applies for forbearance:

$$V^D(x, b, h, r^{\text{lock}}, \eta, s, d, f, a) = \max \left(\{V^{\text{cure}}, V^{\text{forecl}}\} \cup \{V^{\text{app}} : f = 0, e(\Omega) = 1\} \right). \quad (27)$$

Cure pays the arrears plus a late fee ϕ_{late} on the missed payment and returns to good standing ($d' = 0$):

$$\begin{aligned} V^{\text{cure}} &= \max_{c, x'} u^{\text{own}}(c, h) + \mathbb{E}V'^O \\ \text{s.t.} \quad c + x' &= \overbrace{x(1 + r^f) + y}^{\text{resources}} - \overbrace{(1 + \phi_{\text{late}}) M(b, r^{\text{lock}}, n)}^{\text{arrears + late fee}} - \overbrace{\delta_m P^h(s) h}^{\text{maintenance}}. \end{aligned} \quad (28)$$

Foreclose, V^{forecl} , is the delinquent-state counterpart of default (26): surrender, non-recourse release, penalty ψ , lockout $\ell' = T_{\text{lock}}$, continue as a non-owner. The *application* value (14) is the approval-weighted average

$$V^{\text{app}} = \overbrace{p^{\text{forb}}(s)}^{\text{approval prob.}} V^{\text{forb}} + \overbrace{(1 - p^{\text{forb}}(s))}^{\text{rejection prob.}} \overbrace{\max\{V^{\text{def}}, V^{\text{bdel}}\}}^{\text{fallback if rejected}}. \quad (29)$$

Approved forbearance defers the fraction $1 - \xi = \mathcal{D}/12$ of the payment (equation 15), so the household remits only ξM plus maintenance; the deferred interest capitalizes into the balance, the term extends so the post-forbearance payment is unchanged, and the once-only option is spent ($f' = 1$):

$$\begin{aligned} V^{\text{forb}} &= \max_{c, x'} u^{\text{own}}(c, h) + \mathbb{E}V'^O \\ \text{s.t.} \quad c + x' &= \overbrace{x(1 + r^f) + y}^{\text{resources}} - \overbrace{\xi M(b, r^{\text{lock}}, n)}^{\text{partial (paid) payment}} - \overbrace{\delta_m P^h(s) h}^{\text{maintenance}}, \\ &\quad \overbrace{b' = b(1 + r^{\text{lock}}) - \xi M(b, r^{\text{lock}}, n)}^{\text{deferred interest capitalized}}, \quad f' = 1. \end{aligned} \quad (30)$$

Terminal condition and solution. At the terminal age the value equals the bequest of net worth,

$$V(\Omega, T) = \text{Beq} \left(\overbrace{x + P^h(s) h - b}^{\text{terminal net worth}} \right). \quad (31)$$

The problem is solved by backward induction from T , and the resulting age- and state-contingent policy functions are simulated forward on a panel of households; Appendix D sets out the discretized state space, the solve, and the simulation. The exact cash-flow accounting for the refinance, trade-up, and forbearance regimes is given in Appendix A.

5 Calibration and Identification

5.1 Datasets

I use four, publicly available datasets. The **Survey of Consumer Finances** (2019 wave) disciplines steady-state moments: the homeownership rate, the distribution of liquid wealth-to-income ratios, and the distribution of LTV and PTI at origination. The **Panel Study of Income Dynamics** identifies the idiosyncratic income process of (6); the deterministic age-earnings profile χ_a follows Guvenen et al. (2021). The **Fannie Mae Single-Family Loan Performance** data provide the loan-level origination and performance histories used to construct the disjoint-cohort facts of Section 2, to set the crisis-state forbearance-approval probability and the refinancing target. **FRED** monthly series construct the aggregate state vector and the mortgage-rate rule (1). The idiosyncratic income chain is discretized into eleven nodes by the Tauchen (1986) method, and the normal aggregate process is an eight-state chain estimated from the macro series; both constructions are detailed in Sections 4.6 and 4.3.

5.2 Calibration

The model is calibrated rather than estimated. Risk aversion is $\sigma = 2$. The minimum down payment is $\delta = 0.20$, the pre-boom standard documented by Landvoigt et al. (2015). The payment-to-income ceiling is set to $\text{PTI}_{\max} = 0.60$, deliberately above any observed underwriting standard—Greenwald (2026) calibrates 36% to the pre-boom conventional book and the GSE limit has run in the 45–50% range—so that the test does not bind mechanically on households facing higher mortgage-rate regimes. This is the conservative choice for the question at hand: a loose ceiling lets the payment-to-income test bind only where income has genuinely collapsed, so the segmentation the model produces cannot be an artefact of a tight cap excluding the forbearance cohort by construction. Section 6.7 reports what removing the test entirely does. The idiosyncratic income parameters are estimated from PSID; the deterministic age profile follows Guvenen et al. (2021). The discount factor β , the consumption share α , the ownership premium θ^{own} , the bequest weight B , and the default disutility ψ are set to match SCF 2019 moments—the homeownership rate, the working- and late-life wealth-to-income ratios, and an owner default rate. Table 6 lists the parameters. The default configuration is non-recourse and the mortgage term is remaining-life; the risk-free rate is 2%, and the crisis cuts the short rate to zero, transmitting through the loading $\beta_r \approx 0.55$.

5.3 The Length of a Deferral, and Why It Is a Fixed Point

One parameter is neither assigned nor calibrated to a data moment, and it deserves its own paragraph because it is solved for rather than chosen. A model period is a year, but a forbearance spell is measured in months, so the backward induction has to compress the spell into a single number: the fraction of the year’s scheduled payment an approved household defers, $1 - \xi = \mathbb{E}[\mathcal{D}]/12$. It has to be a single number because realized spell length cannot be carried as a state variable without

Table 6: Model Parameters

Parameter	Description	Value	Source
<i>Demographics and preferences</i>			
T	Horizon (ages 25–84)	60 periods	Set
a^{ret}	Retirement age	65	Set
β	Discount factor	0.94	Standard
σ	Risk aversion	2.0	Standard
α	Consumption share	0.85	Standard
B	Bequest weight	20.0	Calibrated
θ^{own}	Ownership premium	2.0	Calibrated
ψ	Default disutility	20.0	Calibrated
T_{lock}	Post-default lockout	5 yrs	Data
repl	Pension replacement rate	1.0	Set
<i>Income process</i>			
ρ_η	Persistence (annual)	0.961	PSID
σ_η	Innovation s.d. (annual)	0.131	PSID
n_η	Idiosyncratic nodes (Tauchen)	11	Set
(χ_1, χ_2, χ_3)	Output-exposure loadings	(0.807, 11.11, −3.31)	estimated
<i>Housing and mortgage market</i>			
δ	Min. down payment	0.20	Landvoigt et al. (2015)
δ_m	Maintenance rate	0.03	set
Φ^{new}	Purchase fee	0.03	calibrated
Φ^{refi}	Refinance fee ($\times P^h h$) + fixed	$0.03 + \$2\text{k}$	calibrated
ϕ^{sell}	Selling cost ($\times P^h h$, on trade-up)	0.09	set
$\underline{h}^{\text{rent}}$	Rental services floor	0.10	set
$\bar{h}$	Housing size ceiling (owned and rented)	2.0	set
PTI_{max}	Max payment-to-income	0.60	set
λ	Recourse (0 = non-recourse)	0.0	set
n	Mortgage term	$\min(30, T - a)$	set
<i>Mortgage rate, forbearance, delinquency</i>			
$(r_0^m, \beta_r, \beta_g)$	OLS rule (1)	(.0385, .552, −.051)	FRED/BEA
p_f^{crisis}	Crisis forbearance approval	1	CARES/FNM
τ^{pti}	Forbearance eligibility: M/y floor	0.20	set (see text)
π^{fast}	Share in the temporary-layoff regime	0.78	CPS/BLS
$h^{\text{fast}}, h^{\text{slow}}$	Monthly re-employment hazards	0.24, 0.07	CPS/BLS
N_{sub}	Within-year sub-periods (months)	12	set
ϕ_{late}	Late fee on cure	0.05	Loewenstein et al. (2025)
N_A, N_M	Asset/mortgage grid points	15	set
n_{agg}	Normal aggregate states	8	FRED/BEA

Note: Baseline crisis is `covid_effect` (output −10%, short rate to zero, house prices +8%). One model unit $\approx$ \$60,000.

adding a dimension to the grid.

The simulator is finer. Each approved household draws a re-employment month from the two-regime hazard and then decides, month by month, whether to keep deferring or resume paying, so its realized $\mathcal{D}$ is an outcome that varies across households. Internal consistency requires that the $\mathbb{E}[\mathcal{D}]$ the value function prices equal the mean the simulation delivers; otherwise households are making every decision in the model—whether to apply, whether to refinance, whether to default—while valuing a policy that does not occur.

That is a fixed point, and I solve it by iteration: solve at m , simulate, read back the realized mean, re-solve at that value. Starting from the hazard-implied expectation the iteration converges quickly. The reported economy prices $\mathbb{E}[\mathcal{D}] = 2.87$ months and its simulation realizes 2.67 over 424 spells, a residual of 0.20 months, or 1.7 percentage points of the deferred fraction. Every counterfactual arm in Sections 6–8 is solved at the same value, so the arms price deferral identically and differ only in the lever each one moves.

The number matters for magnitudes. Deferral is short: the median spell is two months and 81% of spells end within three, so forbearance buys roughly a quarter of a year’s payment, not a year’s. That is what makes the option worth applying for less often than the 2020 data show, and it is the honest version of a parameter that earlier drafts of this model inherited at a much larger value.

5.4 Which Moments Are Targeted

Because the paper leans on model fit, it matters exactly which moments the calibration used and which it did not. Table 7 makes the mapping explicit. Every parameter is either assigned from an external source (panel a) or calibrated to a stated target moment (panel b); only the moments in panel (c) are genuinely untargeted and therefore available as validation. The distinction is not cosmetic. The crisis refinancing share is a calibration input, so the model reproducing it is a restatement of the calibration and is reported nowhere as evidence of fit. The forbearance take-up rate is *not*: the specification in Section 4.8 replaced an affordability screen with the eligibility rule (13), whose threshold τ^{pti} is set rather than fitted, so take-up is an outcome and belongs in panel (c). It is reported below as validation, and the model undershoots it.

5.5 Model Fit

Life-cycle profiles against the SCF and PSID. Figure 5 overlays the model’s pre-crisis life-cycle profiles on the corresponding data moments: income and total assets against the 2019 SCF (mean and median) and the 2018 PSID, non-housing (liquid) assets against both, and the homeownership rate against the SCF mean $\pm$ one standard error. The calibration is disciplined to the SCF cross-section, and the fit is close on the moments that matter for the crisis experiments. The homeownership rate reproduces the life-cycle ramp and the old-age plateau, at 66.9% in the model against 64.9% in the 2019 SCF; the income profile tracks the hump-shaped SCF/PSID path and peaks in the late forties; and total assets follow the SCF median closely, cresting near retirement. Two gaps are worth naming. The model’s ownership curve climbs somewhat too steeply and runs

Table 7: Parameter-to-target mapping: what the calibration used, and what it did not.

Parameter	Target moment	Source
<i>(a) Assigned from external evidence—not fitted to any moment here</i>		
$\beta = 0.94$	—	standard
$\sigma = 2.0$	—	standard
$\alpha = 0.85$	—	standard
$(\rho_\eta, \sigma_\eta) = (0.961, 0.131)$	—	PSID
(χ_1, χ_2, χ_3)	—	estimated (Section 4.6)
π_a (mortality)	—	SSA life tables
$\delta = 0.20$	—	Landvoigt et al. (2015)
$\delta_m = 0.03$	—	set
$\text{PTI}_{\max} = 0.60$	—	set (see text)
$\phi_{\text{late}} = 0.05$	—	Loewenstein et al. (2025)
$(r_0^m, \beta_r, \beta_g)$	—	OLS on FRED/BEA
$T, a^{\text{ret}}, \text{repl}, \lambda, n, T_{\text{lock}}$	—	set by design
<i>(b) Calibrated to a stated target—these moments are inputs, not tests</i>		
$\theta^{\text{own}} = 2.0$	aggregate homeownership rate	2019 SCF
$B = 20.0$	old-age asset holdings	2019 SCF
$\psi = 20.0$	lifetime foreclosure rate	Fannie Mae
$\Phi^{\text{new}} = 0.03$	purchase/turnover frequency	industry estimates
$\Phi^{\text{refi}} = 0.03 + \$2k$	crisis refinancing share	Fannie Mae, 2020
$p_f^{\text{crisis}} = 1$	CARES approval-on-request	CARES Act / Fannie Mae
<i>(c) Untargeted—available as validation</i>		
—	life-cycle income and asset profiles	2019 SCF, 2018 PSID
—	life-cycle homeownership profile	2019 SCF
—	LTV profile of mortgaged owners by age	2019 SCF
—	crisis forbearance take-up rate	Fannie Mae, 2020
—	composition of crisis decisions	Fannie Mae, 2020
—	forbearance duration distribution	Sánchez and Wilkinson (2022)

Notes: Panel (b) lists the moments the calibration was fitted to. They are *not* reported as model successes anywhere in the paper: in particular the crisis refinancing share is a calibration input, so the model matching it is a restatement of the calibration rather than evidence for it. Forbearance take-up used to sit in this panel, matched by a HAMP-style affordability gate τ^{aff} ; that gate has been replaced by the eligibility rule (13), whose threshold is set and not fitted, so take-up has moved to panel (c) and is now a test the model can fail—and does, in the direction reported in Section 5.5. Only panel (c) is used as validation there. Three rows in panel (a) carry qualifications. $\delta = 0.20$ is the pre-boom down-payment standard in Landvoigt et al. (2015), whose own point is that the constraint loosened during the boom; the model holds it fixed. $\phi_{\text{late}} = 0.05$ follows the cure cost assumed by Loewenstein et al. (2025), where it is a modelling assumption applied to missed payments rather than an estimated fee schedule. The eligibility threshold $\tau^{\text{pti}} = 0.20$ is set, not fitted; it sits below the 31% affordability standard HAMP applied (Agarwal et al., 2017), since CARES forbearance was self-certified hardship rather than an underwritten affordability test. The aggregate homeownership *rate* is targeted; its *profile* over the life cycle is not, and the two are reported separately for that reason. The near-zero 2020–21 foreclosure rate is excluded from panel (c) and is not used as validation at all: the federal moratorium made it administratively determined, so reproducing it tests the moratorium’s presence in the model rather than the model’s behaviour—see the discussion below.

above the SCF cross-section through the middle of the working life before the two meet at retirement, and the model *under*-accumulates non-housing liquid assets relative to the SCF median. Both are life-cycle-calibration features governed by the ownership premium and the age-earnings profile, and neither bears on the incidence result, which turns on leverage and flow income rather than on the level of liquid buffers (Section 8.8).

Leverage over the life cycle. The remaining-life mortgage term delivers a loan-to-value profile that declines toward zero in old age (Figure 6), the object the crisis experiments act on; the mean loan-to-value ratio of mortgaged owners in the calibrated economy is 0.31. The model reproduces the downward-sloping LTV profile of mortgaged owners in the 2019 SCF, tracking the SCF mean $\pm$ one standard error through the working life and staying below the 0.80 origination ceiling throughout; it declines a little faster than the SCF mean at mid-life, so the calibrated economy is if anything slightly *less* levered in middle age than the data, which makes the crisis refinancing and forbearance responses documented below conservative rather than inflated.

The 2020 mortgage market. Of the three crisis margins in the Fannie Mae data, one is a calibration input and two are not, and it is worth separating them before reading the fit.

The refinancing share is the input. On impact 4,777 of the 20,029 mortgaged owners in the simulated crisis period refinance, a share of 23.9% against 26.8% in the Fannie loan-level data. The refinance cost Φ^{refi} was set to that moment (Table 7, panel b), so the model matching it restates the calibration rather than validating it.

Forbearance take-up is not an input, and here the model *undershoots, by a factor of about two and a half*. On impact 445 of the same 20,029 mortgaged owners are approved, 2.2% against the 5.4% of loans that entered forbearance in 2020; over the whole life cycle 0.58% of households ever use the option. Nothing in the calibration is free to close that gap. Eligibility is the set threshold $\tau^{\text{pti}} = 0.20$, approval is certain, and the value of deferral is pinned by the fixed point of Section 5.3, so take-up is an outcome of how many owners the crisis income drop pushes across a payment burden of one fifth of income and of what a short deferral is worth to them once there.

Three features of the environment plausibly account for the shortfall, and all three cut the same way. First, the model’s owners are less levered at mid-life than the SCF cross-section, so fewer carry a payment large enough to clear the threshold. Second, the spell the model prices is short—an expected 2.87 months of deferred payment, which is what the re-employment hazard and the household’s own month-by-month exit decision generate—so forbearance buys about a quarter of a year’s payment rather than a year’s, and is correspondingly less attractive to apply for. Third, the option is once-only and capped at twelve months, whereas CARES allowed re-entry and extension, so a household that would have taken a short precautionary spell in the data has more to lose by spending the option here. A model that reaches fewer than half the households the program reached understates its aggregate footprint, so the counterfactuals in Section 6 should be read as a lower bound on what forbearance does, not an inflated one. The incidence results those sections turn

Model vs SCF 2019: Lifecycle Profile Comparisons

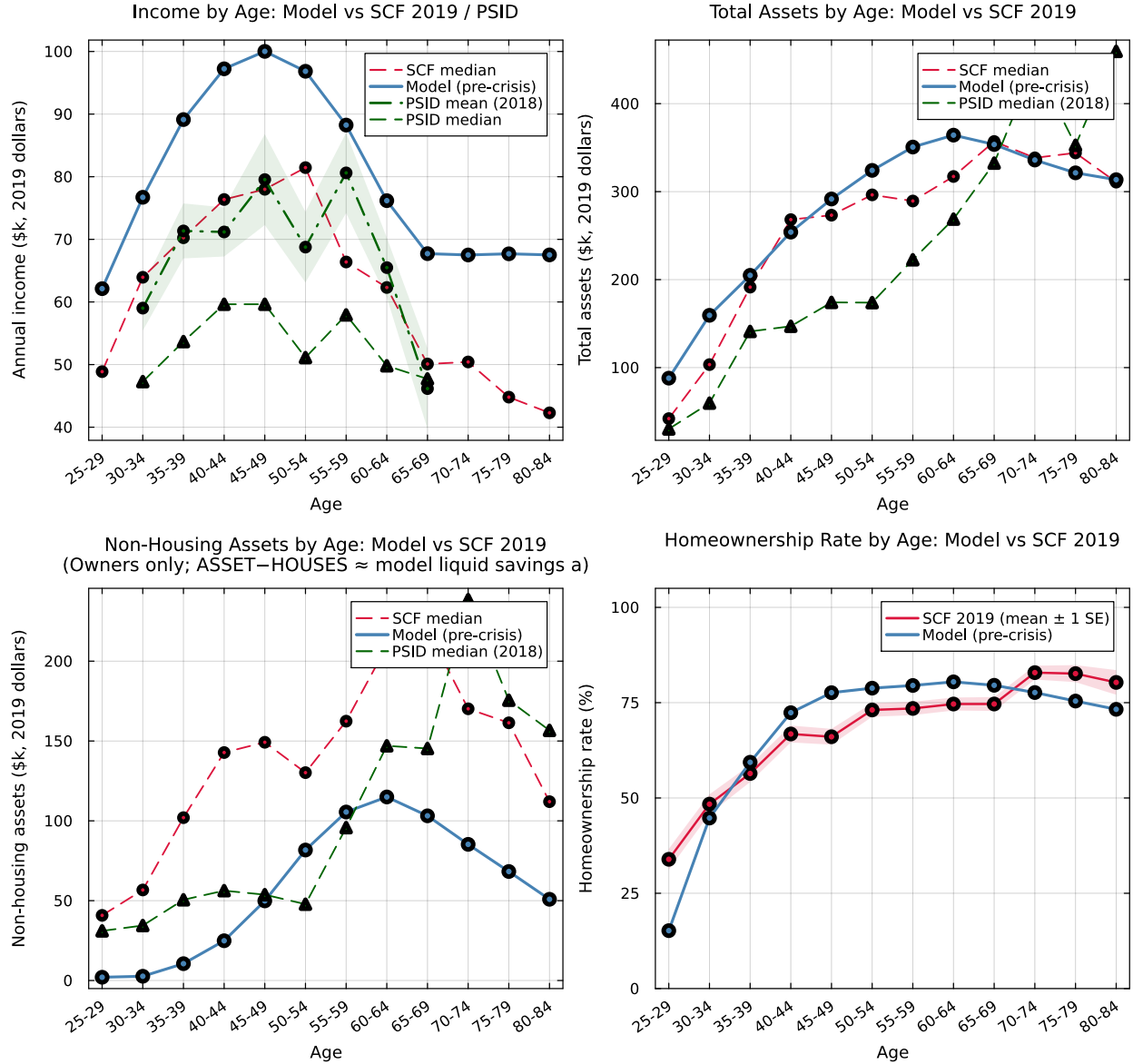


Figure 5: Model versus data over the life cycle: income, total assets, non-housing (liquid) assets, and the homeownership rate—model (pre-crisis) against the 2019 SCF (mean $\pm$ 1 SE and median) and the 2018 PSID. Levels are in 2019 dollars, scaled to the model by the SCF/model income ratio at ages 25–34; ratios (homeownership) are unscaled.

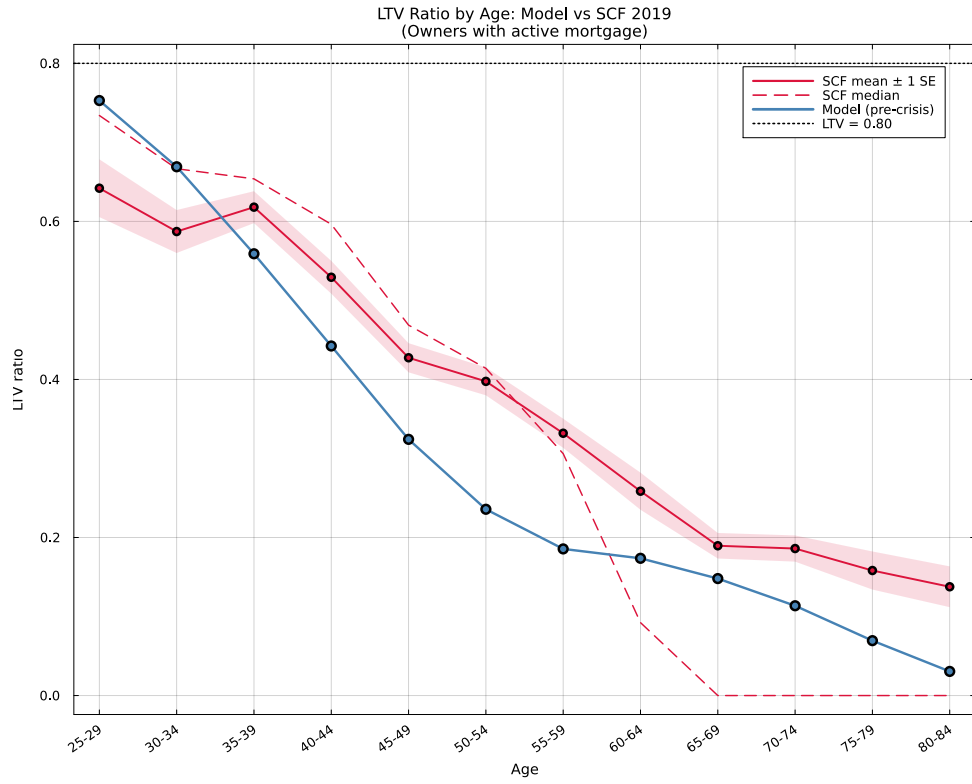


Figure 6: Loan-to-value ratio of mortgaged owners by age: model (pre-crisis) versus the 2019 SCF (mean $\pm$ 1 SE and median). The remaining-life term removes the spurious old-age rise a working-life-only horizon generates, and the profile declines toward zero as in the data.

on are about *which* owners the instrument reaches, and are not mechanically tied to the level of take-up; but the level is a genuine miss and is reported as one.

The crisis foreclosure rate is also untargeted, and sits on the floor ($\approx 0.06\%$), matching the near-zero completed foreclosure of the moratorium period; the default disutility ψ is set to the *lifetime* foreclosure rate and carries no information about its crisis-period value. For the reason given below, I do not count that match as validation either. Figure 7 places the model’s decision shares beside the Fannie Mae shares period by period: on impact, refinancing rises to 24% in the model against 26.8% in the data, forbearance appears only in the crisis, and the foreclosure bar stays near the floor throughout. The untargeted objects the fit rests on are those of panel (c) of Table 7—the life-cycle profiles, the crisis take-up rate just discussed, the composition of crisis decisions, and the forbearance-duration distribution examined in Section 8.4.

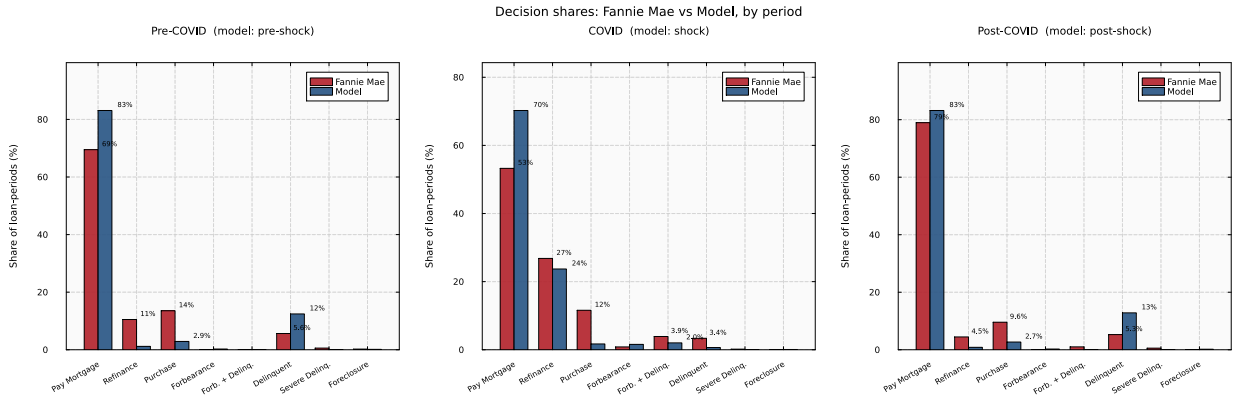


Figure 7: Decision shares of mortgaged owners, Fannie Mae versus model, by period (pre-COVID, COVID, post-COVID). The COVID panel shows the refinancing share (data 27%, model 24%), the appearance of forbearance, and a foreclosure bar on the floor; the percentage is labeled on each bar.

Why the foreclosure floor is not used as validation. The model produces a near-zero crisis-period foreclosure rate, which matches the observed 2020–2021 collapse in completed foreclosures. I do not count this as validation of the model’s default margin. The federal foreclosure moratorium administratively suppressed completed foreclosures on federally backed loans for much of the period, so the empirical moment is a policy artifact rather than an equilibrium outcome of borrower choice. Matching it is uninformative about whether the model’s default decision is correctly specified. The moratorium enters the analysis as a policy lever in Section 8.1.

6 Policy Incidence

This section reports the paper’s quantitative results. It opens with the baseline COVID experiment, in which forbearance and the rate cut are both active against the appreciating-price backdrop, and documents how the bundle reshapes household decisions and the default margin, which borrowers each instrument reaches, how the forbearance and refinance populations relate, and the welfare each

action delivers. It then prices the cost of forbearance implied by the cohort attribution, and closes by showing how the incidence result depends on the design of the instrument, the aggregate state the crisis strikes from, the recourse assumption, and the housing backdrop—and by defending the exclusion of a downward-adjustment margin. All experiments are read within the scope conditions of Section 8.8.

6.1 Simulation Design

I run the model forward through a long burn-in to a stationary age distribution, pin the aggregate state at a fixed normal state for a pre-shock window, apply the unanticipated `covid_effect` shock (a 10% output contraction, the short rate cut to zero, and house prices +8%), and return the economy to the normal state. The panel is 30,000 agents, and all idiosyncratic draws (mortality, income transitions, forbearance approvals) and the aggregate path are seeded identically across scenarios, so that a counterfactual re-run isolates policy from simulation noise (common random numbers). The crisis window reported below is the last pre-shock period, the impact period, and the first post-shock period; Figures 8–9 show the full path.

6.2 What Each Instrument Does: A Single-Lever Decomposition

The clearest way to see what each instrument does is to impose the levers one at a time on top of a common output shock, holding the idiosyncratic draws fixed across arms (Table 8). The rate cut is what drives refinancing: crisis refinances rise from 443 with the output shock alone to 3,439 once the rate cut is added, and to 4,777 in the full package as the 8% house-price appreciation relaxes the loan-to-value and no-cash-in constraints and admits more owners to the refinancing valve. Forbearance is what drives take-up: crisis approvals rise from 2 to 1,570 when the option is switched on.

The pattern that carries the incidence result is what happens to take-up as the refinancing valve opens. Forbearance take-up is *highest* when forbearance is the only instrument on offer (1,570 approvals), falls to 1,047 once the rate cut is added, and falls again to 445 in the full package—a decline of 72%. Owners who can clear the payment-to-income test refinance rather than forbear, so the residual forbearance is the income-stressed segment that cannot. The decomposition shows the segmentation of Section 6.6 operating on the extensive margin: the same crisis, the same households, and forbearance shrinks to under a third of its size as soon as the price instrument becomes available to those who qualify for it.

The default margin tells a more even-handed story, and it is worth stating plainly. Relative to the output shock alone (82 crisis foreclosures), the rate cut on its own removes 19 and forbearance on its own removes 20; imposed together they remove 39. The two instruments are close to indistinguishable in how much foreclosure each suppresses, and their effects are close to additive. Forbearance is not the margin that contains default—it is the margin that reaches a *different* population, which is a claim about incidence rather than about aggregate effectiveness, and it is the claim the rest of this section develops.

One row runs against the grain and should not be smoothed over. The full package produces *more* crisis foreclosure than the rate cut and forbearance without house-price appreciation (53 against 43), though the only difference between those two arms is that prices rise 8%. Appreciation widens the refinancing valve (3,431 to 4,777 refinances) and pulls more than half the forbearance cohort out of forbearance (1,047 to 445); the marginal households who can do neither are left without the bridge they would otherwise have taken. Rising collateral values help the owners who can be re-underwritten and do nothing for the owners who cannot.

Table 8: Single-lever decomposition of the COVID package: mortgaged-owner decisions in the shock period.

Scenario (imposed on the output shock)	Refinance	Forbearance	Default
Output shock only	443	2	82
+ Rate cut	3,439	1	63
+ Forbearance	426	1,570	62
+ Rate cut & forbearance	3,431	1,047	43
Full package (+ house-price rise)	4,777	445	53

Notes: Shock-period counts of mortgaged-owner decisions in the single-lever experiment, which imposes each lever on the common 10% output contraction and shares idiosyncratic draws across arms. “Forbearance” is crisis-period approvals; “Default” is foreclosures. The full-package row is this experiment’s `covid_effect` arm and is the same simulation as the baseline of Table 9, so the two agree exactly. All five arms reproduce the pre-shock period identically—8,837 renters, 19,117 payers, 122 refinancers—which is the diagnostic that the common random numbers held rather than merely being similarly seeded. Panel of 30,000 simulated households.

6.3 The Crisis Path and the Aggregate Response

Figure 8 plots the aggregate conditions the households face: the mortgage rate falls as the short-rate cut transmits, house prices appreciate, the forbearance-approval probability jumps to near one, and average income drops on impact and recovers. Figure 9 summarizes the household-side impulse responses relative to the pre-shock mean.

6.4 The Decision Response

Table 9 reports the share of living households in each state across the crisis window, with the multi-period delinquency bridge now shown explicitly rather than folded into a residual. The two active instruments show up as two different actions. The rate cut sets off a *refinancing wave*: refinancing jumps from 0.4% of households to 16.2% on impact, 4,777 choices in a panel of 30,000. CARES *forbearance switches on*: approvals rise from near zero to 1.5%, or 445 households, counting both direct approvals and delinquent owners approved that period. And in contrast to both surges, the *foreclosure margin stays near the floor*: default is at or below 0.2% throughout. The delinquency bridge behaves differently on either side of the shock. It is nearly empty in normal times (1.2% still delinquent) and the shock empties it further, to 0.9% on impact, as distressed owners are routed into forbearance or cure. It then *refills*: 2.3% are delinquent in the first post-shock period, roughly

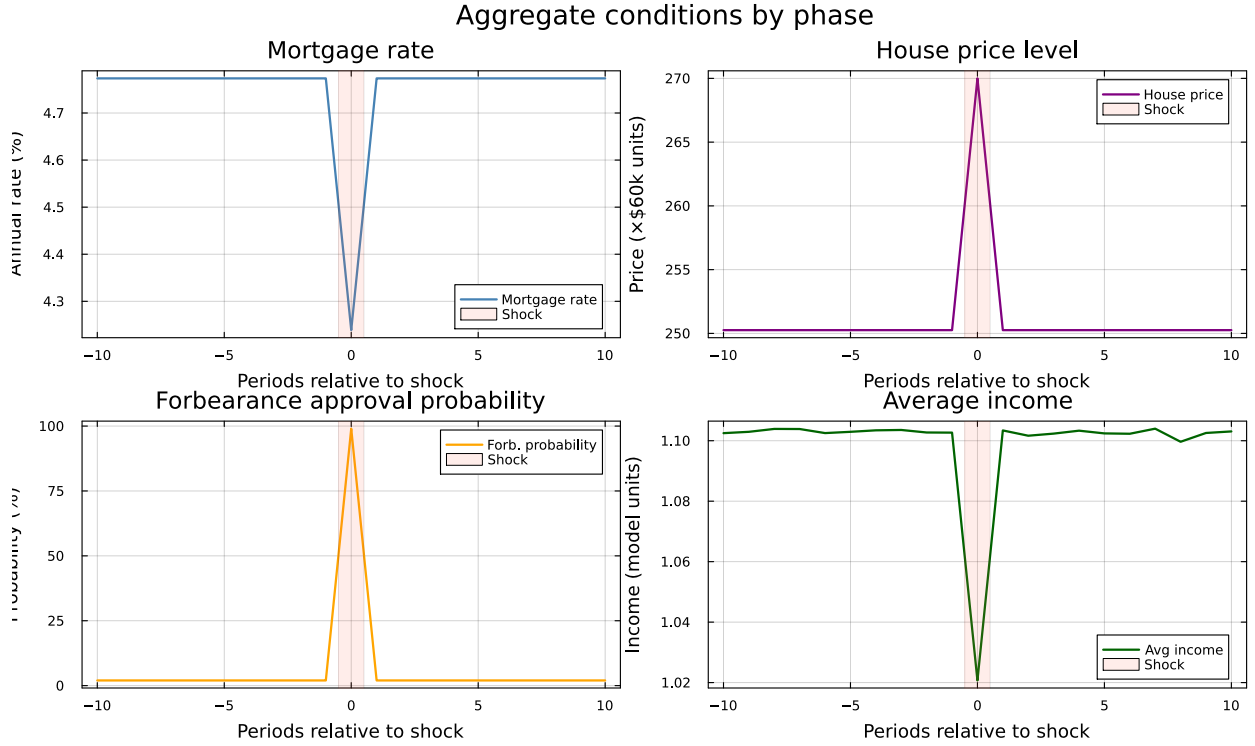


Figure 8: Aggregate conditions across the crisis: mortgage rate, house prices, forbearance-approval probability, and average income, by period relative to the shock.

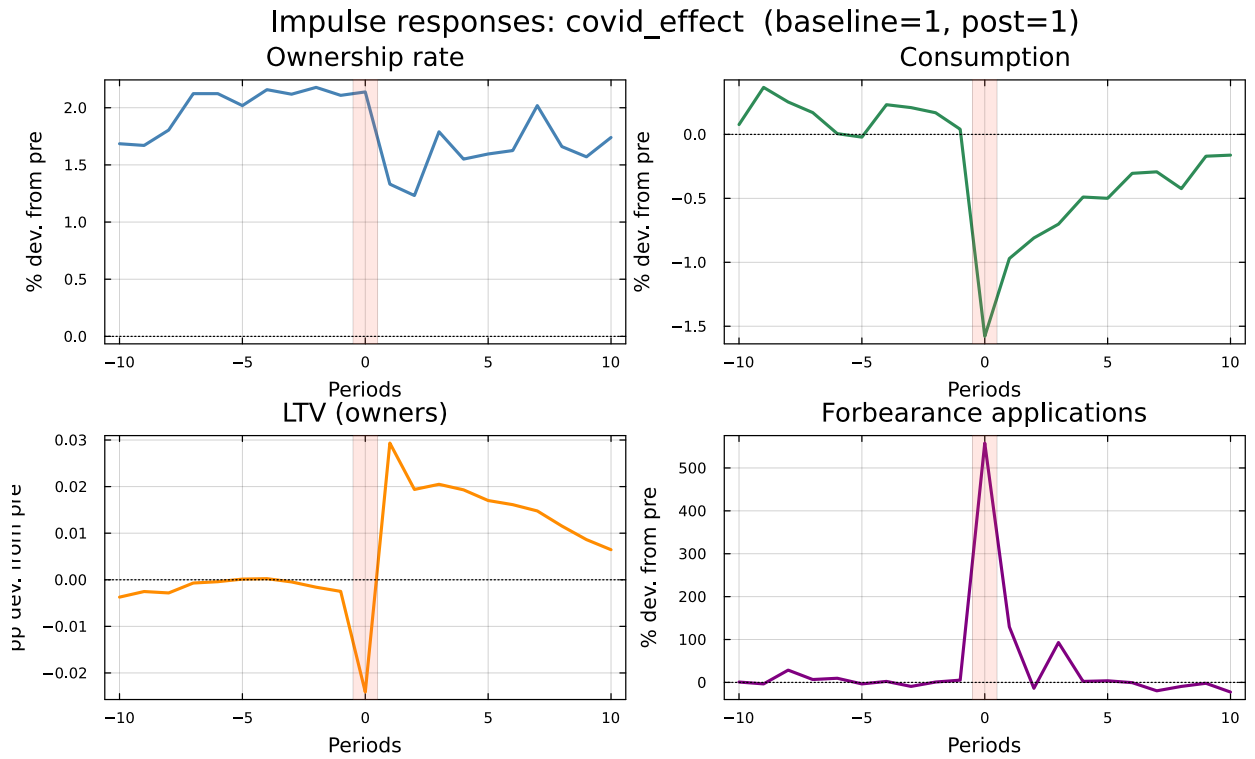


Figure 9: Impulse responses to the crisis: homeownership rate, consumption, owner loan-to-value, and forbearance applications, deviations from the pre-shock mean.

twice the pre-shock share, because the spells opened during the crisis close into delinquency rather than into payment. That reversal is the first sign of the postponement result of Section 7. With both valves open the baseline economy still produces no foreclosure wave on impact. The refinancing and forbearance responses are the object of the incidence analysis that follows; the flat foreclosure margin is what the two instruments jointly deliver in the crisis period itself.

Table 9: Household decision distribution around the COVID shock.

Decision (% of living households)	Pre-shock	Shock	Post-shock
Rent	30.1	30.6	30.7
Purchase	1.8	1.3	1.8
Pay mortgage	65.1	49.2	63.9
Refinance	0.4	16.2	0.1
Trade up	0.2	0.0	0.3
Forbearance approved	0.0	1.5	0.0
Default	0.1	0.2	0.1
Cured from delinquency	1.1	0.1	0.8
Delinquent (in bridge)	1.2	0.9	2.3
Total	100.0	100.0	100.0
<i>N</i> (living)	29,368	29,406	29,392

Notes: Each column is the distribution of living households—of the 30,000 simulated, net of the 632, 594, and 608 deaths—across states in the last pre-shock period, the impact period, and the first post-shock period; columns sum to 100 up to rounding. “Forbearance approved” counts both direct approvals and delinquent owners approved that period. “Cured from delinquency” and “Delinquent (in bridge)” make the multi-period delinquency bridge, previously folded into the residual, explicit. Under the eligibility specification of Section 4.8 a household with no balance left to defer can no longer be labelled a forbearer or a delinquent when its position is payoff-identical to paying, which is why the pre-shock bridge is small here.

Figure 10 shows the same distribution as shares of all decisions, by phase. The crisis bar is the refinancing wave; forbearance approval appears only in the crisis; and the default and delinquent-to-default bars sit on the floor in every phase.

Figure 11 traces the forbearance and default dynamics together—applications spike while the default rate falls to zero—and Figure 12 shows take-up concentrated among working-age, high-leverage owners.

The refinancing wave is the mirror image of the forbearance response. Figure 13 plots the model’s refinancing rate over the crisis window, by age, and by phase. Refinancing is negligible in normal times—well under 1% of owners per period—spikes sharply on impact when the rate cut lands, and subsides once the low rate has been captured. In the cross-section of age it is concentrated among younger owners and declines toward retirement—the mirror image of the near-retirement concentration of forbearance (Figure 12), and a preview of the ten-year age gap between refinancers and forbearers documented in Section 6.5.

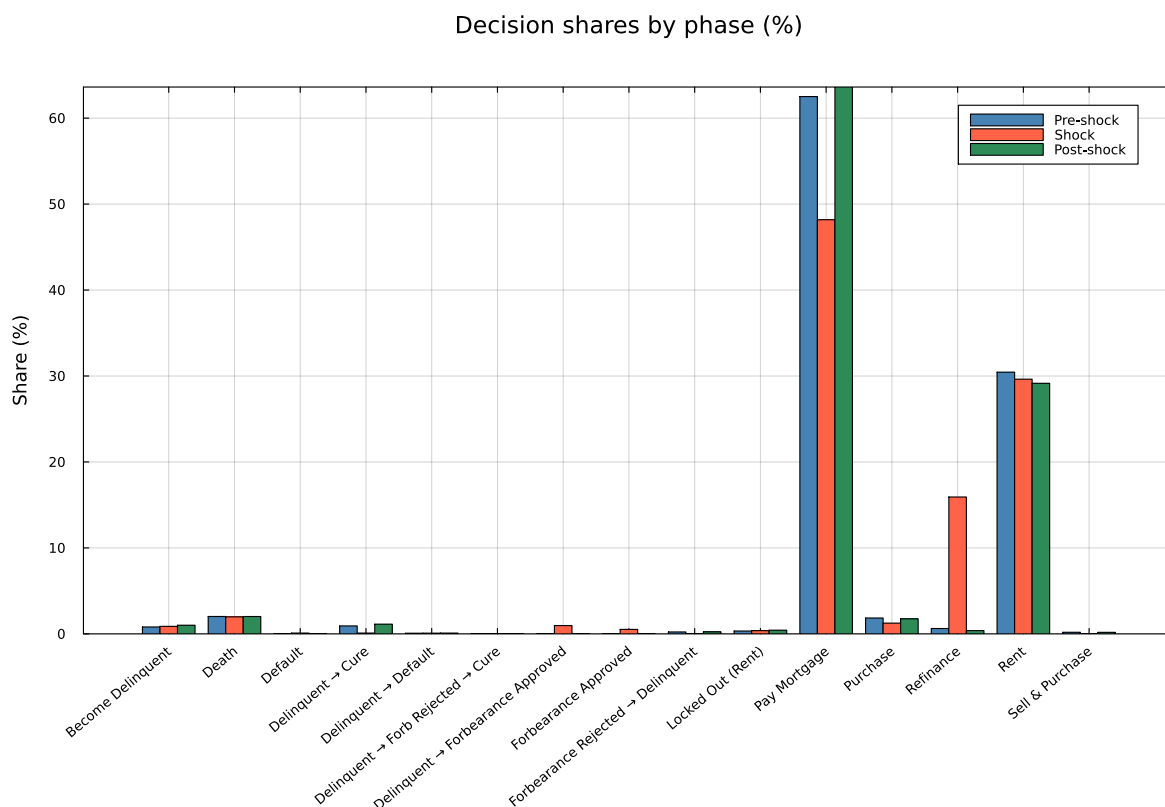


Figure 10: Decision shares by phase (percent of all decisions), pre-shock, shock, and post-shock. The refinancing share rises from under 1% to about 16% on impact and falls back afterwards; forbearance approval appears only in the crisis; the default bars remain on the floor throughout.

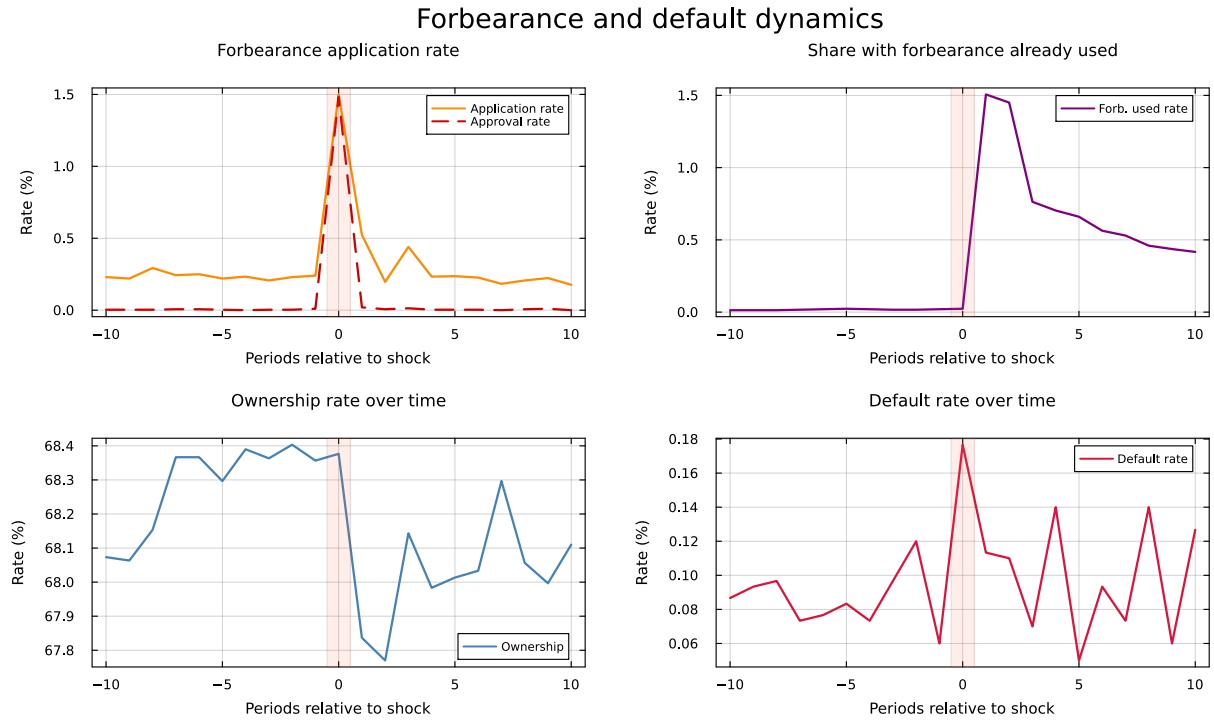


Figure 11: Forbearance and default dynamics: application and approval rates, the active-forbearance share, and the default rate over the crisis window.

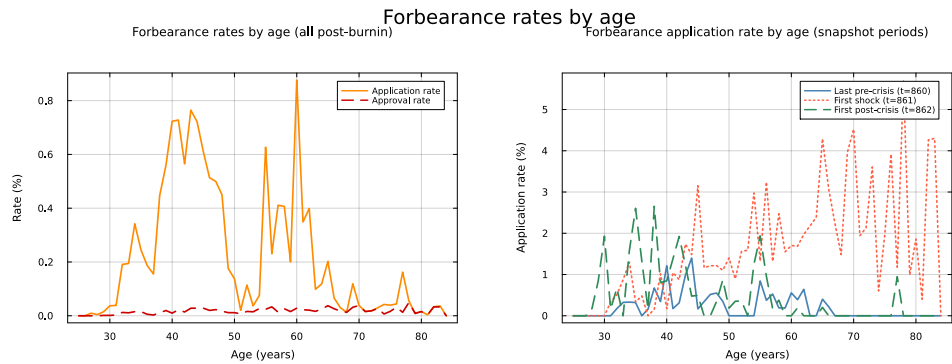


Figure 12: Forbearance application and approval rates by age. Take-up concentrates among working-age, high-leverage owners.

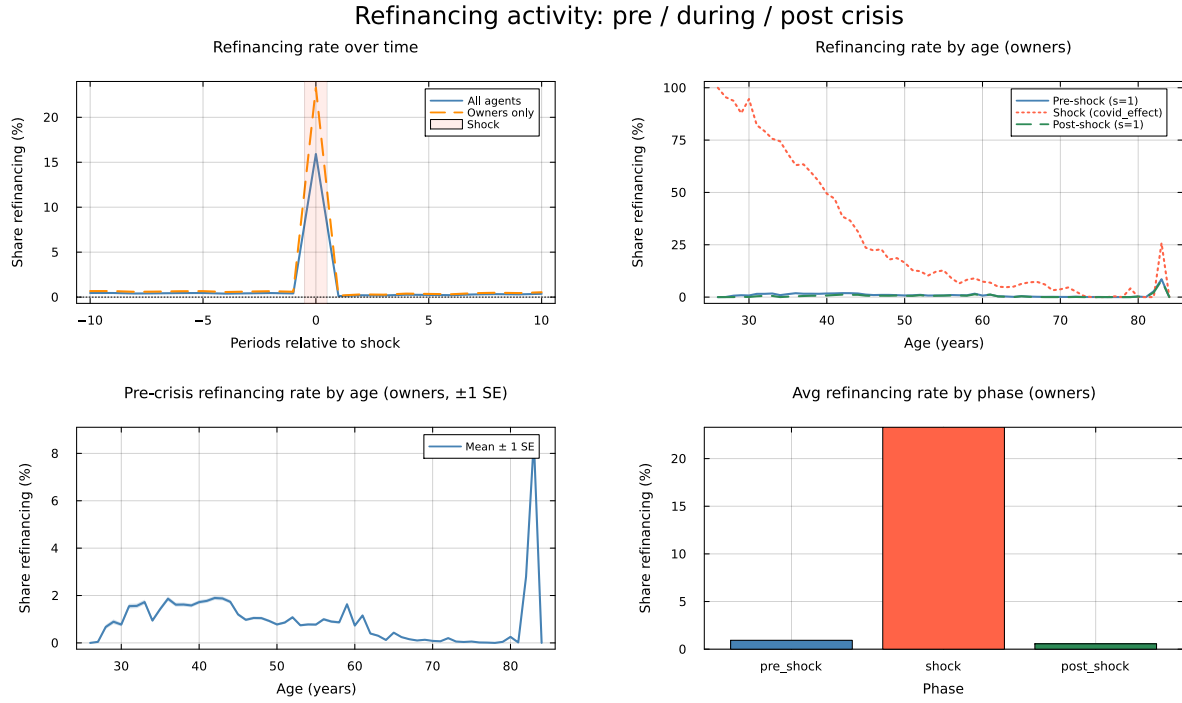


Figure 13: Model refinancing activity around the COVID shock. **Top left:** the share of agents, and of owners, refinancing by period relative to the shock—a sharp on-impact spike over a near-zero baseline. **Top right:** the refinancing rate by age in the pre-shock, shock, and post-shock phases; during the shock it concentrates among younger owners and declines toward retirement. **Bottom left:** the pre-crisis refinancing rate by age (± 1 standard error). **Bottom right:** the average refinancing rate by phase. The on-impact response is the 4,777 refinances of Table 9.

6.5 Who Takes Each Action: Selection into Decisions

The model places the instruments on different segments of the pre-crisis distribution, and the margin that separates them is income. Table 10 reports pre-crisis characteristics by the action a household takes in the impact period. The forbearance-approved enter with high leverage and collapsing income (pre-shock LTV 0.66, mortgage 2.75, crisis income 0.52), an older, near-retirement group (mean age ≈ 61). Refinancers are *comparably* leveraged (LTV 0.60, mortgage 2.48) but keep their income (crisis income 0.99, nearly twice the approved cohort’s) and are two decades younger (mean age ≈ 42): they exercise the option the historic-low rate opens. Those who keep paying are the low-leverage, high-income owners (LTV 0.11, income 1.35), and the fifty-three shock-period defaulters are the most levered group of all (LTV 0.71) with the lowest income (0.27). The critical comparison is forbearers versus refinancers: they sit at nearly the same leverage—0.66 against 0.60, six hundredths apart against the 0.11 of the payers—and are separated by flow income, which differs by a factor of two. The payment-to-income constraint is what makes this margin bite: an owner whose income collapses to 0.52 cannot clear the PTI test that refinancing into the low rate requires, so the price instrument cannot reach her, and forbearance is the only instrument that can. Section 6.7 tests that reading directly by removing the test, and Section 6.6 takes up the two cohorts’ twenty-year age gap, which is the other thing this table shows.

Table 10: Pre-crisis characteristics by crisis-period decision.

(mean)	Refinance ($N=4,777$)	Forbearance ($N=445$)	Pay mtg. ($N=14,454$)	Default ($N=53$)	Rent ($N=8,388$)
LTV (pre-shock)	0.60	0.66	0.11	0.71	0.00
Mortgage (pre-shock)	2.48	2.75	0.46	2.95	0.01
Liquid assets (pre-shock)	0.10	0.37	0.69	0.21	0.36
Income (crisis)	0.99	0.52	1.35	0.27	0.54
Age (approx. years)	42	61	60	45	44

6.6 Segmented Incidence: The Cohort-Overlap Diagnostic

Whether forbearance and refinancing reach separate populations is the paper’s central positive question. For each shock-period owner I compute whether they are ever forbearance-approved (FA) and whether they ever refinance, and form the *refinancing rate ratio* $P(\text{refi} \mid FA)/P(\text{refi} \mid NA)$. Table 11 reports $P(\text{refi} \mid FA) = 17.8\%$ against $P(\text{refi} \mid NA) = 27.6\%$, an aggregate rate ratio of 0.64: the forbearance cohort refinances at roughly two-thirds the rate of non-applicants. That gap is real, and Section 6.7 shows it is generated by the underwriting tests. But the form it takes is compositional, and saying so precisely matters more than claiming it is large.

The refinancing indicator records whether a household refinances at any point between the shock and the end of its life, so it accumulates with remaining lifespan: among never-applicants it falls from 93% in the 25–29 bin to 10% after age 80. The two cohorts sit at opposite ends of that gradient.

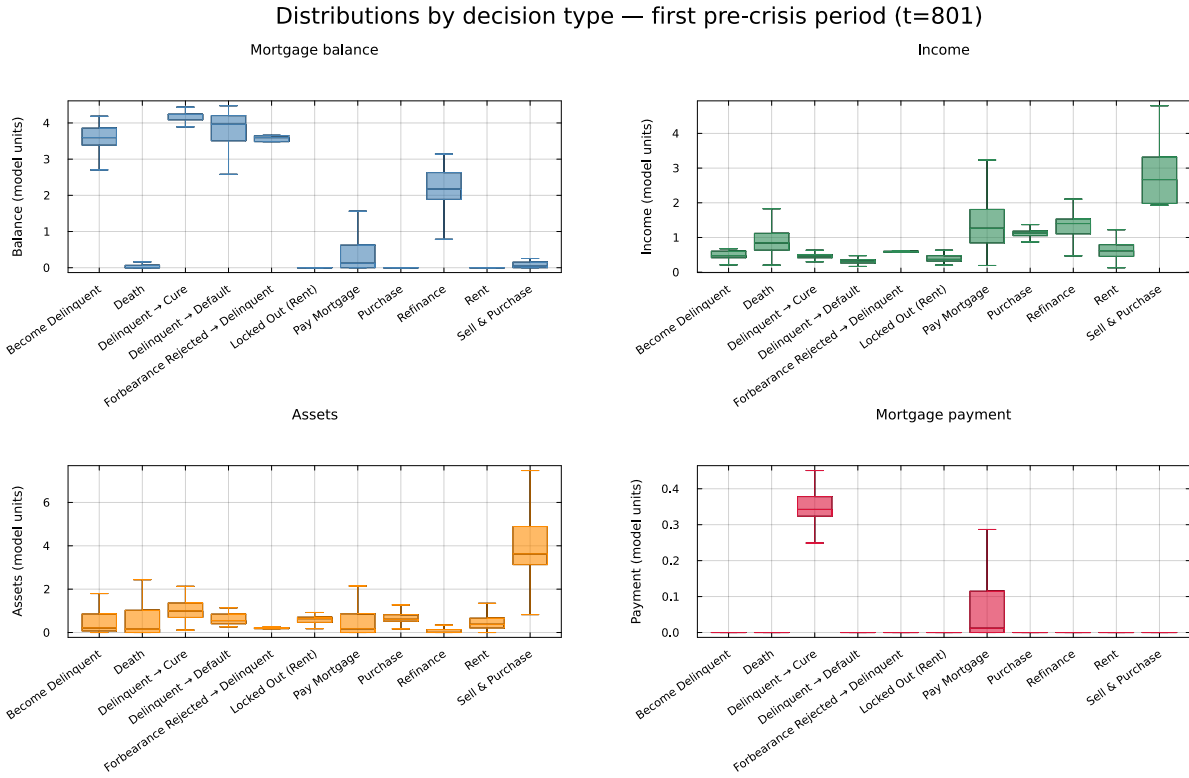


Figure 14: Pre-crisis distributions by crisis-period decision: income, mortgage balance, LTV, and assets in the first pre-shock period.

Not one of the 445 approved households is under 30, against 451 never-applicants in that bin; the approved modal bin is 65–69, and their mean age is 61 against the refinancers’ 42. Applying the age-specific refinancing rates of non-applicants to the age distribution of the approved, the approved would be expected to produce 83.7 refinances against the 79 observed: an *age-standardised rate ratio* of 0.94, with a Poisson interval of [0.74, 1.15] on the observed count. Conditional on age, the approved refinance at essentially the same rate as everyone else.

The honest reading is that the segmentation is a segmentation *by age*, and that this is a statement about the mechanism rather than a retraction of it. The maturity rule ties the term to remaining life, $n = \min(30, T - a)$, so an older owner refinancing into the same rate faces a shorter amortisation window, a higher payment, and a payment-to-income test that tightens mechanically with age (Section 8.5 quantifies how much). Age is therefore not a confounder sitting beside the underwriting channel; it is one of the routes through which the underwriting channel operates. Standardising on it removes part of the mechanism along with the composition, which is why the standardised figure should be read as a lower bound on the constraint’s role rather than as the corrected estimate. Both numbers are reported, and neither is the whole statement: the crude ratio says the two programmes reached different people, the standardised ratio says the difference runs through who was old and levered rather than through a residual difference among otherwise identical owners.

What the crude gap is *not* is an artefact of who happened to apply. Section 6.7 removes the underwriting tests one at a time and the rate ratio moves above one—the approved cohort, once allowed to be re-underwritten, refinances *more* than non-applicants, because it is the high-leverage segment with the most to gain from a rate cut. That is the sharpest available evidence that the constraint, and not selection, is what separates the two populations.

6.7 Which Constraint Generates the Segmentation

Section 6.5 shows that the forbearance-approved and the refinancers differ on two margins at once: the forbearance-approved enter the crisis with much lower flow income *and* with higher leverage. Because the model imposes a payment-to-income test, a loan-to-value limit, a no-cash-in restriction, and a retirement-tied maturity, the segmentation it produces is generated by construction; the substantive question is not whether the mechanism operates but how much of the observed incidence gap each restriction accounts for.

I answer that by relaxing the restrictions one at a time and recomputing the overlap statistics of Section 6.6. Each arm re-solves the household problem with a single restriction lifted and re-simulates the `covid_effect` experiment. The arms are: the baseline; no payment-to-income test; a loan-to-value limit relaxed from 80% to 95%; cash-in refinancing permitted out of liquid assets at the usual transaction cost; a fixed thirty-year refinance term; and no refinance transaction cost.

One feature of the design bounds what the exercise can establish. Each knob enters the household problem over the whole life cycle, not only in the crisis, so the arms are not one economy with a crisis channel switched off but six economies whose households reach the crisis with different balance sheets. The forbearance-approved cohort itself changes size across arms, from 356 households when

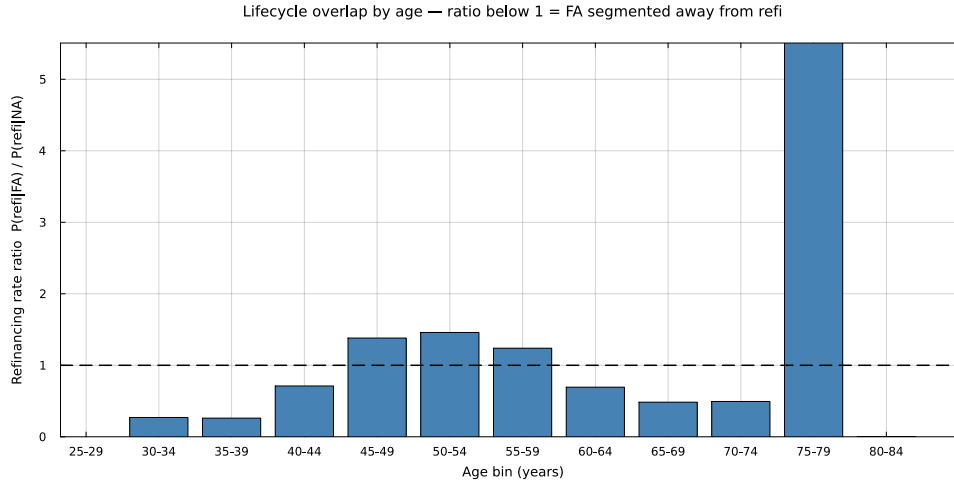


Figure 15: Lifecycle decomposition of the $FA \times$ Refinance rate ratio. The dashed line at ratio = 1 separates segmentation (below) from common participation (above). The profile is non-monotone: the rate ratio is well below one before age 40 and between 60 and 75, and rises above one in the mid-life bins, where refinancing among non-applicants falls faster with age than it does among the approved. Much of this shape is exposure rather than behaviour: because the indicator records refinancing at any point up to the end of life, $P(\text{refi} \mid NA)$ falls from 0.93 among the youngest to 0.10 after age 80, and the approved cohort is concentrated at the older end. Standardising for age raises the aggregate rate ratio from 0.64 to 0.94 (Table 11). Figures above each bar are the number of forbearance-approved households in that bin; they range from none below age 30 to 68 in the 65–69 bin, so the profile is informative about shape rather than about any individual bin—the 75–79 ratio of 5.5 rests on 18 refinances among 45 approved households and should not be read as a finding.

Table 11: FA $\times$ Refinance cohort-overlap diagnostic.

Statistic	Value
N forbearance-approved (FA)	445
N never-applicant (NA)	20,063
N refinancers (shock + post)	5,607
$P(\text{refi} \mid FA)$	0.178
$P(\text{refi} \mid NA)$	0.276
Rate ratio $P(\text{refi} \mid FA)/P(\text{refi} \mid NA)$	0.64
FA refinances observed	79
FA refinances expected at NA age-specific rates	83.7
Age-standardised rate ratio (observed / expected)	0.94

Notes: A ratio below one indicates the FA cohort refinances *less* than non-applicants. The age-standardised rate ratio applies the age-specific refinancing rates of non-applicants to the age distribution of the approved cohort (indirect standardisation over the twelve five-year bins of Figure 15); it removes the part of the crude gap that comes from the approved being an older cohort with less remaining lifespan over which to refinance. As Section 6.6 argues, that removal is not innocuous: the maturity rule makes the payment-to-income test tighten with age, so age is one of the routes the constraint operates through, and the standardised figure is a lower bound on the constraint’s role rather than a corrected estimate. This is the baseline simulation used throughout the paper and the source of the figures, on a panel of 30,000 households; the constraint arms of Table 12 are the same panel and the same draw, so its baseline row and this table are the same simulation and the two are directly comparable.

refinancing is costless to 1,167 under the relaxed loan-to-value limit, against 445 in the baseline, so each arm’s $P(\text{refi} \mid FA)$ conditions on a different population. Differences across rows therefore combine the direct effect of relaxing the restriction with a compositional effect, and the exercise ranks restrictions rather than decomposing the gap additively. All six arms are run on the same 30,000-household panel under the same draw as Table 11, so its baseline row and this one are the same simulation and the two tables can be read against each other directly.

The result is sharper than a narrowing of the gap: three of the five restrictions *reverse* it. Removing the payment-to-income test takes the rate ratio from 0.64 to 1.74; permitting cash-in refinancing takes it to 1.17; relaxing the loan-to-value limit to 1.11. In each of those economies the forbearance-approved refinance at a *higher* rate than never-applicants, not merely at a comparable one. That is what one should expect if the tests are the binding object: the approved cohort is the high-leverage, high-payment segment with the most to gain from a rate cut, and it is exactly that segment the tests exclude. Take away the test and the ordering flips.

The two *price* margins do nothing of the kind. A fixed thirty-year term leaves the ratio at 0.62 and removing the refinance transaction cost leaves it at 0.59, both statistically indistinguishable from the baseline’s 0.64 and neither showing any sign of the reversal the underwriting arms produce. Whatever role the maturity rule plays in the *age* profile—and Section 8.5 shows it plays a large one—it is not what generates the aggregate incidence gap.

Among the three underwriting margins the payment-to-income test dominates, and by a wide margin: it moves the ratio 1.10 points where cash-in moves it 0.53 and the loan-to-value cap

0.47. The earlier version of this exercise could not separate income underwriting from the cash-in restriction; on the corrected specification it can, and the income test is roughly twice either collateral margin. Section 8.6 reports the across-replication variation that applies to statistics computed on cohorts of a few hundred, and the gaps here are large relative to it; the ranking of the three, and the flat response of the two price margins, are what the exercise establishes.

The no-cost arm illustrates why the rate ratio must be read together with its components. Setting the refinance transaction cost to zero drives crisis refinancing from 15.9% to 70.4% of households and pushes $P(\text{refi} \mid NA)$ to 0.96. With the unapproved cohort nearly saturated the ratio is compressed against its ceiling, and its value of 0.59 reflects that arithmetic rather than any strengthening of the segmentation—note that $P(\text{refi} \mid FA)$ rises to 0.57 in that arm, more than three times its baseline value.

Table 12: Constraint decomposition: the incidence gap with one restriction relaxed at a time.

Arm	$P(\text{refi} \mid FA)$	$P(\text{refi} \mid NA)$	Rate ratio	n_{FA}
Baseline	0.178	0.276	0.64	445
No payment-to-income test	0.629	0.362	1.74	545
Cash-in refinance permitted	0.499	0.427	1.17	864
Loan-to-value limit relaxed to 95%	0.379	0.343	1.11	1,167
Fixed 30-year refinance term	0.247	0.395	0.62	531
No refinance transaction cost	0.570	0.965	0.59	356

Notes: Each row re-solves the household problem with one restriction relaxed and re-simulates the `covid_effect` experiment. All six rows are the same 30,000-household panel under the same draw, and the baseline row is the same simulation as Table 11, so the two tables are directly comparable. The rate ratio is $P(\text{refi} \mid FA)/P(\text{refi} \mid NA)$; values below one indicate segmentation and values above one indicate the approved cohort refinancing more than non-applicants. The horizontal rules separate the three underwriting margins, all of which reverse the ratio, from the two price margins, neither of which moves it. Section 8.6 reports the across-replication variation for statistics computed on cohorts of this size. Because each knob applies over the whole life cycle, the arms differ in their pre-crisis distributions and in the size of the forbearance-approved cohort (n_{FA}); the table ranks restrictions and does not decompose the gap additively. In the final row $P(\text{refi} \mid NA)$ is near its ceiling, which compresses the ratio mechanically.

7 Prevention versus Delay, and Welfare

Section 6 establishes what each instrument does and whom it reaches. This section asks what the eligibility instrument accomplishes for the households it reaches: whether it prevents foreclosure or postpones it, and what it is worth to them.

7.1 Cohort Attribution: Who Is Saved, Who Is Delayed

To attribute default outcomes I track three cohorts through the crisis: the forbearance-approved (FA), those who applied and were rejected (FR), and the never-applicants (NA). Table 13 reports cohort sizes and eventual-default outcomes. The FA cohort (445 owners) defaults at 36.6% eventually, against 3.4% for the NA control—an eleven-fold gap. Because crisis approval is certain on request, the FR cohort is empty, so the naive rejected-then-default counterfactual is not populated in this

run. The FA cohort is the structurally fragile segment: the gap between its near-zero shock-period default and its 36.6% eventual default is the delayed default that forbearance postpones rather than prevents—of the 445 forbearance-approved owners, 163 eventually default and 282 never do, so on the cohort accounting the bridge is followed by no foreclosure for a little under two thirds and by a later one for the rest. Section 7.2 shows why that two thirds is not a count of foreclosures prevented. Section 7.2 shows that the first half is not the same as a foreclosure prevented. Forbearance is thus insurance against the shock-period liquidity event for this segment, not a durable reduction in its lifetime default risk. Figure 16 shows the FA and NA event-time trajectories, and Figure 17 the lag between approval and eventual default among FA defaulters, most of which occur within a few periods.

Table 13: Cohort sizes and eventual-default outcomes.

Cohort	N	Ever default (%)	Saved share (%)
FA (forbearance-approved)	445	36.6	63.4
FR (applied, rejected)	0	—	—
NA (never-applicant)	20,063	3.4	96.6

Notes: FA = forbearance-approved during the shock; FR = applied but rejected; NA = never-applicant. “Ever default” tracks any default over the whole horizon.

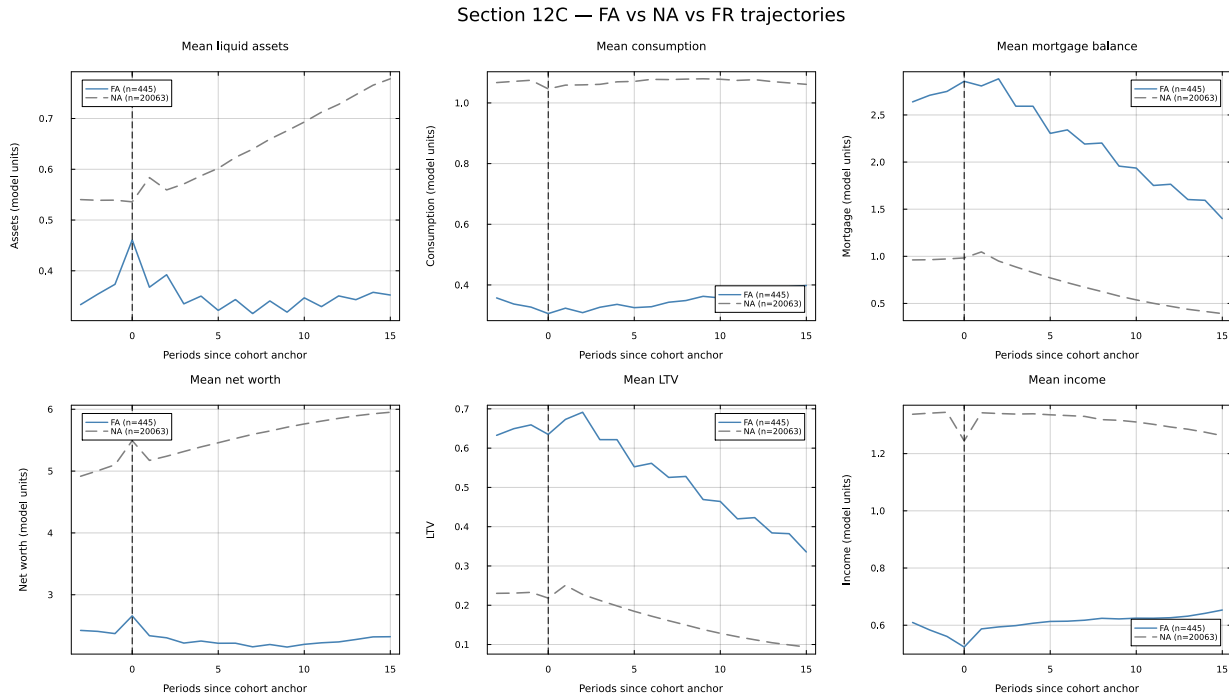


Figure 16: Event-time trajectories of the FA and NA cohorts: mean assets, consumption, mortgage, net worth, LTV, and income around the shock.

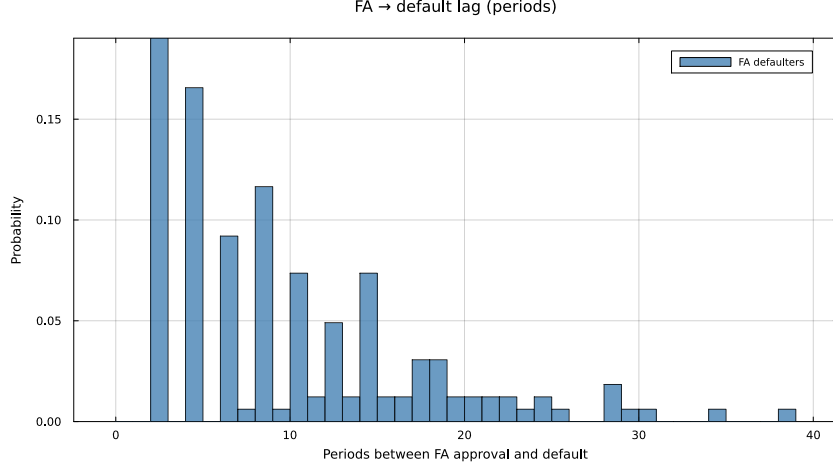


Figure 17: Periods between forbearance approval and eventual default among FA agents who default. Most delayed defaults occur within a few periods.

7.2 Prevented or Postponed?

The cohort shares of Section 7.1 describe what happens to the forbearance-approved; they do not identify what forbearance *did*. A household that does not foreclose after receiving forbearance need not have been saved by it, and a household that eventually forecloses need not have been failed by it. One minus the treated cohort’s default rate is a descriptive statistic, not a treatment effect. Separating the two requires the counterfactual the structural model is uniquely able to supply: the same households, under the same shocks, without the option.

I therefore solve the model twice, identically except that forbearance is unavailable in every aggregate state, and simulate both economies under common random numbers, so that each household faces the same mortality, income, and approval draws in both. Households approved for forbearance during the crisis in the first economy are then matched to themselves in the second and classified by the pair of realized outcomes. The matching is exact: every one of the 102,794 matched households lives the same span in both arms, which is the diagnostic that the draws are genuinely shared rather than merely similarly seeded. Pre-shock outcomes differ for 0.02% of them, as they must—the option is removed in every aggregate state, not only in the crisis, so the two economies are not identical before the shock either.

Table 14 reports the classification at four horizons, and two things stand out. The first is how little of the cohort forbearance moves at all. Over the remaining lifetime, 62.5% of the approved never foreclose in either arm and a further 27.3% foreclose at the same date in both: for nine in ten of the households the instrument reaches, it changes the tenure outcome not at all. Whatever forbearance did for them—and Section 7.3 shows it smoothed consumption during the hardship year—it did not alter whether or when they lost the house.

The second is that among the tenth it does move, delay dominates prevention. Over the lifetime 1.1% of the cohort avoided a foreclosure they would otherwise have had, while 5.2% merely

postponed one: relief delayed foreclosure roughly five times more often than it prevented one. And the horizon manufactures the appearance of prevention. At twelve months forbearance looks purely protective—3.6% prevented and no other category populated, because no delayed foreclosure has had time to arrive—and by the lifetime horizon two thirds of that apparent prevention has resolved into delay or into no effect at all. A study that stops at twelve months would report the 3.6% as lives saved.

This is the same-agent version of the timing result that Section 7.1 infers from cohort shares, and it is what the reduced-form evidence of Section 2.3 cannot deliver: the 1.5% against 17.2% three-year foreclosure gap there is consistent both with prevention and with postponement, and only the counterfactual separates them.

Two smaller categories deserve to be reported rather than absorbed. Foreclosure is *brought forward* for 3.4% of the cohort—the household defaults earlier with the bridge than without it. The mechanism is that deferral lets a household run its liquid assets down before defaulting, arriving at the default decision with a thinner buffer than it would have had under immediate pressure to cure or sell. And 0.5% are *adverse* in the strict sense of foreclosing with forbearance while avoiding foreclosure without it. Both are small, and both are estimated on a handful of households.

The counts behind these shares are thin and the paper should not pretend otherwise. The classification cohort is 443 households; the lifetime prevented share rests on 5 of them and the delayed share on 23. The binomial standard error is 0.5 percentage points on the former and 1.1 on the latter, and Section 8.6 reports the wider across-replication variation that applies to statistics computed on cohorts of this size. What survives that uncertainty is the ordering and the magnitude of the gap—delay exceeds prevention several times over, and the great majority of the treated cohort is unaffected either way—not the individual cells.

The policy reading is not that forbearance failed. A postponed foreclosure is worth something: it buys time in a labor market that was recovering fast, and it moves the terminal event away from the moment of maximum aggregate distress, which is the externality a moratorium is designed to address. Consumption smoothing during the hardship year is a real benefit and is where the welfare gains of Section 7.3 sit. But forbearance should not be credited with durably removing default risk from a leveraged, income-stressed cohort, because for the great majority of that cohort it did not change the outcome, and where it did change it, it more often moved the date than removed the event.

7.3 Welfare by Age $\times$ LTV

The overlap statistics measure whether the FA and refinance cohorts share a population; they do not measure the welfare each instrument delivers. For each shock-period agent choosing Refinance or Forbearance I compute the consumption-equivalent variation (CEV) of the chosen action against two counterfactuals, Pay Mortgage and Default, evaluated one period after the shock (Appendix B). The closed-form CEV assumes the value function is homogeneous in consumption. This one is not—the default penalty enters additively and the bequest term does not scale—so alongside it I

Table 14: Same-agent classification of the forbearance-approved cohort.

With forbearance	Without forbearance	1 year	3 years	5 years	Lifetime
No foreclosure	Foreclosure	3.6	2.0	1.8	1.1
Foreclosure later	Foreclosure earlier	0.0	2.0	2.5	5.2
Foreclosure, same	Foreclosure	0.0	4.1	8.8	27.3
Foreclosure earlier	Foreclosure later	0.0	0.2	0.9	3.4
Foreclosure	No foreclosure	0.0	0.5	0.7	0.5
No foreclosure	No foreclosure	96.4	91.2	85.3	62.5

Notes: Rows are, in order: prevented, delayed, little or no effect, brought forward, adverse, and infra-marginal.

Shares of the 443 households approved for forbearance during the crisis in the forbearance-available economy, matched to themselves in an otherwise identical economy in which forbearance is unavailable in every aggregate state. Both arms are the same 30,000-household panel under the same seed as the rest of the paper, and all 102,794 matched agents have identical lifespans across arms—the hard diagnostic that the draws are shared. Of the 445 approved, 2 had already foreclosed before the crisis in the no-forbearance arm and are excluded, since for them the second arm is not a counterfactual for the crisis. Horizons are measured from the shock period; “lifetime” runs to the terminal age. Columns sum to 100 up to rounding. The counts are small: the lifetime row holds 5 prevented, 23 delayed, 121 unchanged, 15 brought forward, 2 adverse and 277 infra-marginal, so the binomial standard error is 0.5 percentage points on the prevented share and 1.1 on the delayed share. Section 8.6 reports the wider across-replication variation that applies to cohort statistics of this size; the ordering, not the individual cells, is what those intervals leave standing.

solve the comparison numerically, for the resource transfer that would leave a household as well off under the counterfactual as under the action it took (Appendix B). The numerical version needs no homogeneity assumption, uses the model’s own value function including the penalty and the bequest, and never divides by a value, so it is free of the sign degeneracy that forces the closed form to be read ordinally. I report medians and the share of agents with a positive gain (Table 15) and read the age $\times$ LTV heatmaps (Figure 18) ordinally.

Against *continuing to pay* the two instruments look very different, and the forbearance column needs a caveat before it can be read. Refinancing gains at the median—a closed-form CEV of +0.3%, or +0.08 years of income numerically—and 64.5% of refinancers gain at all. Forbearance shows a median gain of exactly zero, with only 13.9% of the approved better off and 49.7% worse off. But 36.4% of the cohort sits at an *exact* tie: deferring roughly a quarter of one year’s payment does not move those households off their mortgage grid node, so the pay and forbear continuations are the same number by construction. That is a resolution limit at $N_M = 15$, not an economic finding, and it is why the median is 0.000 rather than a small positive or negative value. The against-pay comparison should be read as *small and not well resolved*; Section 8.6 flags the grid refinement this calls for.

Against *default* there are no ties at all and the comparison is clean. Refinancing dominates default for 94.8% of refinancers, by 1.06 years of income at the median. Forbearance does not: only 40.4% of the approved end the year after the shock in a better position than if they had surrendered the house, and the median is -0.62 years of income.

That sign deserves care, because it is not a violation of optimality. These values are continuations evaluated one period after the shock, not total values including the crisis-period flow. A forbearer trades a worse position next year—a balance grown by capitalized interest, in a house whose payment

it may still not be able to carry—for consumption during the hardship year itself, and chooses forbearance over default on the sum of the two. What the negative figure says is that for three fifths of the approved, the position forbearance leaves them in is worse than the one default would have left them in. That is the postponement result of Section 7.2 read off the value function rather than off realized foreclosure dates, and the two agree: an instrument that leaves most of its recipients in a worse continuation than surrender, and that changes the tenure outcome for one in ten of them, is doing its work somewhere other than the foreclosure margin. The heatmaps place what gains forbearance does deliver against default in the high-LTV, near-retirement cells—the segment for which foreclosure is the binding alternative—and refinancing’s gains at the higher-income cells.

The numerical comparison preserves every ordering the closed form gives—refinancing’s advantage is overwhelmingly over default, forbearance’s over continuing to pay—and the two disagree only on magnitude, most sharply for refinancers against default, which is exactly where the additive default penalty breaks the homogeneity assumption. Neither measure is clean everywhere, and the failures are not in the same places. The closed form discards 10.4% of refinancers and 13.7% of the approved on the default comparison, where V^{chosen} and V^{def} carry opposite signs or the denominator approaches zero. The numerical solver never divides by a value but can run out of grid: it censors at the lowest feasible asset level for 1.7% of refinancers against default—a clear improvement—but for 39.6% of the approved against default, 30.3% of refinancers against paying and 24.7% of the approved against paying. Censoring below means the counterfactual is better than the chosen action even stripped to zero assets, so those entries are bounds and the true transfers are more negative still. The censored households are the same ones carrying a negative ΔV ; the reported medians sit above the censored block in each case, so they are conservative rather than misleading, but the tails should not be read cardinally.

One margin is worth stating plainly rather than leaving to the table. Refinancing beats default for almost every refinancer, but it beats simply continuing to pay for only 65% of them: for about a third, the transaction cost of refinancing exceeds the payment relief it buys, and they would have done better to keep paying. The rate cut is therefore not a free option even for the households that can clear underwriting.

8 Robustness, Cost, and Scope

This section asks how far the incidence result of Section 6 depends on the design of the instrument, on the aggregate state the crisis strikes from, on the modelling assumptions most open to question, and on the numerical implementation. It closes with the cost of the instrument and the scope conditions under which every counterfactual in the paper should be read.

8.1 The Design of the Instrument: Moratorium and Eligibility

The eligibility instrument is a bundle of design choices, and the model lets me vary three. First, crossing forbearance with a temporary foreclosure moratorium (a 2×2) shows the two are strong

Table 15: Welfare gain from the chosen action vs. counterfactuals (crisis period).

Chosen action	Median CEV (%)		Median transfer (yrs inc.)		Share with $\Delta V > 0$	
	vs Pay	vs Default	vs Pay	vs Default	vs Pay	vs Default
Refinance ($N = 4,777$)	+0.3	+15.6	+0.08	+1.06	64.5%	94.8%
Forbearance ($N = 445$)	0.0	-28.6	-0.00	-0.62	13.9%	40.4%

Notes: Median consumption-equivalent variation of the chosen action against each counterfactual, and the share of agents with a positive value gain. Means are omitted because the closed-form CEV is ill-defined where the continuation value changes sign. The closed-form percentages should be read ordinally—signs and rankings—rather than cardinally, because the additive default penalty and the bequest term do not scale with consumption. The final two columns report the numerically solved resource transfer, in years of household income, which requires no homogeneity assumption; see Appendix B. The two measures agree on every ranking and differ in magnitude, most sharply for refinancers against default. All six columns are continuations evaluated one period after the shock, so a negative entry against default says the chosen action leaves the household in a worse position next year, not that it was dominated at the moment of choice: the crisis-period flow, which is where deferral pays off, is not in these values. The transfer columns are censored at the lowest feasible asset level for 30.3% of refinancers against paying and 39.6% of the approved against default; those entries are bounds and the medians reported here sit above the censored block. The forbearance row against paying is dominated by exact ties—36.4% of the approved have identical pay and forbear continuations because deferring about a quarter of one year’s payment leaves them on the same mortgage grid node at $N_M = 15$ —so that cell is a statement about grid resolution as much as about welfare and should not be read as a point estimate.

Section 22 — CEV by (age quintile $\times$ LTV quintile)

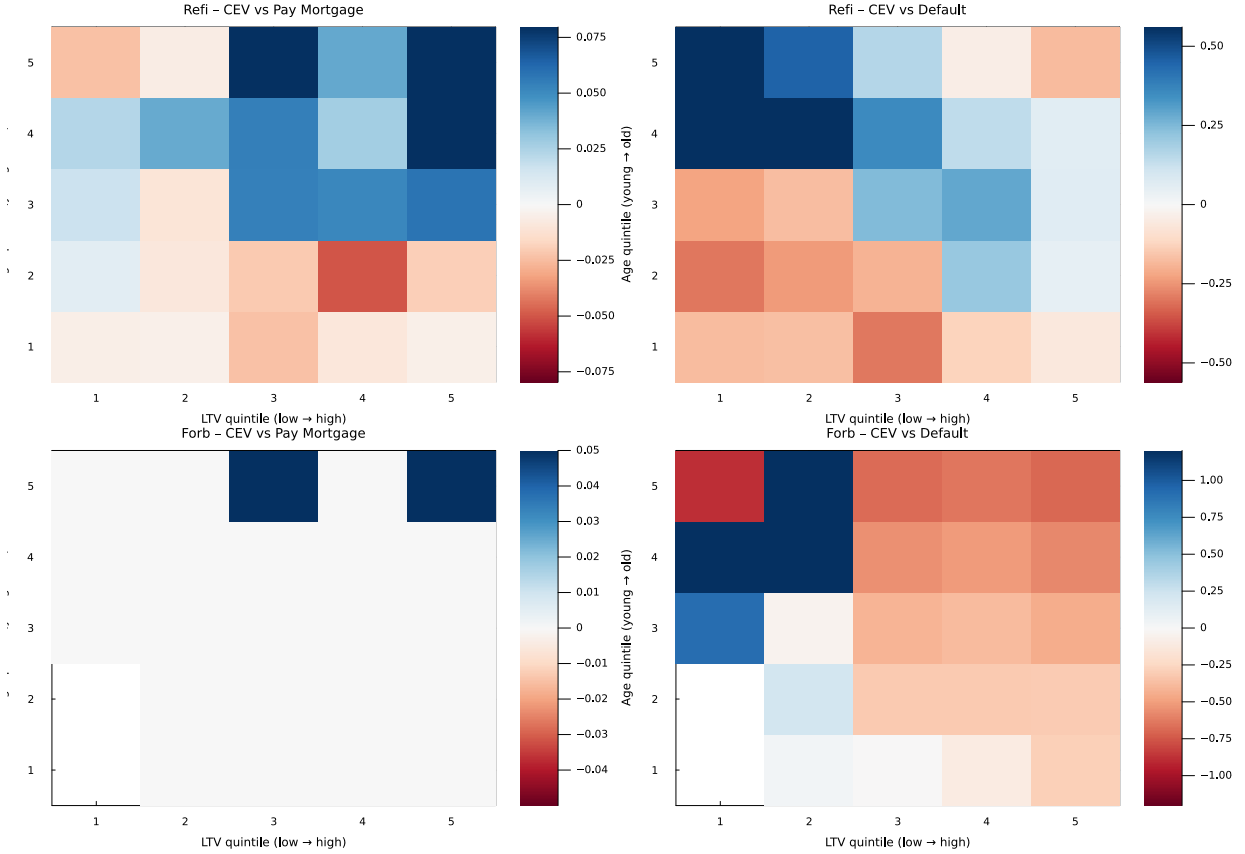


Figure 18: Consumption-equivalent welfare by age and pre-shock LTV. Rows: chosen action (Refinance, Forbearance). Columns: counterfactual (Pay Mortgage, Default). Positive means the chosen action dominates the counterfactual.

substitutes on the foreclosure margin: with neither instrument the COVID shock produces 69 crisis-period foreclosures; forbearance alone cuts this to 53, the moratorium alone to 0, and the two together to 0. The moratorium is the stronger instrument within the crisis period, but it is so largely by construction—it bans the action rather than changing the household’s incentive to take it—and almost all of what it achieves there is timing. Foreclosures in the first post-crisis period run 34 with neither instrument and 41 under the moratorium alone. Over the lifetime the differences are too small to read: lifetime default is 3.14% with neither instrument, 3.13% with forbearance alone and 3.10% with the moratorium, against an across-replication standard deviation of 0.05 percentage points on an identical economy (Section 8.6). The crisis-period effect is large and unambiguous; the lifetime effect is not separately identified at this panel size. Both instruments move *when* foreclosure happens considerably more than *whether* it happens, which is the conclusion of Section 7 arrived at from the aggregate rather than the household side. Second, the CARES repayment regime—how the deferred balance is settled on cure—moves take-up while leaving the foreclosure floor intact: the crisis forbearance-approval rate ranges from 1.48% of households under term extension, the most attractive of the four regimes, to 0.39% under reinstatement on cure, with crisis default at 0.18% and lifetime default at 3.13% in every cell.

Third, and most directly, the hardship eligibility threshold τ^{pti} is a pure take-up lever. Sweeping it from 0.10 to 0.40 moves the crisis approval rate monotonically from 3.18% of households to 0.81%, a factor of four, and lifetime take-up from 1.08% to 0.27% (Table 16). Nothing else moves. Crisis default is 0.177% in every one of the seven cells, lifetime default 3.13% in every one, and the crisis refinancing share sits between 15.85% and 15.94% across the whole sweep. Widening eligibility fourfold changes who is admitted to the bridge and changes nothing at all about what happens on the foreclosure margin. How widely the bridge is extended is a design choice; whether it holds for those who cross it is not.

Table 16: The hardship eligibility threshold as a take-up lever.

τ^{pti}	Crisis appr. (%)	Lifetime take-up (%)	Crisis default (%)	Lifetime default (%)	Crisis refi. (%)
0.10	3.18	1.08	0.177	3.13	15.85
0.15	2.01	0.73	0.177	3.13	15.85
0.20 (baseline)	1.48	0.58	0.177	3.13	15.92
0.25	1.31	0.53	0.177	3.13	15.92
0.30	1.10	0.45	0.177	3.13	15.92
0.35	0.96	0.36	0.177	3.13	15.92
0.40	0.81	0.27	0.177	3.13	15.94

Notes: Each row re-solves and re-simulates the baseline `covid_effect` experiment with the eligibility threshold τ^{pti} of equation (13) set to the stated value and every other lever held fixed, on the same 30,000-household panel under the same draw. A household may apply only if its scheduled payment at the locked rate is at least τ^{pti} of current income. “Crisis forb. appr.” is the shock-period approval share of households; “Lifetime take-up” the ever-forbearance share. The default and refinancing columns are reported precisely because they do not move: the threshold governs admission to the instrument and nothing else.

8.2 The Repayment Regime

Forbearance defers a payment; it does not forgive one. How the deferred balance is settled once the household cures is therefore a design choice in its own right, and one the CARES Act left to servicers. The model implements the four regimes that account for the observed exits, and re-solves the household’s problem under each, because the settlement terms enter the continuation value and so change the decision to apply.

Deferral to loan-end moves the missed payments into a non-interest-bearing balloon due at the loan’s original maturity, leaving both the balance and the scheduled payment untouched in the interim; this was the modal CARES exit. *Term extension*, the baseline used elsewhere in the paper, capitalizes the unpaid interest and lengthens the term so that the post-forbearance payment matches the pre-shock payment. *Re-amortization* capitalizes the same interest but holds the term fixed, so the payment resumes strictly higher than before. *Reinstatement* capitalizes the full deferred amount immediately on cure and re-amortizes it over the remaining term, concentrating the repayment burden at exactly the moment the household exits hardship; the FHFA barred servicers from *requiring* it in April 2020.

Table 17 imposes each regime in turn. Term extension and re-amortization are indistinguishable: they share a continuation value in the backward induction and are separated only by the post-cure payment path, and at the model’s deferral intensity that path is too short to tell them apart. Both approve 445 households in the crisis period and both carry a lifetime take-up of 0.58%. The other two regimes move take-up, and both move it down—deferral to loan-end approves 277 households (0.92%) and reinstatement 118 (0.39%).

That ordering reverses the one this table carried under the paper’s earlier deferral calibration, where loan-end deferral was the most attractive regime and term extension the middle one, and the reversal is worth stating rather than absorbing. The likely reason is the size of the deferred balance. At $\mathbb{E}[D] = 2.87$ months the arrears are small enough that what separates the regimes is not the interest cost of carrying them—which loan-end deferral minimizes—but when the household is asked to produce them. Term extension never asks for a lump at all; loan-end deferral asks for the whole balance in one payment at maturity; reinstatement asks for it at the moment of cure. Under the earlier calibration the deferred balance was more than three times larger and its carrying cost dominated. I have not separately verified this channel, and it should be read as the reading the ranking suggests rather than as a result the experiment establishes.

What the regime does *not* move is the foreclosure margin, and the corrected specification makes this sharper than the earlier one did. Lifetime default is 3.13% in all four cells—identical to three decimal places—and crisis default is 0.177% in three of them and 0.180% in the fourth. A design change that cuts crisis approvals by 73% leaves the number of foreclosures where it found it. The repayment regime is a lever over how many households the bridge carries, not over whether it holds. The paper’s baseline is term extension; the institutional Enterprise benchmark is closer to deferral to loan-end, which Table 17 shows would carry roughly 38% fewer households at the same foreclosure count.

Table 17: Forbearance take-up and default under the four repayment regimes.

Repayment regime	Crisis appr. (%)	Take-up (%)	Crisis default (%)	Ever default (%)
Term extension (baseline)	1.483	0.581	0.177	3.132
Re-amortization	1.483	0.581	0.177	3.132
Deferral to loan-end	0.923	0.363	0.177	3.132
Reinstatement on cure	0.393	0.117	0.180	3.132

Notes: Each row re-solves and re-simulates the baseline `covid_effect` experiment under one settlement regime, holding every other lever and all idiosyncratic draws fixed. “Crisis appr.” is the shock-period forbearance-approval share of households; “Take-up” is the lifetime ever-forbearance share. Term extension and re-amortization share a continuation value in the backward induction and are separated only by the post-cure payment path in the simulation, which is why their crisis-period approval rates coincide. Crisis approvals are 445, 445, 277 and 118 households respectively. All four rows and the baseline are simulated on the same panel of 30,000 households at seed 1234 with common random numbers, so counts as well as shares are directly comparable.

8.3 How Deep the Crisis Cuts

Section E.1 varies where in the cycle the crisis lands; this subsection varies how hard it hits. I sweep the size of the output contraction from -5% to -30% , holding every other lever of the `covid_effect` bundle fixed—the same 2 percentage point cut in the short rate, the same 8% house-price appreciation, the same CARES approval-on-request—so that the only thing changing across rows is the depth of the income shock the relief has to bridge. The paper’s baseline is the -10% cell, so the sweep nests it. Because the shock is an unanticipated (MIT) event reached with zero probability from the normal states, each severity is a separate crisis state whose value function the backward induction solves independently of the others; the six cells are one solve, but they are six genuinely re-solved economies, not one solution re-simulated at different income levels. All six rows share the same pre-shock economy, so homeownership (67.6%) and mean owner LTV (0.31) agree across them to within simulation noise and the differences below are the crisis response alone.

Refinancing collapses monotonically with severity: the crisis refinancing share falls from 18.41% of households at a -5% contraction to 3.25% at -30% , a factor of nearly six, even though the rate cut driving it is identical in every row. The same underwriting test that admits the distressed to forbearance excludes them from refinancing, and as the contraction deepens it begins to exclude the undistressed as well.

Forbearance scales with the shock, but not without limit, and this is where the corrected specification departs from a simple monotone story. Crisis approvals rise from 1.28% of households at -5% to a peak of 5.30% at -20% , and then fall back to 3.68% at -30% ; lifetime take-up traces the same hump, from 0.53% up to 1.68% and back to 1.21%. The turning point is not simulation noise—the across-replication standard deviation of the crisis approval rate is 0.08 percentage points (Section 8.6), and the differences here are an order of magnitude larger. Past roughly a -20% contraction the marginal distressed household is not being routed into forbearance but carried past it: crisis delinquency decisions rise from 263 households at the baseline to 4,015 at -30% , and crisis foreclosures from 53 to 172. The bridge has a capacity, and a deep enough contraction delivers

households to the far bank of it rather than onto it.

The collapse in refinancing is what makes the sweep more than a robustness check, because it overturns the sign of the incidence result at sufficient depth. The rate ratio rises from 0.59 at -5% through 0.64 at the baseline -10% , crosses one between -15% and -20% , and settles near 2 across the three deepest cells. What produces the crossing is almost entirely one of its two components: $P(\text{refi} \mid NA)$ falls by more than half, from 0.308 to 0.117, while $P(\text{refi} \mid FA)$ moves only from 0.182 to 0.225. The approved cohort’s refinancing rate is close to flat across the whole sweep; it is the unapproved who lose access. In a mild contraction the payment-to-income test binds selectively, on the income-collapsed, and the forbearance-approved are the households it shuts out of refinancing. In a severe one it binds nearly universally, the unapproved lose the income advantage that distinguished them, and the ratio inverts—not because forbearance begins reaching refinancers, but because refinancing stops reaching everyone else.

The paper’s segmentation result is therefore a statement about moderate contractions, and the -10% COVID calibration sits closer to the sign change than is comfortable: one cell further down the sweep the ratio has already risen to 0.94, and the cell after that it is 1.95. I read this as a boundary condition rather than a threat to the empirical finding, which is estimated on a -10% episode and matched by the -10% cell. But it does mean the mechanism should not be extrapolated to deep recessions. Section E.1 shows the same ratio is also sensitive to the aggregate state the crisis arrives from, and the two results should be read together: the segmentation finding is specific to a moderate contraction arriving from a normal aggregate state, which is what the COVID episode was. Whether the crossing is real or an artifact of the extreme cells, where a -30% annual contraction is far outside the calibration’s support, is a question the model as specified cannot settle.

Table 18: Shock magnitude: the COVID bundle imposed at output contractions from -5% to -30% .

Output contraction	Crisis appr. (%)	Crisis refi. (%)	Take-up (%)	$P(\text{refi} \mid FA)$	$P(\text{refi} \mid NA)$	Rate ratio
-5%	1.28	18.41	0.53	0.182	0.308	0.59
-10% (baseline)	1.48	15.92	0.58	0.178	0.276	0.64
-15%	2.26	12.23	0.80	0.215	0.228	0.94
-20%	5.30	4.54	1.68	0.249	0.128	1.95
-25%	4.54	3.78	1.47	0.240	0.120	2.01
-30%	3.68	3.25	1.21	0.225	0.117	1.92

Notes: Each row imposes the `covid_effect` bundle—a 2 percentage point cut in the short rate, house prices $+8\%$, CARES forbearance on request—with only the output contraction changed, from the pre-shock normal state 1 used elsewhere in the paper. The six severities are grafted as separate crisis states in a single backward induction; with the crisis reached at probability zero from the normal states, that is equivalent to six separate solves.

Homeownership (67.6%) and mean owner LTV (0.31) are common to all rows up to simulation noise—the six severities share one pre-shock economy—and are omitted. “Crisis forb. appr.” is the shock-period forbearance-approval share of households, “Crisis refi.” the shock-period refinancing share, “Take-up” the lifetime ever-forbearance share. The rate ratio is $P(\text{refi} \mid FA)/P(\text{refi} \mid NA)$; it crosses one between -15% and -20% . The approved cohort numbers 384 households in the -5% cell and 1,590 in the -20% cell, so the deeper rows are the more precisely estimated ones. The across-replication standard deviation of the rate ratio at this panel size is 0.046 (Section 8.6). Idiosyncratic draws and the aggregate path are seeded identically across rows, and all six are simulated on the same panel of 30,000 households as the baseline.

8.4 Further Robustness

Five further checks confirm the incidence result without changing it, and are reported in Appendix E to keep the main text on the results: imposing the crisis from each of the eight normal aggregate states (Section E.1); full recourse rather than non-recourse default (Section E.2); a 2008-type house-price collapse in place of the 2020 appreciation (Section E.3); permitting downward housing adjustment (Section E.4); and extending the forbearance grace window (Section E.5). In each the composition of who each instrument reaches is unchanged; what moves is the level of the response, and in the grace-window case its duration.

8.5 Mortgage Maturity and the Age Profile

The baseline ties the mortgage term to remaining life, $n = \min(30, T - a)$, so that no new loan extends past the terminal age. This is a substantive assumption for the age results. A household near the terminal age is forced into a short amortization window, which raises the payment on any new loan and so makes the payment-to-income test progressively harder to clear with age. Three findings could in principle inherit from that assumption: that refinancing concentrates among the young, that forbearance concentrates near retirement, and that forbearance’s welfare gains are largest near retirement.

Actual thirty-year mortgages need not mature before the borrower dies; the estate settles the outstanding balance. I therefore re-solve with a fixed thirty-year term for all originations, with any outstanding balance settled against terminal net worth in the bequest, and re-run the incidence diagnostics. The arm appears as the fixed-term row of Table 12: in the aggregate it barely moves the segmentation, raising the rate ratio from 0.57 to 0.68, about one and a half seed standard deviations and not distinguishable from simulation noise. The aggregate incidence result does not depend on the maturity rule.

The age profile is another matter, and here the assumption does most of what one would fear. Refinancing among the unapproved, which in the baseline collapses after age 50 and never recovers, instead plateaus under a fixed term: $P(\text{refi} \mid NA)$ at ages 70–74 is 0.41 against the baseline’s 0.10, and at 75–79 it is 0.39 against 0.12. Measured against the young-age peak, about a third of the baseline’s decline in refinancing by these ages is attributable to the shrinking amortization window rather than to economic forces, and the late-life level is roughly four times higher under a fixed term. The distribution of forbearance shifts with it: the approved cohort peaks at ages 60–64 in the baseline, with 31% of approvals falling at age 65 or above, and peaks at 50–54 under a fixed term, with 13% above 65.

I therefore restrict the age claims accordingly. That refinancing is concentrated among the young survives in direction but not in magnitude; the steepness of the baseline profile past midlife is a modelling artifact and should not be read as an economic finding. That forbearance concentrates near retirement is weakened enough that I do not rest any argument on it. What the maturity arm leaves intact is the aggregate composition result—who each instrument reaches, pooled across ages—which is what Sections 6.6 and 7.1 actually use. The age-binned profiles are reported for

description, with the caveat of Section 8.6 that individual bins of the approved cohort are too thin to interpret one at a time.

8.6 Numerical Accuracy and Simulation Uncertainty

The headline statistics rest on a finite simulated panel, and several of them are computed on small cells—the forbearance-approved cohort numbers a few hundred households, and crisis-period foreclosure counts are in the dozens. That uncertainty should be measured rather than assumed away, so I repeat the baseline experiment across ten independent seeds, holding the solved policy functions fixed and redrawing only the idiosyncratic histories.

Table 19 reports the result, and the two panels behave quite differently. The aggregate statistics are tight. The standard deviation across replications is 0.09 percentage points for homeownership, 0.02 for lifetime take-up and 0.05 for lifetime default, so the calibration targets and the headline take-up and default numbers are reproducible to the precision at which the paper quotes them. Crisis-period statistics computed on the full panel are nearly as stable: crisis refinancing has a standard deviation of 0.20 percentage points against a mean of 15.88.

One crisis-period statistic is not. Crisis default has a mean of 0.195% and a standard deviation of 0.033—seventeen percent of its own mean—because it is a count of roughly sixty foreclosures out of thirty thousand households. Differences in crisis default smaller than about 0.07 percentage points between two arms of this model are within one replication’s noise, and I do not rank arms on that margin anywhere in the paper.

The overlap diagnostic is another matter, and the contrast is instructive. $P(\text{refi} \mid NA)$, computed on roughly twenty thousand unapproved owners, has a standard deviation of 0.004. $P(\text{refi} \mid FA)$, computed on roughly four hundred and fifty approved owners of whom some seventy-five refinance, has a standard deviation of 0.012 around a mean of 0.166—a range across seeds spanning nearly a quarter of its own mean. The rate ratio inherits this almost entirely, with a standard deviation of 0.046 around a mean of 0.600 and a range from 0.54 to 0.70.

The pattern is not a property of the seed but of the conditioning event. The panel holds thirty thousand households; the forbearance-approved cohort holds about four hundred and fifty, and it varies from 413 to 494 across replications on an otherwise identical economy. Sampling error scales with the cohort rather than the panel, which is why aggregates computed on the whole panel are stable to the precision this paper quotes while everything conditioned on approval is an order of magnitude noisier. I report the statistics at the current panel size and flag the ones it cannot yet support, rather than quoting a precision the cohort does not have.

Two consequences follow, and I apply them throughout. First, the segmentation result is robust: every one of the ten replications puts the rate ratio below one, the largest being 0.70, so the qualitative finding does not depend on the draw even though its second decimal does. Second, the rate ratio is not presently precise enough to support fine distinctions. Every arm in the paper is now simulated on the same 30,000-household panel, so comparisons across the constraint arms of Section 6.7 are read against the standard deviation reported here (0.046), and differences smaller

than about two of them are reported as indistinguishable rather than ranked. The same caution applies with more force to the age-binned decomposition of Figure 15: with an approved cohort of a few hundred households spread across twelve age bins, individual cells contain as few as one household, and the profile should be read as a shape rather than as a sequence of estimates. It applies equally to the classification shares of Table 14, which are computed on a cohort of the same size. Cell counts are reported alongside every age-bin and age-by-LTV statistic so that the precision implied by a heat map is not mistaken for the precision the underlying cell can support.

Table 19: Simulation uncertainty: the baseline experiment across ten independent seeds.

Statistic	Mean	SD	MC s.e.	Min	Max
Homeownership (%)	67.72	0.090	0.028	67.63	67.91
Mean owner LTV	0.310	0.001	0.000	0.309	0.311
Lifetime take-up (%)	0.599	0.024	0.008	0.561	0.644
Lifetime default (%)	3.108	0.054	0.017	3.020	3.183
Crisis forbearance approvals (%)	1.494	0.082	0.026	1.377	1.647
Crisis refinancing (%)	15.88	0.200	0.063	15.63	16.30
Crisis default (%)	0.195	0.033	0.010	0.140	0.257
$P(\text{refi} \mid FA)$	0.166	0.012	0.004	0.152	0.192
$P(\text{refi} \mid NA)$	0.277	0.004	0.001	0.273	0.284
Rate ratio	0.600	0.046	0.014	0.542	0.699
n_{FA}	448	25	8	413	494

Notes: Ten replications of the baseline `covid_effect` experiment off a single solved policy, varying only the seed of the idiosyncratic draws; the crisis path and the aggregate sequence are held fixed. “MC s.e.” is the standard deviation divided by $\sqrt{10}$. The lower panel is the overlap diagnostic of Table 11. The contrast between the two conditional probabilities is the point of the table: $P(\text{refi} \mid NA)$ is estimated on roughly 20,000 households and $P(\text{refi} \mid FA)$ on roughly 450, and the rate ratio inherits the latter’s imprecision. All ten replications place the rate ratio below one. Every table in the paper, this one included, is simulated on the same 30,000-household panel, so the standard deviations reported here are the ones against which differences elsewhere should be judged.

8.7 Who Bears the Cost of Forbearance?

The incidence analysis identifies which borrowers each instrument reaches; it is silent on who bears the cost. Forbearance is not free even when it prevents no aggregate price decline: deferral converts a stream of scheduled payments into a capitalized balance repaid later, so the lender bears the time-value of the deferred cash flow and the incremental default risk on the fraction of the forbearance-approved who eventually default anyway. The cohort attribution of Section 7.1 is what makes this cost legible. Because the forbearance-approved cohort eventually defaults at 36.6%, the lender’s cost of extending the bridge is not the full deferred balance on every approved loan but the deferred cash flow plus the loss-given-default on the roughly one third of the cohort for whom the bridge only postpones foreclosure. The incidence result sharpens the accounting in a second way: because the instrument reaches the income-collapsed, high-leverage segment rather than the equity-rich, the marginal forbearance dollar is spent on exactly the borrowers whose collateral cushion is thinnest, so the realized loss-given-default on the delayed-default tail is larger than a random-borrower calculation would imply. A full cost–benefit statement prices the lender’s discounted cash flow under the deferral schedule against the counterfactual of immediate foreclosure, borrower by

borrower; the structural model supplies the objects that pricing requires—the deferred schedule, the delayed-default hazard, and the collateral position of the treated cohort—and locates the cost at the high-leverage, near-retirement cells, which are also the cells where what the instrument does deliver against default is concentrated (Section 7.3). The comparison of that cost against a foreclosure moratorium, which shifts the same forbearance-eligible borrowers off the foreclosure margin at no deferred-interest cost to the lender but at an administrative and property-market cost the model does not price, is the natural next step for a welfare-ranking of the two instruments.

8.8 Scope and Limitations

The model is a disciplined laboratory for one question—the incidence of two crisis instruments—and its design choices are what make that question answerable. Several of those choices are also its limitations, and stating them before the counterfactuals clarifies what the experiments in Section 6 do and do not identify. The theme common to them is that the model isolates *who is reached, when, and on which margin*, at the cost of the general-equilibrium and higher-frequency machinery a full welfare accounting of the 2020 or 2008 episodes would require.

The most consequential limitation is that the analysis is *partial equilibrium*. House prices follow the imposed crisis path rather than responding to the wave of household decisions the crisis sets off. This is deliberate: holding the price path fixed is exactly what isolates the incidence of each instrument—which borrowers refinance, which take forbearance, at what age, on which income margin—from the price feedback that would otherwise confound it. But it omits two channels. The first has a known sign: by preventing foreclosures, forbearance supports house prices and thereby further suppresses default, so the general-equilibrium complementarity between the two instruments is, if anything, stronger than the partial-equilibrium version documented here. The second is the price-spiral feedback central to a full 2008 counterfactual—foreclosures depressing prices, pushing more owners underwater, and inducing further foreclosures—which the model cannot generate. This is why, when Section E.3 imposes a 2008-type house-price collapse, the model does *not* reproduce a 2008-scale foreclosure wave: the level of the foreclosure response is muted by construction. The incidence pattern survives the change in backdrop, but the model should not be read as a quantitative account of the *magnitude* of the Great Recession’s foreclosure crisis.

Three further modeling restrictions bound the exercise. Default is *non-recourse* in the baseline, so the default decision is an option to walk away from negative equity rather than a choice weighed against a deficiency judgment; Section E.2 re-solves under full recourse and finds the character of the incidence result unchanged, but the level of default and its age profile do shift with the recourse assumption, which varies across U.S. states in a way the single-economy calibration does not capture. Housing adjustment is *upsized-only*: households may hold or trade up but not trade down. The restriction is empirically appropriate for the 2020 episode and, as Section E.4 shows, the households who would use a downward-adjustment margin are equity-rich optimizers essentially disjoint from the leveraged, income-stressed households the paper studies—but admitting the margin would be necessary to speak to episodes or populations in which trading down is a genuine alternative to

default. Finally, the model is solved at *annual frequency*: one period is one calendar year, the delinquency bridge is a one-year state, and forbearance duration enters as a within-year deferred fraction. This is adequate for the tenure and default margins that are the paper’s object but abstracts from the higher-frequency cure-redefault-re-delinquency dynamics that a monthly servicing model would resolve.

A fourth restriction concerns the *application* margin. Applying for forbearance is costless in the model and a rejected applicant returns to her unconstrained best alternative, so applying is weakly dominant in every state. That is why the pre-shock application rate sits near 8–9% even in aggregate states where the approval probability is close to zero: the model is reporting willingness to apply, not a considered decision, and any pre-crisis application series should be read accordingly. Approvals, which is what every result in this paper is computed on, are unaffected—the gate binds regardless of how many households present at it—but a genuine application cost, whether pecuniary, documentary, or a stigma or inattention cost, would separate willingness from take-up and is a channel the constraint decomposition of Section 6.7 does not test. Adding it requires a new state variable and a re-solve of every cell, so I flag it as an untested channel rather than estimate it.

A concrete illustration is the length of forbearance spells. The model draws the number of months in hardship from a re-employment hazard, so realized forbearance duration is a within-year object bounded above by twelve months and concentrated at short spells—a median of two months, a mean of 2.67, with about 81% of forbearers exiting within three months. The COVID-19 data are longer-tailed. Table 20 and Figure 19 set the model beside two benchmarks: in the overall mortgage market 51% of forbearances lasted three months or less and 85% a year or less, leaving roughly 15% beyond twelve months, with spells still being observed at twenty-one months, the limit of that study’s window (Sánchez and Wilkinson, 2022); and among Ginnie Mae (FHA/VA) borrowers—closer to the leveraged, income-stressed segment the model targets—only about half had exited by six months and some 40% were still in forbearance in their eleventh month (Loewenstein and Njinju, 2022). The model captures that most spells are short but understates duration on two margins: it places too much mass at one- and two-month spells, and, as a one-year state, it cannot represent the long-duration tail—the borrowers who stayed past twelve months—which is exactly the segment for which the delayed-default cost of Section 8.7 is largest. Matching the full duration distribution, and the CARES extension structure behind the long tail, would require a monthly model of the forbearance spell; the annual model captures the *incidence* of forbearance but not the fine structure of its *length*.

Two features of the calibration and the experiment temper how precisely some numbers should be read. The model is calibrated to match SCF and Fannie moments rather than formally estimated, and it does not fully reproduce the *shape* of the mid-life homeownership rise—the ownership curve climbs too steeply and runs above the SCF cross-section through the middle of the working life before the two meet at retirement (Section 5.5)—so the middle-age cross-section is fit less tightly than the population rate and the old-age plateau. And because the crisis-state forbearance-approval probability is near one, the applied-but-rejected (FR) cohort is essentially empty in the baseline run

Table 20: Forbearance-spell length: model versus COVID-19 data.

	Model	Overall market ^a	FHA/VA loans ^b
Median spell (months)	2	≈ 3	≈ 6
Exit within 3 months (%)	81	51	20
Exit within 6 months (%)	96	70	50
Exit within 12 months (%)	100	85	—
Lasting beyond 12 months (%)	0	15	—

Notes: Model figures from the baseline `covid_effect` run; because the model treats forbearance as a one-year state, spells are capped at 12 months by construction. ^a Overall mortgage market, April 2020–December 2021 (Sánchez and Wilkinson, 2022). ^b Ginnie Mae (FHA/VA) loans (Loewenstein and Njinju, 2022); entries read off the reported survival profile ($\approx 50\%$ still in forbearance at 6 months, $\approx 40\%$ at 11 months). Dashes mark thresholds the source does not report.

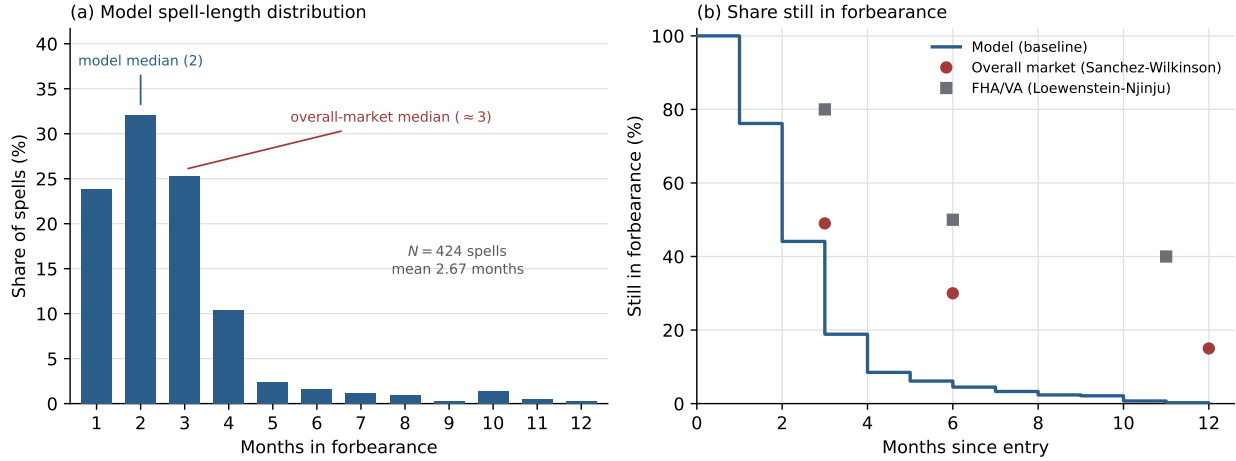


Figure 19: Forbearance-spell length: model versus COVID-19 benchmarks. **Panel (a):** the distribution of months in forbearance in the baseline `covid_effect` run (424 spells, mean 2.67 months), with the model’s median of two months and the overall market’s median of roughly three marked. The model over-concentrates at one- and two-month spells and, as a one-year state, is truncated at twelve months, whereas the overall market peaks at two to three months and still shows spells running at twenty-one, where that study’s observation window ends. **Panel (b):** the share of forbearers still in forbearance by month—the model as a step function against the survival points reported for the overall mortgage market ($\approx 49\%$ still in forbearance at three months, $\approx 30\%$ at six, $\approx 15\%$ past twelve; Sánchez and Wilkinson 2022) and for Ginnie Mae FHA/VA loans ($\approx 80\%$ at three months, $\approx 50\%$ at six, $\approx 40\%$ into the eleventh; Loewenstein and Njinju 2022). The model empties out markedly faster than either benchmark and cannot reach the beyond-twelve-month tail. The benchmarks are the survival points those studies report, not a digitized full distribution.

(Section 7.1); the model therefore cannot populate a rejected-then-default counterfactual, and the incidence result rests on the contrast between the forbearance-approved and the never-applicants rather than on rejected applicants.

The model is also simplistic in its setting as it does not include any non-pecuniary frictions in households’ decisions. There are no psychic or attention costs related to refinancing, moving or applying for forbearance, while there is evidence in the literature that many households do not refinance optimally Berger et al. (2025). Meanwhile, surveys conducted by the Philadelphia Federal Reserve Bank showed that many households that would have qualified for forbearance did not apply, either because they did not know about the program or had concerns regarding repayment structures³. The model’s simplicity also ignores adjustable rate mortgages and choice set of different mortgage term-lengths. Because the model does not explicitly include any spill-over effects from default, the social costs of foreclosure are likely understated, as the literature shows that one’s home foreclosure raises the probability of their neighbors foreclosure and their property values.

A final caveat is measurement, not modeling. The consumption-equivalent welfare comparison of Section 7.3 uses a closed-form CEV that is ill-behaved in the mean at $\sigma = 2$, where the normalized value function can change sign; I therefore report medians and the share of agents with a positive gain and read the age $\times$ LTV heatmaps ordinally rather than cardinally. The welfare rankings are robust to this—both instruments dominate the default counterfactual by a wide margin—but the point estimates of the consumption-equivalent gain should be read as ordinal comparisons across cells, not as precise cardinal magnitudes. Several of these assumptions may affect the *magnitude* of the segmentation result, and the exercises in this section quantify them one at a time. What they do not overturn is the qualitative finding, which is a structural consequence of the underwriting constraints: the two instruments partition the distressed population along the income margin. They bound instead the *quantitative* and *general-equilibrium* claims the model can support, and they mark the extensions Section 9 identifies as next steps.

9 Conclusion

Emergency mortgage policy in 2020 bundled a price instrument—a rate cut transmitted through refinancing—with an eligibility instrument—forbearance granted without underwriting. This paper has asked, in a life-cycle heterogeneous-agent model disciplined to the 2019 SCF and the 2020 mortgage market, which borrowers each instrument reaches and on which margin of the default process. The model reproduces the joint signature of the episode—a refinancing wave of about a quarter of mortgaged owners, forbearance take-up near the observed rate, and a foreclosure non-event—and, within that disciplined laboratory, delivers three results.

First, the two instruments are *substantially segmented*, and what separates them is the requirement to be re-underwritten at all. Forbearance-approved owners refinance at little more than half the rate of non-applicants (a refinancing rate ratio of 0.67). Both groups are leveraged well above

³Recent Data on Mortgage Forbearance: Borrower Uptake and Understanding of Lender Accommodations. - Lauren Lambie-Hanson, James Vickery, and Tom Akana.

the routine payer and the forbearance-approved are the more leveraged of the two, but they differ far more sharply on flow income, and the payment-to-income test that refinancing must clear is exactly the test the income-collapsed borrower fails. Section 6.7 quantifies how much of the gap each restriction generates: removing the payment-to-income test and permitting cash-in refinancing each close most of it by indistinguishable amounts, while the loan-to-value cap, the maturity rule and the transaction cost account for considerably less. The binding object is re-underwriting itself, on income or on the ability to turn liquid wealth into repaired collateral. The rate cut reaches the high-leverage owner only when her income is intact; the owner whose income the shock destroys is the owner forbearance must catch, because the price instrument structurally cannot. Second, forbearance shifts the *timing* of default more than its level. The forbearance-approved cohort eventually defaults at roughly one in two, against one in twenty-three for non-applicants; but that is a cohort share, and the same-agent counterfactual of Section 7.2 is what turns it into a statement about the program. Simulating the identical households in an economy without the option, only 1.1% of those the program reached avoided a foreclosure they would otherwise have suffered, while 5.2% merely postponed one—relief delayed foreclosure some six times more often than it prevented it. The horizon does most of the work in how that looks: at twelve months 3.6% of the cohort appears to have been saved, and two-thirds of that share is borrowed from the future. The bridge postpones foreclosure for a structurally fragile, near-retirement, high-leverage segment rather than durably removing its default risk. Third, a consumption-equivalent welfare comparison places both instruments above the default counterfactual at the median—though the gain is near-universal for refinancers and holds for a smaller majority of the forbearance-approved, the same delayed-default tail seen in the cohort attribution—with forbearance’s largest gains at the high-leverage, near-retirement cells for which foreclosure is the binding alternative.

The results have a direct policy reading. A universal, self-certified payment-relief program is not a substitute for a rate cut but a complement to it: it reaches precisely the income-stressed, high-leverage segment that the refinancing channel cannot, and it does so without the underwriting that would exclude them. The design details that dominate public debate—how the deferred balance is repaid, how tight the eligibility test is—govern how widely the bridge is used far more than whether it holds for those who cross it; and a foreclosure moratorium is a close substitute for forbearance on the foreclosure margin, though the two differ in who bears the cost. But because forbearance postpones default for a fragile segment rather than curing it, a relief program that ends abruptly returns that segment to the foreclosure margin, and the delayed-default tail is where the policy’s residual risk lives.

The model’s scope conditions (Section 8.8) are also its research agenda. The partial-equilibrium structure isolates incidence but omits the general-equilibrium feedback by which forbearance, in preventing foreclosures, supports prices and further suppresses default—a channel whose sign is known and which would strengthen the complementarity found here—as well as the price-spiral feedback central to a full 2008 counterfactual. Embedding the delinquency bridge and the endogenous-duration forbearance technology in a model with an endogenous house-price response, and using

the cohort attribution developed here to price the lender’s cost of the delayed-default tail against a foreclosure moratorium, are the natural next steps. What the present model establishes is the incidence structure any such extension must respect: the two instruments of crisis mortgage policy are complements that reach differently selected segments of the distressed population, and payment relief is the instrument for the segment the price instrument leaves behind.

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A The Recursive Problem: Cash-Flow Detail

This appendix collects the accounting the main model section (Section 4.9) states in words. Throughout, $P^h(s)$ is the house price in aggregate state s , δ_m the maintenance rate, δ the minimum down payment, and $\mathbb{E}V'$ the survival-weighted continuation operator of Section 4.9.

Purchase and trade-up feasibility. A household buying house h' with new loan b' must finance the down payment and the proportional purchase fee out of current resources. Writing the house cost as $H = P^h(s)h'$, feasibility requires

$$x(1 + r^f) + y - [(H - b') + \Phi^{\text{new}}H] - c - x' \geq 0, \quad x' \geq 0, \quad c \geq c_{\min},$$

with b' constrained by the LTV and PTI conditions (11). A trade-up is the same problem restricted to $h' > h$, with the outstanding balance on the old loan retired from the proceeds and the new loan re-originated at the current market rate.

Refinance cash flow. A refinancing owner keeps house h , retires the old balance b , and originates b_{new} at the current market rate, paying a refinance cost that is proportional to house value plus a fixed component, $\Phi^{\text{refi}} = \phi_r P^h(s)h + F_r$. The net cash raised is $b_{\text{new}} - b - \Phi^{\text{refi}}$, which the no-cash-in restriction (12) ($b_{\text{new}} \geq b$) holds at or below zero net of fees, so refinancing lowers the payment rather than extracting equity. The new loan must satisfy the LTV and PTI constraints (11) at b_{new} .

Delinquency and cure. A become-delinquent owner pays only maintenance $\delta_m P^h(s)h$, lets the balance accrue at the locked rate, and advances the delinquency count. On cure, the household pays the missed payment plus a late fee ϕ_{late} assessed on the missed amount and returns to good standing; foreclosure from the delinquent state is the non-recourse surrender of Section 4.7. The delinquency state persists at most one year before it must resolve.

Forbearance mechanics. An approved forbearer defers a fraction $1 - \xi = \mathcal{D}/12$ of the year's payment (equation 15), where $\mathcal{D}$ is the drawn number of months in hardship. Deferred interest is capitalized into the balance and the term is extended so that the scheduled payment after forbearance matches the payment before it; the forbearance-used indicator is set to $f' = 1$. The four repayment regimes that settle the deferred balance on cure are described, and varied, in Section 8.2.

B Consumption-Equivalent Welfare

The welfare comparison of Section 7.3 asks, for each shock-period owner who chose Refinance or Forbearance, how much better off that action left her than two counterfactual actions—Pay Mortgage and Default—in consumption-equivalent terms. All three objects are continuation values *one period after the shock*, evaluated from the household's crisis-period state at the same realized next-period income draw, so that the only thing that differs across them is the action taken.

The three continuation values. For a shock-period owner in state (h, a, b, η, f, s) , the chosen-action value V^{ch} is the stored owner value V^{own} evaluated at the next-period state the simulation actually realized. The two counterfactuals re-run the one-period transition from the *same* crisis-period grid state under an alternative decision rule: the pay-mortgage value V^{pay} applies the pay

policies (amortization net of any prepayment) and reads V^{own} at the resulting balance and assets, with the forbearance-used flag and locked rate left unchanged; the default value V^{def} applies the default policy, which surrenders the home under the non-recourse rule of Section 4.7, and reads the non-owner value V^{non} . All three are evaluated at the identical realized income innovation η' drawn in the simulation, so the comparison isolates the action rather than the luck of the subsequent draw.

The consumption-equivalent variation. Flow utility is $u(c, h) = (c^\alpha (\theta^{\text{own}} h)^{1-\alpha})^{1-\sigma} / (1-\sigma)$. Because consumption enters the kernel as c^α and the kernel is raised to the power $1-\sigma$, scaling *all* future consumption by a common factor $1+\omega$ multiplies each period's utility—and hence the entire discounted value—by $(1+\omega)^{\alpha(1-\sigma)}$. The consumption-equivalent variation of the chosen action over a counterfactual is the transfer ω that equates the counterfactual value to the chosen value, $(1+\omega)^{\alpha(1-\sigma)} V^{\text{alt}} = V^{\text{ch}}$, i.e.

$$\omega = \left(\frac{V^{\text{ch}}}{V^{\text{alt}}} \right)^{1/(\alpha(1-\sigma))} - 1, \quad \text{alt} \in \{\text{pay}, \text{def}\}. \quad (32)$$

A positive ω says the chosen action delivers the welfare of a uniformly $(1+\omega)$ -times-larger consumption stream under the counterfactual. Equation (32) is exact for the CRRA–Cobb–Douglas kernel absent a bequest motive and is the standard lifecycle approximation once the bequest term is present.

The numerical alternative. Equation (32) is exact only if every term in V is homogeneous of the same degree in consumption. Two are not: the default penalty ψ enters additively, and the bequest term is a function of terminal net worth rather than of the consumption stream. The approximation error is largest precisely where the comparison matters most—close to the default margin—so I also solve the comparison numerically. For each household I find the transfer Δ satisfying

$$V^{\text{alt}}(a^{\text{alt}} + \Delta) = V^{\text{ch}}, \quad (33)$$

by bisection with linear interpolation in assets: the resources the household would have to be given under the counterfactual action to be left as well off as under the action it took. This is a compensating variation in resources rather than in consumption. It requires no homogeneity assumption, it evaluates the model's own value function including the default penalty and the bequest, and—because it solves an equation in Δ rather than taking a ratio of two values—it cannot suffer the sign degeneracy described below. It is reported in years of household income.

Equation (33) has no interior solution when the answer lies outside the asset grid. It is censored *above* when the household is better off under the chosen action than under the counterfactual at any feasible wealth, so the required transfer exceeds $a^{\text{max}} - a^{\text{alt}}$, and *below* when the reverse holds at a^{min} . Both are reported at the relevant grid bound rather than dropped: they are by construction the extreme-gain and extreme-loss cells, and discarding either would bias the median. Censoring is not rare, and it is mostly from below. In the baseline, censoring above does not occur in any

of the four comparisons; censoring below is 30.3% of refinancers against pay-mortgage, 1.7% of refinancers against default, 24.7% of the forbearance-approved against pay-mortgage and 39.6% of the forbearance-approved against default. The reported statistic is the median, which is invariant to how far a censored cell lies beyond its bound so long as fewer than half the cells are censored on one side—as is the case in all four comparisons—so the medians of Table 15 are unaffected. The means would not be, which is a further reason they are not reported.

Sign degeneracy and robust reporting. At the calibrated $\sigma = 2$ the normalized value function is negative, and near grid boundaries—an almost-defaulting owner, a near-worthless counterfactual— V^{alt} can approach zero, so the ratio in (32) explodes and can even change sign. The formula is therefore applied only where V^{ch} and V^{alt} carry the same sign, and is left undefined otherwise; cells with $|\omega|$ beyond a wide economic band are discarded as well. The resulting loss of sample is concentrated in the comparison against default: 2.5% of refinancers and none of the forbearance-approved are dropped from the pay-mortgage comparison, and 10.4% and 13.7% respectively from the default comparison. This is the reason the numerically solved transfer of (33)—which is defined for every household and censors rather than discards—is reported alongside the closed form, and the reason the closed-form column should be read ordinally. Because the surviving distribution is heavy-tailed, the paper reports the *median* ω by group and by cell rather than the mean, alongside the sign-only statistic $\Pr(V^{\text{ch}} > V^{\text{alt}})$ —the share of agents made better off, which needs no ratio and is immune to the degeneracy (Table 15). The age $\times$ LTV panels of Figure 18 bin agents into quintiles and are read ordinally, with symmetric colour limits set at the 90th percentile of $|\omega|$.

C Reduced-Form Specifications

The linear probability models of Section 2.2 regress indicators for ever reaching 90+ delinquency, ever entering forbearance, and—conditional on serious delinquency—ever foreclosing, on standardized origination characteristics (loan size, LTV, DTI, note rate, credit score), a 30-year-term indicator and loan-purpose indicators, with origination-year (vintage) fixed effects and HC1-robust standard errors. Each specification is estimated twice, once on all vintages and once on the post-2010 book, so that the post-crisis underwriting regime can be read separately. Tables 21 and 22 report the full coefficient set behind Figures 1 and 2. The hazard models of Section 2.3 estimate time-to-foreclosure from first serious delinquency during COVID, controlling for origination LTV, DTI, credit score, note rate, log balance, and loan purpose. Four specifications are reported in Table 4: the original design, with an ever-forbearance indicator and prepayment treated as independent censoring; a cause-specific model whose exposure switches on at the observed forbearance entry month, which removes immortal-time bias; a Fine–Gray subdistribution model treating prepayment as a competing risk; and the two corrections combined with a six-month landmark. The preferred estimate is the landmark plus Fine–Gray specification, a hazard ratio of 0.126 (95% CI [0.105, 0.151]). Cumulative incidence functions are reported in Table 5; because the Grambsch–Therneau test rejects

proportionality for the forbearance term, those are the preferred summary.

Because all three outcomes are rare, each specification is re-estimated by logit as a check on the linear-probability functional form. The logits are fitted by Newton–Raphson (iteratively reweighted least squares) on the same design matrices, with White–Huber robust standard errors, and are summarised by the average marginal effect $\overline{p(1-p)} \cdot \hat{\beta}_j$, which is on the same probability scale as the LPM coefficient and so is directly comparable to it. Tables 23 and 24 place the two side by side. Models 1 and 2 converge cleanly. The two model-3 logits stop at the iteration cap, but for a reason that leaves the estimates of interest intact: one origination-year dummy, the 2024 cohort, is perfectly separated—the direction of the divergence identifies it as a cell with no foreclosures among its seriously delinquent loans, which is expected given how little time those loans have had to reach termination—so its coefficient runs toward $-\infty$ and never clears the step tolerance. It reaches -24 , against a maximum absolute value of 1.9 among all substantive coefficients. When an indicator is separated in this way the remaining parameters converge to the maximum-likelihood estimates of the sample that excludes that cell, so the origination characteristics, the main effects and the interaction are unaffected by the cap. McFadden pseudo- R^2 runs from 0.12 to 0.29. Across the thirty standardised origination coefficients—loan size, LTV, DTI, credit score and note rate in each of the six specifications—the linear coefficient and the logit marginal effect agree in sign in every case, and the two main effects of the foreclosure model, the ever-forbearance and COVID-onset indicators, agree in sign and significance as well.

The single exception is the forbearance $\times$ COVID interaction of model 3. The linear model estimates it at +23.7 percentage points on all vintages and +7.1 on the post-2010 book, both precisely estimated; the logit puts the all-vintage interaction at 0.05 log-odds with a standard error of 0.16—indistinguishable from zero—and the post-2010 interaction at -0.68 with a standard error of 0.21, significantly negative. The disagreement is not an artefact of the marginal-effect approximation, which is only first-order for indicators and their products: the sign reverses in the logit coefficient itself. It is instead what one expects when a linear model is asked to fit a probability near the boundary of the unit interval. The interaction identifies COVID-onset delinquencies that entered forbearance, and that group foreclosed at 1.0% (Table 3, Group B), so the linear specification is assigning a seven-point shift to a probability of one point. We therefore report the interaction for completeness but place no interpretive weight on it; nothing elsewhere in the paper depends on it, and the protective association it is sometimes read as measuring is estimated properly in Section 2.3 by the landmark Fine–Gray hazard.

D Computation and Reproducibility

The model is used in two stages—a heavy backward-induction *solve* and a lighter forward *simulation*—deliberately separated so that many counterfactuals can reuse a single solve.

Table 21: Linear probability models of serious delinquency and forbearance take-up.

	Pr(ever 90+ dlq.)		Pr(ever forbearance)	
	All vintages (1)	Post-2010 (2)	All vintages (3)	Post-2010 (4)
log(Loan amount) [SD]	-0.583*** (0.022)	-0.102*** (0.024)	0.167*** (0.016)	0.337*** (0.024)
Origination LTV [SD]	0.954*** (0.019)	0.427*** (0.021)	0.225*** (0.014)	0.364*** (0.022)
DTI [SD]	0.825*** (0.020)	0.793*** (0.023)	0.515*** (0.012)	1.044*** (0.024)
Credit score [SD]	-3.289*** (0.027)	-2.650*** (0.034)	-1.138*** (0.017)	-2.088*** (0.032)
Note rate [SD]	1.145*** (0.059)	0.930*** (0.062)	0.887*** (0.044)	0.950*** (0.061)
30-year term	1.102*** (0.046)	0.419*** (0.050)	0.156*** (0.032)	0.241*** (0.052)
Purpose: refinance	0.998*** (0.050)	-0.125* (0.054)	-0.067 (0.036)	0.026 (0.057)
Purpose: cash-out	1.246*** (0.057)	0.084 (0.064)	-0.010 (0.039)	0.017 (0.065)
Constant	-2.208*** (0.373)	3.297*** (0.091)	-2.483*** (0.094)	2.158*** (0.078)
Vintage fixed effects	Yes	Yes	Yes	Yes
R^2	0.065	0.035	0.040	0.040
Observations	1,113,742	669,157	1,113,742	669,157

Notes: Coefficients in *percentage points* (the estimate times 100), with HC1-robust standard errors in parentheses.

Variables marked [SD] are standardised, so the coefficient is the effect of a one-standard-deviation change; the remaining regressors are indicators, so their coefficients are raw percentage-point differences. All models include origination-year (vintage) fixed effects. “Post-2010” restricts to loans originated in 2010 or later. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. These associations are endogenous and selection-laden; the causal weight of the paper rests on the structural counterfactual of Section 6. Estimated on the loan-level 2-percent Fannie Mae sample. Figure 1 plots the standardised origination characteristics of these four columns.

Table 22: Linear probability model of foreclosure conditional on serious delinquency.

	Pr(foreclosure ever 90+ dlq.)	
	All vintages (1)	Post-2010 (2)
log(Loan amount) [SD]	−4.558*** (0.200)	−2.348*** (0.181)
Origination LTV [SD]	7.183*** (0.234)	2.316*** (0.190)
DTI [SD]	−0.233 (0.205)	−0.246 (0.202)
Credit score [SD]	3.200*** (0.182)	0.574*** (0.164)
Note rate [SD]	5.669*** (0.505)	0.741 (0.399)
30-year term	8.757*** (0.553)	2.146*** (0.474)
Purpose: refinance	6.396*** (0.502)	2.258*** (0.414)
Purpose: cash-out	4.959*** (0.490)	2.284*** (0.415)
Ever forbearance	−27.044*** (0.782)	−11.024*** (1.079)
COVID-era 90+ delinquency	−20.898*** (0.729)	−11.736*** (0.832)
Ever forbearance × COVID	23.678*** (0.916)	7.104*** (1.174)
Constant	5.728 (4.131)	21.942*** (1.470)
Vintage fixed effects	Yes	Yes
R^2	0.270	0.119
Observations	47,341	20,338

Notes: Coefficients in *percentage points* (the estimate times 100), with HC1-robust standard errors in parentheses.

Variables marked [SD] are standardised, so the coefficient is the effect of a one-standard-deviation change; the remaining regressors are indicators, so their coefficients are raw percentage-point differences. All models include origination-year (vintage) fixed effects. “Post-2010” restricts to loans originated in 2010 or later. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. These associations are endogenous and selection-laden; the causal weight of the paper rests on the structural counterfactual of Section 6. The estimation sample conditions on ever reaching 90+-day delinquency. Figure 2 plots every slope coefficient of these two columns.

Table 23: Logit robustness: average marginal effects against the linear probability models.

	Pr(ever 90+ dlq.)				Pr(ever forbearance)			
	All vintages		Post-2010		All vintages		Post-2010	
	LPM	AME	LPM	AME	LPM	AME	LPM	AME
log(Loan amount) [SD]	−0.583*** (0.022)	−0.560*** (0.023)	−0.102*** (0.024)	−0.112*** (0.025)	0.167*** (0.016)	0.175*** (0.016)	0.337*** (0.024)	0.355*** (0.026)
Origination LTV [SD]	0.954*** (0.019)	1.553*** (0.027)	0.427*** (0.021)	0.862*** (0.031)	0.225*** (0.014)	0.457*** (0.019)	0.364*** (0.022)	0.707*** (0.031)
DTI [SD]	0.825*** (0.020)	0.975*** (0.019)	0.793*** (0.023)	1.129*** (0.029)	0.515*** (0.012)	0.812*** (0.018)	1.044*** (0.024)	1.373*** (0.031)
Credit score [SD]	−3.289*** (0.027)	−2.502*** (0.018)	−2.650*** (0.034)	−2.115*** (0.023)	−1.138*** (0.017)	−1.118*** (0.015)	−2.088*** (0.032)	−1.700*** (0.024)
Note rate [SD]	1.145*** (0.059)	0.995*** (0.059)	0.930*** (0.062)	0.474*** (0.060)	0.887*** (0.044)	0.399*** (0.043)	0.950*** (0.061)	0.655*** (0.067)
30-year term	1.102*** (0.046)	1.680*** (0.057)	0.419*** (0.050)	0.963*** (0.068)	0.156*** (0.032)	0.539*** (0.044)	0.241*** (0.052)	0.649*** (0.070)
Purpose: refinance	0.998*** (0.050)	0.793*** (0.052)	−0.125* (0.054)	−0.089 (0.060)	−0.067 (0.036)	0.042 (0.038)	0.026 (0.057)	0.046 (0.061)
Purpose: cash-out	1.246*** (0.057)	1.483*** (0.050)	0.084 (0.064)	0.516*** (0.059)	−0.010 (0.039)	0.227*** (0.039)	0.017 (0.065)	0.299*** (0.062)
Vintage fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
McFadden pseudo- R^2	—	0.162	—	0.124	—	0.174	—	0.131
Observations	1,113,742	1,113,742	669,157	669,157	1,113,742	1,113,742	669,157	669,157

Notes: All entries are in *percentage points*. Columns headed LPM repeat the linear-probability estimates of Table 21 (Table 22 for model 3) with HC1-robust standard errors; columns headed AME report the average marginal effect of a logit fitted by maximum likelihood on the identical sample and regressor set, $p(1-p) \cdot \hat{\beta}_j$, with standard errors from the first-order approximation $p(1-p) \cdot \text{SE}(\hat{\beta}_j)$ and significance taken from the logit coefficient. Both estimators are therefore on the same probability scale and are directly comparable. Variables marked [SD] are standardised; the remainder are indicators. All models carry origination-year (vintage) fixed effects. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The two estimators agree in sign on every standardised origination characteristic in all four columns.

Table 24: Logit robustness for foreclosure conditional on serious delinquency.

	Pr(foreclosure ever 90+ dlq.)			
	All vintages		Post-2010	
	LPM	AME	LPM	AME
log(Loan amount) [SD]	−4.558*** (0.200)	−4.674*** (0.208)	−2.348*** (0.181)	−2.163*** (0.159)
Origination LTV [SD]	7.183*** (0.234)	7.839*** (0.270)	2.316*** (0.190)	2.784*** (0.256)
DTI [SD]	−0.233 (0.205)	−0.213 (0.167)	−0.246 (0.202)	−0.103 (0.186)
Credit score [SD]	3.200*** (0.182)	2.571*** (0.163)	0.574*** (0.164)	0.301 (0.156)
Note rate [SD]	5.669*** (0.505)	5.442*** (0.547)	0.741 (0.399)	0.761 (0.392)
30-year term	8.757*** (0.553)	9.063*** (0.562)	2.146*** (0.474)	1.844*** (0.474)
Purpose: refinance	6.396*** (0.502)	4.954*** (0.485)	2.258*** (0.414)	2.072*** (0.407)
Purpose: cash-out	4.959*** (0.490)	4.538*** (0.461)	2.284*** (0.415)	2.288*** (0.397)
Ever forbearance	−27.044*** (0.782)	−25.932*** (1.285)	−11.024*** (1.079)	−4.075*** (0.581)
COVID-era 90+ delinquency	−20.898*** (0.729)	−20.890*** (1.439)	−11.736*** (0.832)	−4.463*** (0.502)
Ever forbearance × COVID	23.678*** (0.916)	0.689 (2.202)	7.104*** (1.174)	−2.585** (0.795)
Vintage fixed effects	Yes	Yes	Yes	Yes
McFadden pseudo- R^2	—	0.294	—	0.255
Observations	47,341	47,341	20,338	20,338

Notes: All entries are in *percentage points*. Columns headed LPM repeat the linear-probability estimates of Table 21 (Table 22 for model 3) with HC1-robust standard errors; columns headed AME report the average marginal effect of a logit fitted by maximum likelihood on the identical sample and regressor set, $p(1-p) \cdot \hat{\beta}_j$, with standard errors from the first-order approximation $p(1-p) \cdot \text{SE}(\hat{\beta}_j)$ and significance taken from the logit coefficient. Both estimators are therefore on the same probability scale and are directly comparable. Variables marked [SD] are standardised; the remainder are indicators. All models carry origination-year (vintage) fixed effects. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The two estimators agree on every origination characteristic and on both main effects, and disagree only on the interaction. The cell that interaction identifies—COVID-onset delinquencies that entered forbearance—has an empirical foreclosure rate of 1.0% (Table 3, Group B), so the linear model is fitting a probability near the boundary of the unit interval there and the logit is the better-behaved estimator: see Appendix C.

State space and grids. The discretized individual state comprises liquid assets on an exponentially spaced grid of $N_A = 15$ points, mortgage balance on $N_M = 15$ points, house size on $n_h = 10$ points over $[1, 2]$, the $n_\eta = 11$ -node idiosyncratic income chain, the eight normal plus the grafted crisis aggregate states, the locked-rate index, a delinquency count, a binary forbearance-used indicator, a post-default lockout indicator, and age. Survival probabilities are taken from a U.S. actuarial table over ages 25–84.

Solve. A single harness file defines the primitives, the constraints, and the Bellman operators; a thin per-cell wrapper sets the grid resolution and the cell’s policy knobs as guarded configuration constants and includes the harness. Backward induction runs from the terminal age to age 25, computing at each age and state the value and the argmax decision across the household regimes; the resulting value and policy arrays are saved to disk.

Simulate. The simulation loads a solved object, draws the panel, burns in to the stationary age distribution under the normal transition matrix, and imposes the realized crisis path. Every scenario is driven by the same seeded draws—mortality, income transitions, forbearance approvals and hardship durations, and the aggregate path—so that differences across scenarios reflect policy alone.

The counterfactual suite. Each experiment is a cell that flips one lever relative to the baseline through a single configuration constant. Cells that change only the simulated path reuse the baseline solve; cells that change the household’s problem re-solve. A summary step stacks the per-cell moments for comparison.

Replication package. The code is organized in two directories. `hpcc_solve/` holds one solve driver per cell together with its cluster job script; each driver is byte-identical to the baseline driver except for the configuration constants named in its header, so the difference between a cell’s results and the baseline’s is attributable to those constants alone. `standalone/` holds the model harness, the simulation harness, the simulation kernel, one simulation driver per experiment, and the table-building scripts, each named for the table it produces. A simulation driver writes one configuration file per cell into `_cells/` and invokes a common runner, so the exact configuration behind every reported number is written to disk rather than held in a script.

Randomness enters at two points and both are pinned. The aggregate path is generated from a fixed seed and held identical across every cell, so that scenarios differ by policy alone. The idiosyncratic draws—mortality, income transitions, forbearance approvals and hardship durations—are taken from (agent, period) tables generated once from the simulation seed before the agent loop begins, so they do not depend on the household’s own decisions, on other households, or on the order in which threads process them. This is what makes the same-agent counterfactual of Section 7.2 well posed, and what makes a given seed reproduce a given table exactly. The seed is a configuration constant; the multi-seed results of Section 8.6 vary it and nothing else.

Three estimated objects are inputs rather than outputs and are supplied as derived data files: the idiosyncratic income process, the deterministic age-earnings component, and the aggregate state space with its transition matrix. The mortgage-rate rule is estimated by OLS of the thirty-year fixed mortgage rate on the short rate and cubically detrended log output, giving $r^m = 0.0385 + 0.552 r^{\text{short}} - 0.051 \log \tilde{Y}$, reported as equation (1) and applied inside the solve at the detrended output of each aggregate state.

E Further Robustness

These checks are referenced from Section 8.4. Each varies one feature of the environment and re-runs the crisis experiment; none changes the composition result.

E.1 Which State the Crisis Strikes From

The economy is not always in the same place when a shock arrives. Imposing the identical shock from each of eight normal aggregate states, which differ in the short rate, the house-price level, and output, shows that both the foreclosure response and forbearance take-up scale with how many leveraged owners are exposed when the shock lands (Table 25). The pre-shock homeownership rate ranges from 37.4% in a low-ownership, high-rate state to 69.6% in a high-ownership state, mean owner leverage from 0.15 to 0.31, and lifetime forbearance take-up moves with them, from 0.03% in the thinnest-exposure state to 0.58% in the deepest, a rank correlation with homeownership of 0.83 across the eight states; lifetime default moves with them too, from 0.90% to 3.13%, at a rank correlation of 0.88. The incidence of a crisis, and of the policy response to it, is contingent not only on the character of the shock but on the point in the cycle at which it lands.

The sweep also carries a result that qualifies the paper’s central finding, and it should be read as a limitation rather than as a confirmation. The rate ratio $P(\text{refi} \mid FA)/P(\text{refi} \mid NA)$ is 0.64 in state 1—the state the baseline uses, and the segmentation result of Section 6.6—and it exceeds one in all seven of the others. Four of those seven rest on approved cohorts of between 11 and 132 households and cannot be interpreted one at a time: the across-replication standard deviation of the ratio is 0.046 on a cohort of about 450 (Section 8.6), and it widens sharply as the cohort thins. But states 3 and 7 carry 336 and 307 approved households, comparable to the baseline’s 445, and both sit at or above one—1.03 and 1.25—many standard deviations away from 0.64. The reversal in those two cells is not a small-sample artefact.

I do not have a verified mechanism for it, and I would rather say so than supply one. The obvious candidate—that the size of the rate cut differs across states—is not it: the crisis bundle applies an additive 2 percentage point cut to the short rate in every state, so the stimulus is identical by construction. What the table does establish is the scope of the claim. The segmentation result is a statement about a moderate contraction arriving from the modal aggregate state, which is what the COVID episode was and what the empirical work of Section 2 estimates. It is not a general property of the two instruments. Section 8.3 shows the same ratio is sensitive to the depth of the

contraction as well, and the two boundaries should be stated together whenever the result is cited. Neither is settled by the model as specified.

Table 25: State dependence: the COVID shock imposed from each of the eight normal aggregate states.

Pre-shock state	Own. (%)	LTV	Crisis appr. (%)	Take-up (%)	Ever def. (%)	Rate ratio	n_{FA}
State 1 (baseline)	67.6	0.31	1.48	0.58	3.13	0.64	445
State 2	60.2	0.31	0.41	0.16	2.21	1.35	124
State 3	69.6	0.30	1.12	0.37	2.38	1.03	336
State 4	59.5	0.25	0.32	0.11	1.38	1.61	97
State 5	53.8	0.25	0.44	0.15	1.66	1.18	132
State 6	37.4	0.15	0.04	0.03	0.98	2.82	11
State 7	56.3	0.24	1.02	0.33	1.66	1.25	307
State 8	43.7	0.18	0.08	0.03	0.90	3.61	25

Notes: Each row imposes the identical `covid_effect` shock from a different pre-shock normal aggregate state (state 1 is the baseline used elsewhere in the paper). “Crisis forb. appr.” is the shock-period forbearance-approval share of households; “Take-up” is the lifetime ever-forbearance share. Crisis take-up and lifetime default rise with the pre-shock homeownership rate and owner leverage—the depth of leveraged-owner exposure when the shock lands. The rate ratio is $P(\text{refi} \mid FA)/P(\text{refi} \mid NA)$ and n_{FA} is the number of forbearance-approved owners it is computed on; the rows with n_{FA} below about 150 are reported for completeness and should not be read individually, since the across-replication standard deviation of the ratio is already 0.046 at $n_{FA} \approx 450$ (Section 8.6). All eight rows are simulated on the same panel of 30,000 households at seed 1234, so counts as well as shares are comparable across rows and with the baseline.

E.2 Recourse

Default is non-recourse in the baseline. Re-solving under full recourse ($\lambda = 1$) lowers lifetime forbearance take-up from 0.58% to 0.51%, crisis approvals from 445 households to 398, and lifetime default from 3.13% to 3.01%, as expected when walking away carries a deficiency. The crisis-period foreclosure count is unchanged at 53. What the recourse assumption does not touch is the character of the incidence result—disjoint cohorts separated by the income test: the refinancing response and the segment each instrument reaches are governed by the underwriting constraints, not by the recourse assumption.

E.3 The Housing Backdrop

The disjointness the underwriting constraints generate is latent under rising prices and can only tighten under falling prices. Replacing the appreciating-price COVID path with a 2008-type 30% house-price collapse narrows the refinancing valve at exactly the high-LTV margin forbearance serves—crisis refinances fall by more than half, from 4,777 to 2,293, as the collapse pushes owners against the LTV and no-cash-in constraints—while forbearance continues to reach the income-stressed segment, with crisis approvals of 618 households against the baseline’s 445. Under the collapse, moreover, the model does not manufacture a 2008-scale foreclosure wave in partial equilibrium: crisis foreclosures rise from 53 to 75 with forbearance in force and from 69 to 98 without it, so the price collapse raises the foreclosure count by roughly forty percent rather than by an order of

magnitude. Forbearance removes somewhat more foreclosure under the collapse (23 crisis foreclosures) than under the COVID path (16), and what it removes is concentrated among the youngest, most-leveraged owners, the segment a price collapse renders underwater. The muted level of the response reflects the partial-equilibrium design (Section 8.8), which holds prices to their imposed path and so omits the price-spiral feedback central to the 2008 experience; the incidence pattern— forbearance reaching the income-stressed, refinancing the income-secure—survives the change in backdrop, which is the object of this paper.

E.4 Excluding Downward Housing Adjustment

The model restricts housing adjustment to holding or trading up. The restriction is empirically appropriate—households did not trade down through the 2020 crisis—and, more tellingly, the households who would use a downward-adjustment margin are not the households the paper studies. Table 26 contrasts, in a downsize-enabled version of the model, the crisis-period profile of the owners who exercise the housing-adjustment margin with those who take forbearance. The movers are equity-rich, near-unlevered, higher-income owners of large homes—optimizing a housing position—while the forbearers are the leveraged, income-stressed households relief is designed for. Mean loan-to-value is 0.03 among the movers against 0.64 among the forbearers, and mean income 1.36 against 0.53; the two distributions barely overlap. Admitting the margin therefore injects a channel absent from the data without touching the forbearance incidence result. Two caveats belong with the table. The mover group pools upward and downward adjustment, which the tabulated output of this arm does not separate, so it is an upper bound on the population a downsizing option would serve. And enabling the margin moves nothing in aggregate: crisis forbearance approvals are 449 households with downsizing permitted against 445 without it, a difference well inside the across-replication standard deviation of 25 reported in Section 8.6.

Table 26: Housing adjusters vs. forbearers in a downsize-enabled model (crisis period).

Characteristic (mean)	Sell-and-repurchase ($N = 70$)	Forbearers ($N = 449$)
Loan-to-value ratio	0.03	0.64
Mortgage balance	0.21	2.86
Income	1.36	0.53
House value	7.70	4.50
Total wealth	9.66	2.66

Notes: Crisis-period means in the `ALLOW_DOWNSIZE` arm, simulated on the same 30,000-household panel as the baseline. “Sell-and-repurchase” is the housing-adjustment decision and pools upward and downward moves, which this arm’s tabulated output does not separate. Balances, income, house value and wealth are in model units of \$60,000.

E.5 The Length of the Grace Window

The baseline treats the crisis as a single period, so a forbearance spell resolves within the year and realized duration cannot exceed twelve months (the ceiling noted in Section 8.8). The CARES grace window was longer—up to roughly eighteen months for many borrowers—and its length is itself a policy choice. I solve an extension in which households *anticipate* a two-period grace window: the backward induction prices a deterministic chain in which the impact period (`covid_effect`) is followed by a second forbearance-eligible period—output recovered, the rate cut and the CARES window still in force—before the economy returns to normal, and the simulation carries an unresolved spell across the annual boundary, so an owner still unemployed at the end of the first period keeps forbearing into the second. Solving the anticipation inside the household’s problem, rather than imposing it ex post in the simulation, lets the length of the grace window shape behavior as households actually experienced it.

The extension shows up as more relief rather than as longer individual spells, and the distinction matters for how the result is read. The number of forbearance spells rises from 424 to 1,577 and lifetime take-up from 0.58% to 1.77%, but most of that is mechanical: the extension supplies two forbearance-eligible periods where the baseline supplies one, so a household that would have taken relief in either can appear twice. Conditional on taking a spell, duration lengthens only modestly—mean realized duration 2.91 months against the baseline’s 2.67, median 3 against 2—and the share of spells resolved within six months is 97.7% against 95.5%.

What the extension does *not* produce is a spell longer than a year. The model records duration within the annual period, so an owner who carries an unresolved spell across the boundary is recorded as two spells rather than one long one, and the twelve-month ceiling noted in Section 8.8 therefore binds in this arm exactly as it does in the baseline. The extension lengthens the *window* over which relief is available; it does not let the model speak to the eighteen-month spells the CARES data contain, and no claim about the duration right tail should be rested on it.

The margin the extension leaves alone is the one the paper is about. Lifetime default is 3.14% against the baseline’s 3.13% and homeownership 67.8% against 67.6%, both inside the across-replication noise of Section 8.6. Anticipating a longer grace window changes how much relief is taken without changing *who* takes it or *whether* they ultimately default: relief reaches the same income-stressed segment and, as in the baseline, postpones default rather than preventing it. The length of the grace window is a lever over the reach of relief, not over its incidence.

F Who Sets the Deferral Intensity

In the baseline the paid fraction ξ is not chosen at application. An approved forbearer faces the within-year re-employment hazard and, month by month, compares remitting that month’s payment against deferring it; the spell ends when paying becomes preferable. Duration is therefore an outcome. The natural alternative is that the household commits at application to a paid fraction for the whole year, picked from a discrete menu. Table 27 solves and simulates that alternative

on a thirteen-point menu $\xi \in \{0, 1/12, \dots, 1\}$ —one menu point per deferred month, so the two formulations span the same set of annual outcomes—holding every other parameter, the eligibility rule, the shock and the random numbers fixed.

Table 27: Who sets the deferral intensity: re-employment hazard versus household choice

	Within-year hazard (baseline)	Annual ξ menu (1/12 steps)
<i>Crisis-period shares (%)</i>		
Forbearance approved	1.48	2.49
Chose delinquency	0.88	0.22
Refinanced	15.92	15.84
Defaulted	0.18	0.10
<i>Realised deferral</i>		
Spells	424	746
Mean months deferred	2.67	10.77
Resolved within 3 months (%)	81.1	1.9
Full twelve months (%)	0.2	72.7
<i>Value to the forbearer</i>		
Median ΔV vs. paying	0.000	−0.000
Median CEV vs. paying (%)	0.000	0.528
Exact ties in the continuation (%)	36.4	0.3

Notes. The two columns share every parameter, the same eligibility rule ($M(b, R^{\text{lock}}, n) \geq 0.20y$ and $b > 0$), the same shock and the same random numbers, and both are simulated on the same panel of 30,000 households. They differ only in who sets the deferral intensity. In the first, the household re-optimises each month and resumes payment once employment returns, so duration is an outcome. In the second it commits to a paid fraction ξ for the year, chosen from a thirteen-point grid in steps of 1/12, so that one menu point equals one month deferred and the two formulations span the same set of annual outcomes. “Chose delinquency” is the shock-period delinquency *decision* as a share of households; the delinquency *state* entering the shock period is predetermined and identical across the two columns, so it is not reported. Aggregate shares are close across the two columns, so the mechanism is not identified by take-up; the duration distributions are far apart, which is what the table is for. The value rows are reported for completeness and should not be used to rank the two mechanisms: at this mortgage-grid resolution more than a third of the hazard column’s forbearers have numerically identical pay and forbear continuations, which pins that column’s median at zero by construction. The same caveat attaches to Table 15.

Two things follow. First, the aggregate crisis-period responses are close across the two columns: forbearance approvals of 1.48% of households against 2.49%, refinancing 15.92% against 15.84%, and 53 crisis foreclosures against 30. Take-up does not sharply discriminate between the two mechanisms, and a model matched on take-up alone would be matched under either. Second, the duration distributions could hardly be further apart. The hazard produces short, targeted spells—mean 2.67 months, 81% resolved within three, one spell in 424 running the full year—while the menu sends the median household to the corner: 72.7% defer all twelve months and the mean is 10.77. The corner is not an artefact of a coarse menu, as it survives refining the menu from six points to thirteen, and it is not opportunistic take-up, since the same hardship eligibility rule screens both columns.

The value comparison is the one part of this table the current solution cannot support, and it is reported with that caveat rather than omitted. At the baseline mortgage-grid resolution, 36.4% of

the hazard column’s forbearers have pay and forbear continuations that are numerically identical—deferring under three months of payments does not move them off their mortgage grid node—so the median ΔV in that column is 0.000 by construction, and it is not comparable with the menu column, where the deferred balance is large enough that only 0.3% of forbearers tie. This is the same tie problem that Section 7.3 flags for Table 15, and until it is settled no ranking of the two mechanisms by the value they deliver should be read off this table. What stands without the value rows is the separation the table was built to show: the two mechanisms reach nearly the same households in nearly the same numbers and differ in how long relief lasts. We report the hazard formulation in the main text because its duration distribution is the one that resembles the servicing data of Section 2; we report this table because the choice matters for duration and not for incidence, and a reader should be able to see how much of the paper’s duration evidence rests on it.

G Additional Loan-Level Evidence

These exhibits support the three facts of Section 2 and are referenced from it. They are reported here rather than in the main text because the argument does not turn on them.

Table 28: Pairwise Differences in Origination Characteristics

Variable	Comparison	Mean A	Mean B	Diff.	<i>p</i> -value
Original LTV (%)	Forbearance vs. Never	77.5	70.3	7.3	< 0.001
	Foreclosure vs. Never	78.1	70.3	7.9	< 0.001
	Forbearance vs. Foreclosure	77.5	78.1	−0.6	< 0.001
DTI (%)	Forbearance vs. Never	38.3	33.6	4.8	< 0.001
	Foreclosure vs. Never	39.0	33.6	5.5	< 0.001
	Forbearance vs. Foreclosure	38.3	39.0	−0.7	< 0.001
Credit Score	Forbearance vs. Never	725.5	748.4	−22.9	< 0.001
	Foreclosure vs. Never	696.6	748.4	−51.8	< 0.001
	Forbearance vs. Foreclosure	725.5	696.6	28.8	< 0.001

Note: 2-percent all-purpose Fannie Mae sample. *p*-values are from two-sample Welch *t*-tests of the mean difference; every difference is significant at $p < 0.001$. The foreclosure group is restricted to post-2010 originations.

H The Delinquency Clock and the Endogenous Risk Set

The landmark and competing-risk corrections of Section 2 address two objections to the design behind Table 4. A third is more serious than either, and no choice of estimator resolves it, because it concerns which loans enter the risk set rather than how their hazards are modelled.

Fannie Mae’s `DLQ_STATUS` counts months of missed payment. A borrower in forbearance has a contractual right to miss payments and is exercising it, so the field continues to accrue while relief is in force. In the 2-percent sample, 72.6% of month-to-month transitions among loans currently in forbearance show `DLQ_STATUS` increasing, against 22.7% unchanged and 4.7% falling. Entry into forbearance also precedes the 90+ flag for 92.0% of the 12,822 ever-forborne loans that reach that

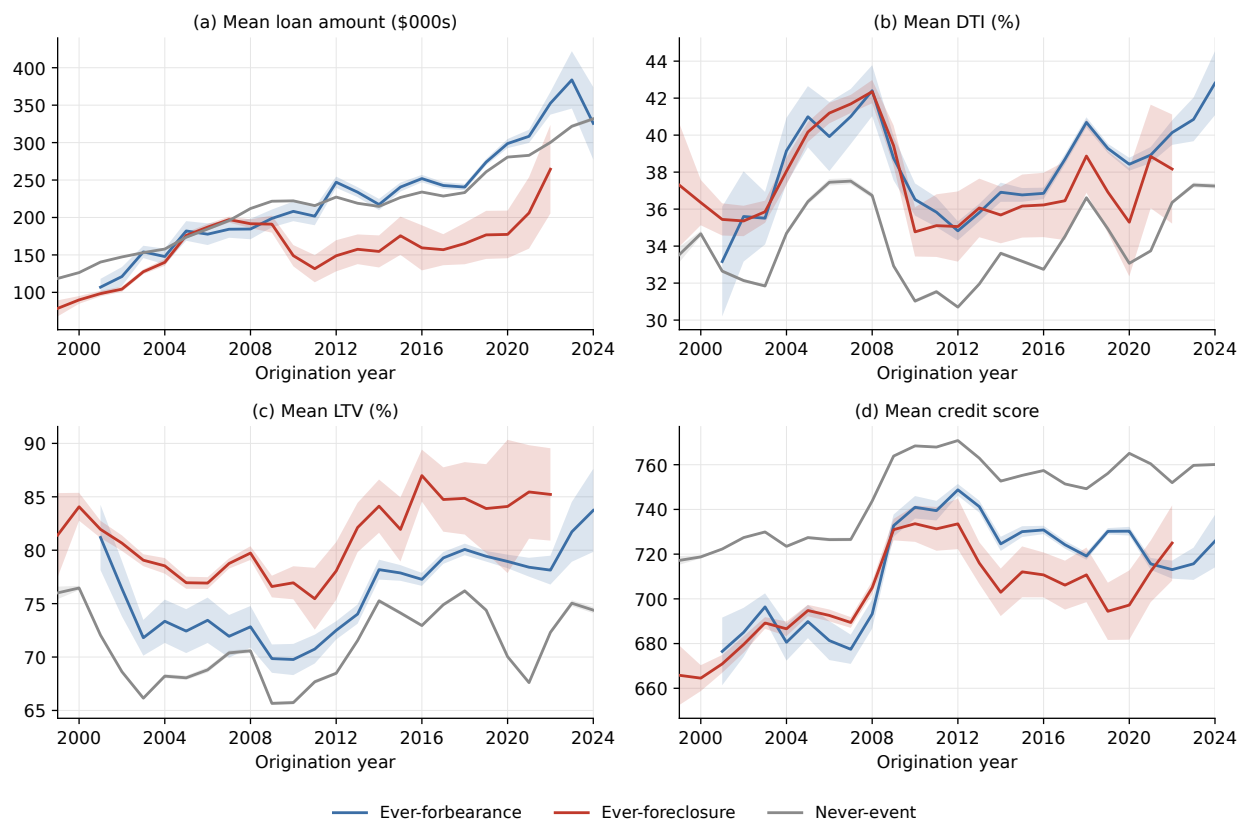


Figure 20: Mean origination characteristics by eventual-outcome group, over origination time. Each panel plots the annual cross-sectional mean of an origination variable for ever-forebearance, ever-foreclosure, and never-event loans (Fannie Mae 2-percent all-purpose sample); years with fewer than 30 loans in a group are omitted, and shaded bands are 95% confidence intervals for the annual mean. The forbearance and foreclosure groups sit above the never-event group in LTV and DTI and below it in credit score at essentially every vintage—the pooled gaps in Table 1 are stable over time. The ever-foreclosure cohorts thin sharply after 2012 (widening bands), because post-QM originations rarely reach foreclosure.

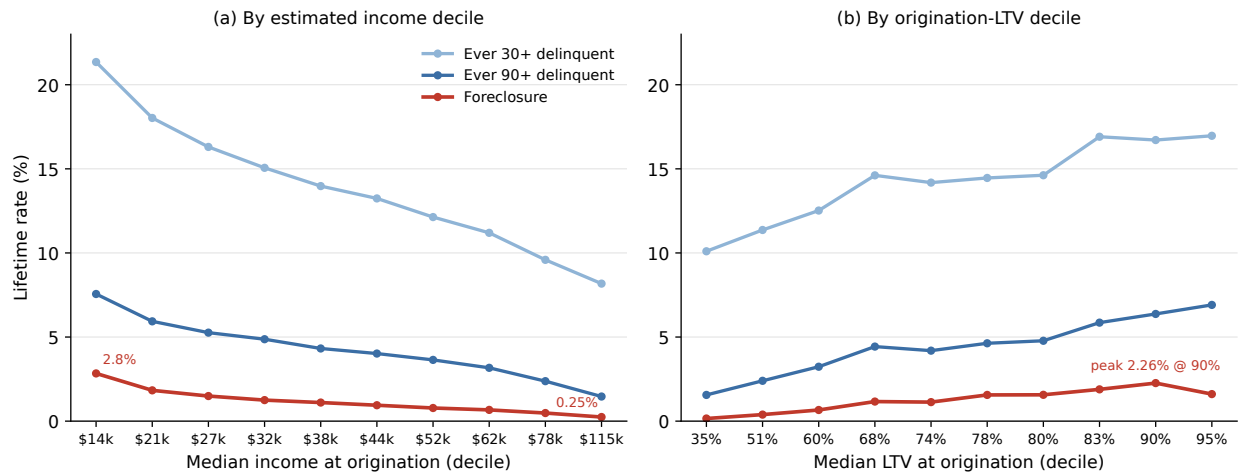


Figure 21: Lifetime delinquency and foreclosure rates by decile of estimated income at origination (panel a) and of origination LTV (panel b), Fannie Mae 2-percent all-purpose sample. Estimated income is imputed from the scheduled principal-and-interest payment and the debt-to-income ratio and is a lower bound. Foreclosure falls by roughly a factor of eleven across the income distribution and peaks at the 90%-LTV decile.

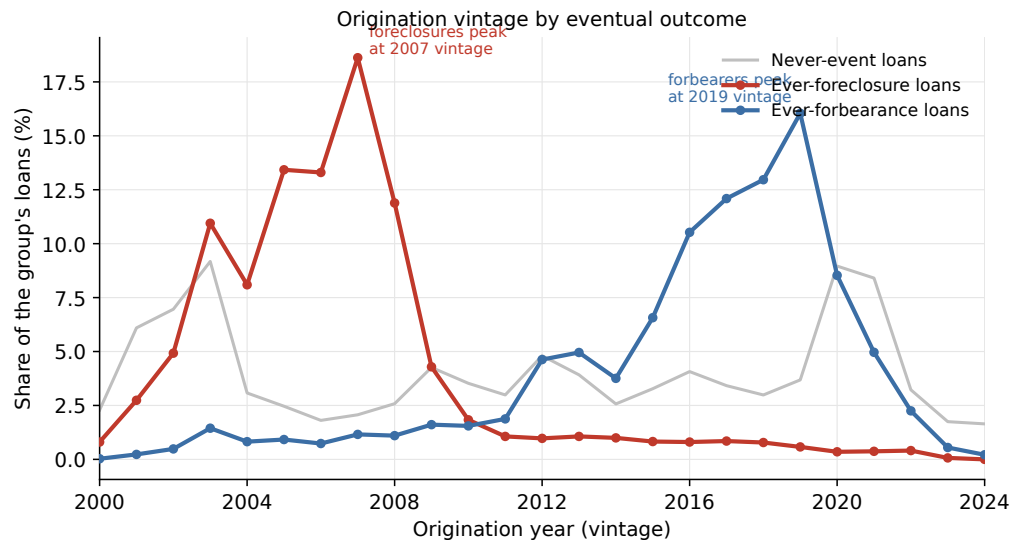


Figure 22: Distribution of each outcome group's loans across origination vintages, Fannie Mae 2-percent all-purpose sample. Ever-foreclosure loans (crimson) concentrate in the 2005–2008 originations; ever-forbearance loans (steelblue) in the 2015–2021 originations; never-event loans (grey) are shown for reference. The two distressed populations—nearly identical on origination leverage and payment stress—are drawn from almost non-overlapping vintages.

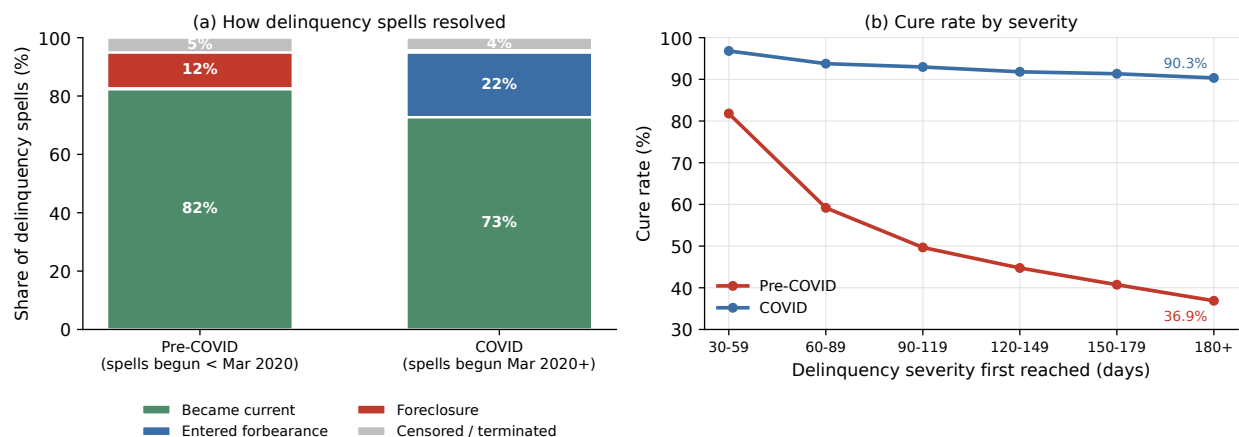


Figure 23: **Panel (a):** composition of delinquency-spell exits by first terminal event, for spells beginning before versus during COVID (Fannie Mae 2-percent sample). The foreclosure exit (12.3% pre-COVID) is replaced by the forbearance exit (22.2% during COVID). **Panel (b):** cure rate by the delinquency severity a loan first reached, pre-COVID versus COVID. The steep pre-COVID severity gradient—cure falling to 37% at 180+ days—flattens under COVID, with even the deepest delinquencies curing about 90% of the time.

threshold during the COVID window, arrives in the same month for a further 2.3%, and follows it for only 5.7%. The 90+ threshold that defines the risk set is therefore, for most treated loans, a consequence of treatment rather than an independent measure of distress. Conditioning on it selects treated units on a post-treatment variable, and the surviving contrast drifts toward sanctioned non-payment versus unsanctioned non-payment—a comparison in which the treated arm cannot foreclose almost by construction.

A clock that does not accrue in forbearance. Define *unexcused* delinquency as months in which the loan is delinquent *and* not in forbearance, counted since the loan was last current. A borrower who enters forbearance current, defers for a year and resumes paying never reaches 90+ on this clock; the raw clock records her as 90+ and not foreclosed. The measure is deliberately conservative: it never counts a month the borrower was entitled to miss. Rebuilding the COVID-era risk set on it retains 28.1% of the treated loans the raw clock admits (3,603 of 12,822); a cruder rule requiring only that the 90+ month itself fall outside forbearance retains 37.7%. The frozen frame is thus a smaller and different population, not a corrected version of the same one.

Two features of Table 29 matter more than the point estimates. Half the sample is lost but 418 of 423 foreclosures are retained: the loans that drop out are those admitted to the risk set only because forbearance placed them there, and they were not foreclosing. And the treated share falls from 88% to 46%, which is why the effective sample size of the reweighted controls rises from 16% to 52%. The raw-clock frame offers so little common support that its inverse-probability estimate rests on a small minority of controls; the frozen frame is the first specification here in which reweighting is defensible.

Table 29: The conditional frame under each delinquency clock.

Clock	Loans	Treated	Foreclosures	Raw diff.	IPW
Raw DLQ_STATUS	14,054	12,379	423	−14.21	−4.83
Frozen (no accrual in forbearance)	6,735	3,094	418	−4.49	−1.82

Notes: Differences in percentage points, treated minus control, outcome is foreclosure within 36 months of a 6-month landmark. Both rows use the same frame builder, the same covariates (origination LTV, DTI, credit score, note rate, log balance, purpose, onset month, state) and the same 1st/99th percentile trimming; applying that builder to the raw clock reproduces the Section 2 estimate of −4.83 exactly, which is what makes the two rows comparable. Effective sample size of the reweighted controls is 276 of 1,675 (16.4%) on the raw clock and 1,892 of 3,641 (52.0%) on the frozen clock.

How much was definitional. The raw difference falls from −14.21 to −4.49 percentage points and the reweighted estimate from −4.83 to −1.82, so roughly two-thirds of the association does not survive the change of clock. For comparison, the immortal-time and competing-risk corrections of Table 4 move the hazard ratio from 0.117 to 0.126, about a tenth.

Selection on unobservables. Applying the Oster (2019) bound to both frames, at the conventional $R_{\max} = 1.3 R_{\text{long}}^2$, gives a bias-adjusted coefficient of −1.78 percentage points on the raw clock and −1.29 on the frozen clock, with $\delta = 1.35$ and $\delta = 2.58$ respectively. The higher δ on the frozen frame is not an artifact of the smaller coefficient: treatment alone explains almost none of the variation there ($R^2 = 0.009$ against 0.073), so the controls do nearly all the work and movement from the short to the long coefficient is cheap in R^2 terms. The result is not robust to a more generous $R_{\max}$: at $2 R_{\text{long}}^2$ the frozen bias-adjusted coefficient turns positive (+0.62 percentage points, $\delta = 0.77$). Taken together with the reweighted estimate, the defensible statement is a reduction of 1.3 to 1.8 percentage points that requires unobserved selection roughly two and a half times the observed selection to eliminate, and that does not survive assuming the observables explain half again as much as they do.

A frame that conditions on nothing post-treatment. Fixing the population before the shock—loans current in February 2020—and comparing foreclosure over the following four years by CARES-window forbearance take-up gives +0.37 percentage points raw and +0.34 reweighted: forborne loans foreclosed at 0.42% against 0.05%, a relative risk of 8.4. The sign is opposite to the conditional estimates, which is the clearest single indication that the conditional frame is measuring selection into a risk set rather than an effect of relief. The price is dilution—most of this population was never at risk—and only 190 foreclosures.

On the same pre-determined population, replacing the outcome with unexcused 90+ delinquency raises the event count from 190 to 6,258. Forborne loans reach it at 17.4% against 1.28%, a difference of +16.1 percentage points raw and +15.3 reweighted, with observables explaining almost none of the gap. Read alongside the foreclosure comparison, this says the program reached borrowers roughly 13.6 times more likely to accumulate three months of unexcused non-payment, whose foreclosure differential was proportionally smaller than their distress differential. It is a statement about who

took up relief, not about what relief did to them: treatment and outcome windows overlap, so nothing here is a pre-post comparison.

Servicer leniency. The instrument is a servicer’s forbearance grant rate among loans in the risk set, so it changes with the clock and must be re-estimated rather than carried over. On the frozen frame it is far better conditioned: grant rates range from 0.32 to 0.70 across 13 named servicers with a standard deviation of 0.124, against 0.74 to 0.96 and 0.077 on the raw clock, where an 88% treated share compresses every servicer against the ceiling. The first-stage F clustered on servicer rises from 14.2 to 34.9, and clustering now costs almost nothing relative to the naive F of 35.9.

This does not rescue the design. Among loans that reached 90+ delinquency *before* the pandemic, when no forbearance program existed, a servicer’s COVID-era grant rate still predicts foreclosure ($\hat{\beta} = -0.433$, $t = -6.11$; -0.449 and $t = -5.17$ on the raw clock). Lenient servicers also modify far more ($t = -11.5$). Both are direct paths from the instrument to the outcome, and they are properties of servicer behaviour rather than of the risk set. A stronger first stage for a violated exclusion restriction is not progress. With 13 clusters, well below the conventional threshold of about 42, even the clustered standard error would require a wild cluster bootstrap before it could be reported.

What this establishes. The loan-level data support a bracket, not a point. Conditional on genuine unexcused delinquency, borrowers who entered forbearance foreclosed 1.3 to 1.8 percentage points less often, on the only frame in this paper with adequate common support; over the whole pre-determined book they foreclosed 0.34 percentage points more often, because take-up concentrated among the far more distressed. Neither number is causal. The binding constraint is events: 197 foreclosures in the analysis frame, and roughly 680 post-2020 foreclosures in the 2-percent sample as a whole. Establishing the counterfactual this paper cares about—whether the rate cut would have reached these same borrowers—is what the structural model is for.